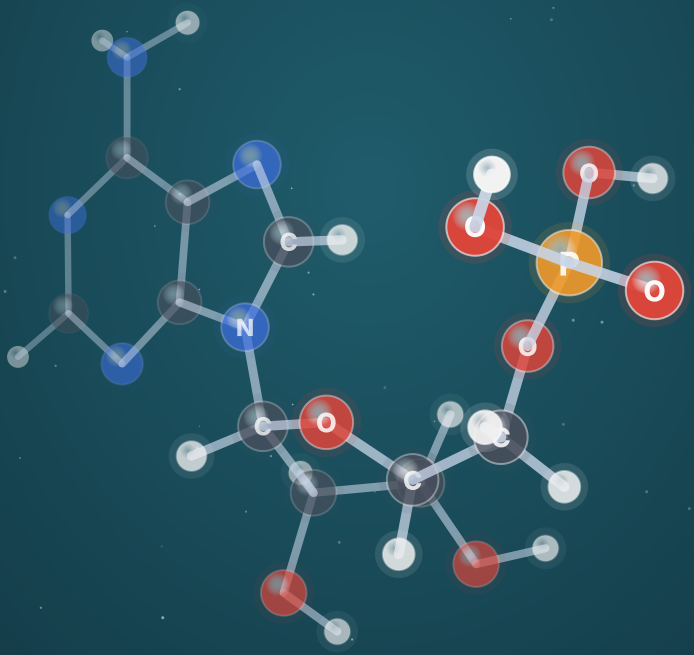
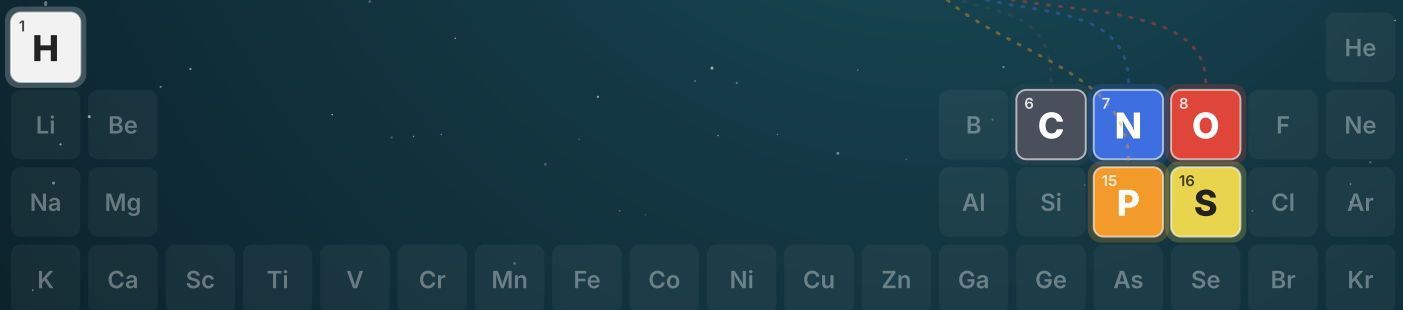
A Chemistry Primer

for the Origins of Life





GOKYONA WAHIRO

A Chemistry Primer for the Origins of Life

Gokyona Wahiro

2026-11-01

A Chemistry Primer for the Origins of Life

Gokyona Wahiro

Version 1.0, 2026

© 2026 Gokyona Wahiro

This work is licensed under the Creative Commons Attribution-NonCommercial-NoDerivatives 4.0 International License (CC BY-NC-ND 4.0). You may copy and redistribute it in any medium or format for non-commercial purposes, provided that you give appropriate credit and do not distribute modified versions. To view a copy of the license, visit https://creativecommons.org/licenses/by-nc-nd/4.0/.

Cite as: Wahiro, G. (2026). *A Chemistry Primer for the Origins of Life* (Version 1.0). Zenodo. https://doi.org/10.5281/zenodo.23857281

ORCID: https://orcid.org/0009-0008-1514-8856
Website: https://gokyona.github.io
Contact: gokyona.wahiro@gmail.com

To my wife, Julie,
and to our three children

Table of contents

Preface **1**
Who this book is for . . . 1
How the book is written . . . 1

How to Use This Book **3**
The path through the book . . . 3
Conventions . . . 3

1. Numbers for Chemistry **5**
The question of this chapter . . . 5
1.1. Very large and very small numbers . . . 5
1.1.1. Powers of ten . . . 5
1.1.2. Scientific notation . . . 5
1.1.3. Calculating with powers . . . 6
1.1.4. Orders of magnitude . . . 7
1.2. Quantities and units . . . 7
1.2.1. A quantity is a number times a unit . . . 7
1.2.2. The International System of Units . . . 7
1.2.3. Prefixes . . . 8
1.2.4. Converting units . . . 8
1.2.5. Temperature . . . 9
1.2.6. Significant figures . . . 9
1.3. Exponential change . . . 9
1.3.1. Growth by a constant factor . . . 9
1.3.2. Decay by a constant factor . . . 10
1.3.3. The number e . . . 11
1.3.4. Everyday example: compound interest . . . 11
1.4. Logarithms . . . 11
1.4.1. The logarithm answers “which power?” . . . 11
1.4.2. Rules for logarithms . . . 12
1.4.3. Logarithmic scales . . . 12
1.4.4. The natural logarithm . . . 13
1.5. Probability . . . 14
1.5.1. Probabilities as numbers between 0 and 1 . . . 14
1.5.2. Independent events multiply . . . 14
1.5.3. Counting possibilities . . . 14

1.6. Reading graphs . . . 15
1.6.1. Axes, variables and slope . . . 15
1.6.2. Graphs with logarithmic axes . . . 16
Key points . . . 16
Self-test . . . 16

2. What Matter Is Made Of **19**
The question of this chapter . . . 19
2.1. Substances and mixtures . . . 19
2.1.1. Pure substances . . . 19
2.1.2. Elements and compounds . . . 19
2.1.3. Atoms and molecules . . . 20
2.1.4. Chemical formulas . . . 20
2.1.5. Solid, liquid and gas . . . 21
2.2. Inside the atom . . . 21
2.2.1. Three particles . . . 21
2.2.2. Mostly empty space . . . 22
2.2.3. Neutral atoms . . . 23
2.3. Atomic number, mass number and isotopes . . . 23
2.3.1. Atomic number defines the element . . . 23
2.3.2. Mass number . . . 23
2.3.3. Isotopes . . . 24
2.3.4. Stable and radioactive nuclei . . . 25
2.4. Ions . . . 25
2.4.1. Gaining and losing electrons . . . 25
2.4.2. Ionic compounds . . . 25
2.4.3. Ions made of several atoms . . . 25
2.5. The periodic table . . . 26
2.5.1. Arranging the elements . . . 26
2.5.2. Metals and non-metals . . . 26
2.5.3. The elements of life . . . 27
2.5.4. Where the elements came from . . . 27
2.6. Weighing atoms . . . 27
2.6.1. The atomic mass unit . . . 27
2.6.2. Standard atomic weights . . . 28
2.6.3. Molecular mass . . . 28
Key points . . . 28
Self-test . . . 29

3. Electrons and Bonds **31**
The question of this chapter . . . 31
3.1. Electrons in shells . . . 31
3.1.1. Energy levels . . . 31
3.1.2. Shells, subshells and orbitals . . . 31

3.1.3. How many electrons fit . . . 32
3.1.4. Electron configurations . . . 32
3.1.5. Valence electrons . . . 33
3.2. Why atoms bond . . . 33
3.2.1. Measuring how tightly electrons are held . . . 33
3.2.2. The octet rule . . . 33
3.3. Ionic bonds . . . 34
3.4. Covalent bonds . . . 35
3.4.1. Sharing a pair of electrons . . . 35
3.4.2. How many bonds each atom forms . . . 35
3.4.3. Lewis structures . . . 36
3.4.4. Phosphorus: an exception that matters . . . 36
3.5. Double and triple bonds . . . 37
3.5.1. More than one shared pair . . . 37
3.5.2. Shorter and stronger . . . 37
3.6. The shapes of molecules . . . 38
3.6.1. Electron pairs keep apart . . . 38
3.6.2. Methane, ammonia and water . . . 38
3.6.3. Flat and straight molecules . . . 38
3.6.4. Hybrid orbitals, briefly . . . 38
3.7. Carbon, the element of chains and rings . . . 39
Key points . . . 40
Self-test . . . 40

4. Reading Molecular Drawings **41**
The question of this chapter . . . 41
4.1. One molecule, many drawings . . . 41
4.1.1. Why the molecular formula is not enough . . . 41
4.1.2. Four ways to draw ethanol . . . 42
4.2. Structural and condensed formulas . . . 43
4.2.1. Structural formulas . . . 43
4.2.2. Condensed formulas . . . 43
4.3. Skeletal formulas . . . 44
4.3.1. The four rules . . . 44
4.3.2. Reading a skeletal formula . . . 44
4.3.3. Rings . . . 45
4.3.4. A worked example: ribose . . . 45
4.4. Charges and lone pairs in drawings . . . 46
4.5. Three dimensions on paper . . . 47
4.5.1. Wedges and dashes . . . 47
4.5.2. Drawing sugar rings: the Haworth projection . . . 47
4.5.3. Ball-and-stick and space-filling models . . . 48
4.6. Flat rings with alternating double bonds . . . 48
4.6.1. Benzene . . . 48

4.6.2. Aromatic rings . . . 49
Key points . . . 50
Self-test . . . 51

5. Polarity, Hydrogen Bonds and Water 53
The question of this chapter . . . 53
5.1. Unequal sharing . . . 53
5.1.1. Electronegativity . . . 53
5.1.2. Polar bonds . . . 54
5.1.3. Polar molecules . . . 54
5.2. Forces between molecules . . . 55
5.2.1. Bonds within molecules and forces between them . . . 55
5.2.2. Dipole-dipole and dispersion forces . . . 55
5.2.3. The hydrogen bond . . . 56
5.2.4. Evidence from boiling points . . . 57
5.3. Water, the solvent of life . . . 57
5.3.1. A small molecule with four hydrogen-bonding sites . . . 57
5.3.2. Solutions . . . 57
5.3.3. Dissolving ions . . . 58
5.3.4. The hydrophobic effect . . . 59
5.3.5. Some unusual properties of water . . . 59
Key points . . . 60
Self-test . . . 60

6. Counting Molecules: Moles and Concentration 61
The question of this chapter . . . 61
6.1. The mole . . . 61
6.1.1. Counting by weighing . . . 61
6.1.2. Definition of the mole . . . 61
6.2. Molar mass . . . 62
6.2.1. From daltons to grams per mole . . . 62
6.2.2. Converting between mass, amount and number . . . 63
6.2.3. How small a molecule is . . . 63
6.3. Concentration . . . 63
6.3.1. Amount concentration . . . 63
6.3.2. Some concentrations for orientation . . . 64
6.3.3. Concentration and the number of molecules in a small volume . . . 65
6.4. Dilution and concentration . . . 65
6.4.1. Dilution . . . 65
6.4.2. Concentration by evaporation . . . 65
6.5. Gases . . . 66
6.5.1. The ideal gas law . . . 66
6.5.2. Mixtures of gases and partial pressure . . . 66
Key points . . . 67

Self-test 67

7. Chemical Reactions and Equations **69**
The question of this chapter 69
7.1. What happens in a reaction 69
7.1.1. Reactants and products 69
7.1.2. Conservation of mass 69
7.2. Chemical equations 70
7.2.1. Writing an equation 70
7.2.2. Balancing an equation 70
7.2.3. State symbols 71
7.2.4. Arrows 71
7.3. Amounts in reactions 72
7.3.1. Coefficients are ratios of moles 72
7.3.2. A worked example 72
7.3.3. The limiting reactant 72
7.4. Transferring protons: acids and bases 72
7.5. Transferring electrons: oxidation and reduction 73
7.5.1. Definitions 73
7.5.2. Oxidation numbers 73
7.5.3. A ladder of carbon compounds 74
7.6. Joining and splitting with water 75
7.6.1. Condensation 75
7.6.2. Hydrolysis 75
7.6.3. Addition 76
Key points 76
Self-test 77

8. Equilibrium, Acids and pH **79**
The question of this chapter 79
8.1. Dynamic equilibrium 79
8.1.1. Two reactions at once 79
8.1.2. Equilibrium is not equality 79
8.2. The equilibrium constant 80
8.2.1. Definition 80
8.2.2. What the size of K means 81
8.2.3. An important consequence: dilution and joining 81
8.3. Shifting an equilibrium 82
8.3.1. Le Chatelier's principle 82
8.4. Water, acids and bases 82
8.4.1. Water ionises itself 82
8.4.2. pH 83
8.4.3. Neutral is not always pH 7 83

8.5. Strong and weak acids . . . 84
8.5.1. Strong acids . . . 84
8.5.2. Weak acids and K_a . . . 84
8.5.3. pH and pK_a: which form is present? . . . 85
8.5.4. Phosphoric acid and phosphate . . . 85
8.6. Buffers . . . 86
Key points . . . 86
Self-test . . . 87

9. Energy and the Direction of Change **89**
The question of this chapter . . . 89
9.1. Energy, heat and temperature . . . 89
9.1.1. Energy and its units . . . 89
9.1.2. Heat and temperature are not the same . . . 90
9.2. Enthalpy: the heat of a reaction . . . 90
9.2.1. Exothermic and endothermic . . . 90
9.2.2. Bonds and enthalpy . . . 90
9.2.3. Enthalpies of formation and Hess's law . . . 91
9.3. Entropy: the number of arrangements . . . 92
9.3.1. Heat alone does not decide . . . 92
9.3.2. What entropy measures . . . 92
9.3.3. The second law . . . 93
9.4. Gibbs energy: one quantity for both . . . 93
9.4.1. Definition . . . 93
9.4.2. Four combinations . . . 94
9.4.3. Two worked examples: water changing state . . . 94
9.5. Gibbs energy and equilibrium . . . 95
9.5.1. The standard Gibbs energy and K . . . 95
9.5.2. Away from equilibrium: the reaction quotient . . . 96
9.5.3. Why water favours hydrolysis . . . 97
9.5.4. How temperature changes K . . . 97
9.6. Driving unfavourable reactions . . . 98
9.6.1. Coupling . . . 98
9.6.2. Activated building blocks . . . 98
9.7. Favourable is not the same as fast . . . 99
Key points . . . 99
Self-test . . . 100

10. How Fast: Reaction Rates **101**
The question of this chapter . . . 101
10.1. The rate of a reaction . . . 101
10.1.1. Definition . . . 101
10.1.2. Rate laws and rate constants . . . 101

10.2. First-order reactions and half-life . . . 102
10.2.1. Exponential decay . . . 102
10.2.2. Half-life . . . 103
10.3. The energy barrier . . . 104
10.3.1. Collisions . . . 104
10.3.2. Activation energy and the transition state . . . 104
10.3.3. Only a few molecules have enough energy . . . 104
10.4. Temperature and the Arrhenius equation . . . 106
10.4.1. The equation . . . 106
10.4.2. Comparing two temperatures . . . 106
10.4.3. The Arrhenius plot . . . 106
10.4.4. An example from the origin of life: the decomposition of ribose . . . 107
10.5. Catalysts . . . 108
10.5.1. What a catalyst does . . . 108
10.5.2. What a catalyst does not do . . . 108
10.5.3. Catalysts in chemistry and in life . . . 108
10.6. Competition between rates . . . 109
10.6.1. Formation against destruction . . . 109
10.6.2. Competing reactions . . . 110
Key points . . . 110
Self-test . . . 110

11. The Carbon Toolkit **113**
The question of this chapter . . . 113
11.1. Skeletons and functional groups . . . 113
11.1.1. Carbon frameworks . . . 113
11.1.2. Functional groups . . . 114
11.2. The functional groups of life . . . 114
11.2.1. Groups containing oxygen . . . 115
11.2.2. Groups containing nitrogen . . . 116
11.2.3. The phosphate group . . . 116
11.2.4. Molecules with several groups . . . 116
11.3. Sugars: groups that react with each other . . . 117
11.3.1. Aldoses and ketoses . . . 117
11.3.2. Ring formation . . . 117
11.4. Isomers . . . 118
11.5. Chirality . . . 118
11.5.1. Left and right hands . . . 118
11.5.2. Enantiomers . . . 119
11.5.3. D and L . . . 119
11.5.4. How many stereoisomers? . . . 119
11.5.5. One hand in life . . . 120
11.6. How reactions begin: nucleophiles and electrophiles . . . 120
11.6.1. Electron-rich and electron-poor sites . . . 120

11.6.2. Curved arrows . . . 121
11.6.3. Substitution and leaving groups . . . 122
Key points . . . 122
Self-test . . . 122

12. The Building Blocks of Life in Miniature 125
The question of this chapter . . . 125
12.1. Polymers and monomers . . . 125
12.2. Nucleobases . . . 126
12.3. Nucleosides and nucleotides . . . 126
12.3.1. Base plus sugar . . . 126
12.3.2. Adding phosphate . . . 127
12.4. Chains of nucleotides . . . 128
12.4.1. The phosphodiester backbone . . . 128
12.4.2. Direction . . . 128
12.5. Base pairing . . . 129
12.5.1. Complementary pairs . . . 129
12.5.2. Complementary and antiparallel strands . . . 129
12.6. RNA and DNA . . . 130
12.7. Amino acids and proteins . . . 131
12.7.1. Amino acids . . . 131
12.7.2. Peptide bonds and proteins . . . 131
12.8. The chemistry of the origin of life, in outline . . . 132
Key points . . . 133
Self-test . . . 133

References 135

Appendices 141

A. Glossary 141

B. Units and Constants 157
B.1. SI units and prefixes . . . 157
B.2. Physical constants . . . 158
B.3. Useful conversions and reference values . . . 159
B.4. Orders of magnitude in this book . . . 159

C. Answers to the Self-Test Questions 161
Chapter 1 . . . 161
Chapter 2 . . . 161
Chapter 3 . . . 161
Chapter 4 . . . 162
Chapter 5 . . . 162

Chapter 6 163
Chapter 7 163
Chapter 8 163
Chapter 9 164
Chapter 10 164
Chapter 11 165
Chapter 12 165

Preface

This is a book about the basic ideas of chemistry: what atoms are, how they join into molecules, how molecules are drawn, how much of a substance there is, which way a reaction runs and how fast it goes. It explains these ideas from the beginning, for readers who have never studied chemistry or who studied it long ago and have forgotten most of it.

The ideas were chosen with one destination in mind: the chemistry of life, and in particular the question of how life began. Many popular accounts of the origin of life stop at the point where the chemistry begins. This book goes the other way. It ends with the molecules from which RNA is built, and it gives the reader the tools needed to follow the research that tries to explain how those molecules first formed and came to copy themselves. The same author's *From Atoms to RNA: An Introduction to the Chemical Origins of Life* tells that story in detail; a reader who has finished the present book will find it much easier to follow, although each book can be read on its own.

Who this book is for

The book is written for adult readers who are curious about chemistry and the chemistry of life. No knowledge of chemistry, physics or biology is assumed. The only mathematics assumed is arithmetic: addition, subtraction, multiplication and division. Powers, exponentials and logarithms, which appear throughout chemistry, are introduced from the beginning in Chapter 1. Readers who already know some school chemistry can use the book selectively, and the next section describes how the chapters depend on each other.

How the book is written

Every technical term is defined when it is first used and collected in the Glossary. Occasionally an everyday example is used where it shows a quantity or an effect more clearly than an abstract statement; such examples are chosen to be physically accurate, and are never a substitute for the definition. Numerical values of physical constants and properties of substances are taken from standard reference sources, which are cited.

Each chapter opens with the question it answers and ends with a list of key points and three to five self-test questions. Answers to the self-test questions are given in Appendix C.

How to Use This Book

The path through the book

The twelve chapters are arranged in the order in which their ideas are needed. Chapter 1 provides the mathematical tools: powers of ten, units, exponentials and logarithms. Chapters 2 to 5 describe atoms, the bonds between them, the way molecules are drawn, and the forces that act between molecules, with water as the central example. Chapters 6 to 10 describe chemical reactions quantitatively: how amounts of substance are counted, how reactions are written, where they stop (equilibrium), which direction they run (energy), and how fast they go (rates). Chapters 11 and 12 introduce the chemistry of carbon compounds and the molecules from which RNA is built.

The chapters build on each other, and a reader new to chemistry should read them in order. A reader who knows some chemistry can begin with any chapter and use the cross-references to fill gaps.

Conventions

Quantities are given in the units of the International System (SI), with a small number of additional units that are standard in chemistry, such as the litre and the degree Celsius; all are listed in Appendix B. British spelling is used throughout. Terms printed in **bold** are defined at that point and appear in the Glossary (Appendix A).

1. Numbers for Chemistry

The question of this chapter

Chemistry describes objects that are far too small to see, present in numbers far too large to count. A single drop of water, about 0.05 millilitres, contains about 1.7×10^{21} water molecules (a calculation shown in Chapter 6), and each molecule is less than a millionth of a millimetre across. The origin of life adds a third extreme: time spans of hundreds of millions of years, set against reactions that are over in a fraction of a second. This chapter asks: *how can such numbers be written, compared and calculated with?* It introduces the tools used for that purpose throughout chemistry: powers of ten, units, exponential change, logarithms and probabilities.

1.1. Very large and very small numbers

1.1.1. Powers of ten

A **power** is a number multiplied by itself a given number of times. The number being multiplied is the **base**, and the number of times it appears is the **exponent**, written as a raised number. Thus 10^3 ("ten to the power three") means $10 \times 10 \times 10 = 1000$, and $2^5 = 2 \times 2 \times 2 \times 2 \times 2 = 32$.

Powers of ten are especially convenient, because the exponent counts the zeros: $10^1 = 10$, $10^2 = 100$, $10^6 = 1\ 000\ 000$. By definition, $10^0 = 1$.

A negative exponent means division instead of multiplication. 10^{-1} is $1/10 = 0.1$, 10^{-2} is $1/100 = 0.01$, and 10^{-6} is one millionth, 0.000 001. For a negative power of ten, the exponent counts the places by which the decimal point moves to the left of 1: $10^{-3} = 0.001$, with the 1 in the third decimal place.

1.1.2. Scientific notation

In **scientific notation**, a number is written as a number between 1 and 10 multiplied by a power of ten. For example:

- 5 970 000 000 000 000 000 000 000 kg, the mass of the Earth, is written 5.97×10^{24} kg (NASA Space Science Data Coordinated Archive 2024);
- 0.000 000 000 052 9 m, a characteristic radius of the hydrogen atom known as the Bohr radius, is written 5.29×10^{-11} m (National Institute of Standards and Technology, n.d.-a).

To convert a number into scientific notation, move the decimal point until a single non-zero digit stands in front of it, and count the moves. Moves to the left give a positive exponent; moves to the right give a negative one. In 52 900 the point moves four places to the left, giving 5.29×10^4; in 0.0072 it moves three places to the right, giving 7.2×10^{-3}.

Scientific notation does two things at once. It makes very large and very small numbers short enough to read, and it separates the two pieces of information that a number carries: the leading part says *how much* within a factor of ten, and the exponent says *how large* overall.

1.1.3. Calculating with powers

Three rules cover almost every calculation with powers of the same base:

$$10^a \times 10^b = 10^{a+b}, \qquad \frac{10^a}{10^b} = 10^{a-b}, \qquad (10^a)^b = 10^{a \times b}$$

Each rule follows from counting factors of ten. For example, $10^3 \times 10^2 = (10 \times 10 \times 10) \times (10 \times 10)$, which contains five factors of ten, so it equals 10^5. Numbers in scientific notation are multiplied by multiplying the leading parts and adding the exponents:

$$(3 \times 10^8) \times (2 \times 10^{-3}) = 6 \times 10^5$$

and divided by dividing the leading parts and subtracting the exponents:

$$\frac{6 \times 10^{23}}{2 \times 10^3} = 3 \times 10^{20}$$

Addition and subtraction do not follow these rules: $10^3 + 10^2$ is 1100, not 10^5. To add numbers written in scientific notation, first rewrite them with the same exponent.

1.1.4. Orders of magnitude

The **order of magnitude** of a quantity is the power of ten nearest to it. Two quantities that differ by a factor of about ten differ by one order of magnitude; quantities that differ by a factor of about a million differ by six. Orders of magnitude are the natural way to compare quantities that span many powers of ten, such as the abundances of the chemical elements in the Sun, or the rates of fast and slow chemical reactions.

Figure 1.1 places some lengths from atomic physics, chemistry and biology on a scale in which each step is one order of magnitude. The span from an atomic nucleus to the Earth covers about 22 orders of magnitude.

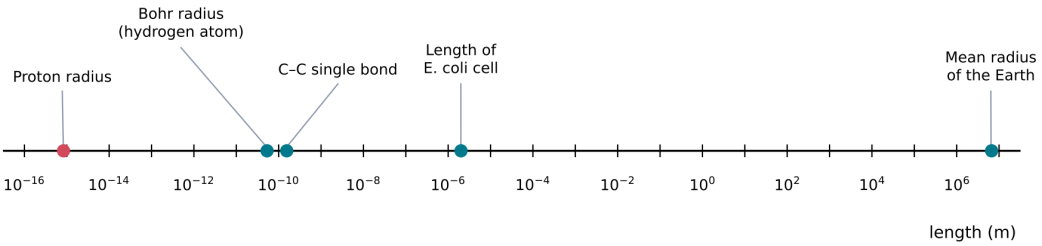


Figure 1.1.: Lengths from the atomic nucleus to the Earth, on a scale in which each step is a factor of ten. Values: proton radius about 8.4×10^{-16} m; Bohr radius 5.29×10^{-11} m; carbon-carbon single bond 1.54×10^{-10} m; length of a cell of the bacterium *Escherichia coli* about 2×10^{-6} m (Grossman et al. 1982); mean radius of the Earth 6.371×10^{6} m.

1.2. Quantities and units

1.2.1. A quantity is a number times a unit

A physical **quantity**, such as a length, a mass or a time, is expressed as a number multiplied by a **unit**. In "2.5 m", 2.5 is the number and the metre is the unit. The number alone is meaningless: 2.5 m and 2.5 km are different lengths. In calculations, units are treated like algebraic factors. They are multiplied, divided and cancelled together with the numbers, which provides a check on every result: if the units of an answer are wrong, the calculation is wrong.

1.2.2. The International System of Units

Science uses the **International System of Units** (SI). It is built on seven base units, of which five appear in this book: the metre (m) for length, the kilogram (kg) for mass,

the second (s) for time, the kelvin (K) for temperature, and the mole (mol) for amount of substance. Since 20 May 2019 every base unit has been defined by fixing the numerical value of a constant of nature (Bureau International des Poids et Mesures 2019); the mole, for example, is defined through the Avogadro constant, introduced in Chapter 6.

Other units are built from the base units and are called **derived units**. The joule (J), the unit of energy, equals 1 kg m^2 s^{-2}; the pascal (Pa), the unit of pressure, equals 1 kg m^{-1} s^{-2}. A negative exponent on a unit means "per": m s^{-1} is read "metres per second", and kJ mol^{-1} is "kilojoules per mole".

1.2.3. Prefixes

A **prefix** placed before a unit multiplies it by a power of ten (Table 1.1). A kilometre is 10^3 m, a nanometre is 10^{-9} m, and a micromole is 10^{-6} mol.

Table 1.1.: The SI prefixes used in this book.

Prefix	Symbol	Factor	Prefix	Symbol	Factor
kilo	k	10^3	milli	m	10^{-3}
mega	M	10^6	micro	μ	10^{-6}
giga	G	10^9	nano	n	10^{-9}
tera	T	10^{12}	pico	p	10^{-12}

Prefixes also apply to non-SI units. In geology, time is measured in years, written "a" (from the Latin *annus*), so that 4.5 billion years is written 4.5 Ga (giga-annum) and a million years as 1 Ma.

1.2.4. Converting units

To convert a quantity from one unit to another, multiply it by a fraction that equals 1, written so that the unwanted unit cancels. Since 1 km = 1000 m, the fraction (1000 m)/(1 km) equals 1, and

$$3.2 \text{ km} \times \frac{1000 \text{ m}}{1 \text{ km}} = 3200 \text{ m}$$

The same method converts compound units. To express 72 km per hour in metres per second:

$$72 \frac{\text{km}}{\text{h}} \times \frac{1000 \text{ m}}{1 \text{ km}} \times \frac{1 \text{ h}}{3600 \text{ s}} = 20 \frac{\text{m}}{\text{s}}$$

1.2.5. Temperature

Temperature has two scales in common scientific use. The **Celsius** scale (°C) was originally defined by the freezing and boiling points of water. The **kelvin** scale (K) measures absolute temperature: its zero, 0 K, is the lowest temperature possible, at which no further energy can be removed from a substance. A step of one kelvin is exactly the same size as a step of one degree Celsius, and the two scales are related by

$$T/\mathrm{K} = t/^{\circ}\mathrm{C} + 273.15$$

where T is the absolute temperature and t the Celsius temperature (Bureau International des Poids et Mesures 2019). Thus 25 °C, a common laboratory temperature, is 298.15 K, and 100 °C is 373.15 K. Equations that contain a temperature, such as those for reaction rates and equilibria, always require the absolute temperature in kelvin.

1.2.6. Significant figures

A measured value is never exact. The number of digits used to report it, its **significant figures**, indicates how precisely it is known. "The mean radius of the Earth is 6371 km" (four significant figures) states the value more precisely than "about 6400 km" (two). A result calculated from measured values should not be reported with more significant figures than the least precise value used. Some values are exact by definition, such as the 273.15 in the temperature conversion and the Avogadro constant; these limit nothing.

1.3. Exponential change

1.3.1. Growth by a constant factor

Many processes in chemistry and biology change by a constant *factor* in each interval of time, rather than by a constant *amount*. A population of bacteria that doubles every 20 minutes has 2 times its starting number after 20 minutes, 4 times after 40 minutes, 8 times after an hour, and 2^{n} times after n doubling times. Change of this kind is called **exponential**, because the elapsed time appears in the exponent.

The difference between constant-amount and constant-factor change is enormous over long times. Adding 2 in each of ten steps gives 20. Multiplying by 2 in each of ten steps gives $2^{10} = 1024$; in thirty steps, 2^{30}, more than a billion.

1.3.2. Decay by a constant factor

When the factor is less than 1, the quantity decreases exponentially. The most precise example in nature is **radioactive decay**. A radioactive nucleus has a fixed probability of decaying in each interval of time, independent of its surroundings and of how long it has already existed. As a result, in a large number of such nuclei, the same fraction decays in every equal interval. The time in which half of the nuclei decay is called the **half-life**, written $t_{1/2}$. After one half-life, 1/2 of the original nuclei remain; after two, 1/4; after three, 1/8; and after n half-lives,

$$\frac{N}{N_0} = \left(\frac{1}{2}\right)^n = \left(\frac{1}{2}\right)^{t/t_{1/2}}$$

where N_0 is the number of nuclei at the start, N the number remaining after time t, and $n = t/t_{1/2}$ the number of half-lives that have passed. Carbon-14, for example, has a half-life of 5700 years (Chechev and Bé 2012): after 11 400 years a quarter of it remains. Figure 1.2 compares exponential growth and decay.

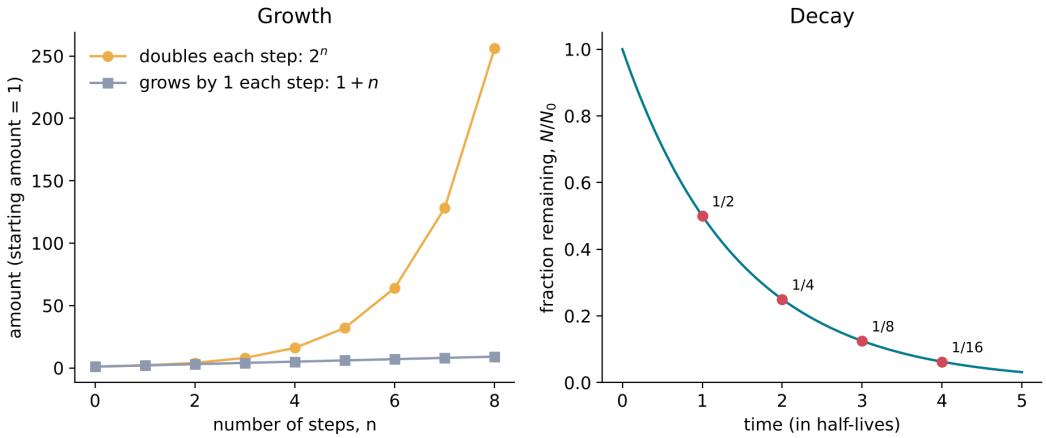


Figure 1.2.: Exponential change. Left: a quantity that doubles in each time step (orange) compared with one that grows by the same amount, the starting value, in each step (grey). Right: exponential decay, in which half of what remains is lost in each half-life; the dots mark 1/2, 1/4, 1/8 and 1/16.

Exponential decay is not limited to nuclei. Whenever each molecule of a substance has the same probability of being destroyed in each second, independently of the others, the amount of substance decreases exponentially. The breakdown of sugars in water, the reaction of many molecules with water (hydrolysis, Chapter 7), and the loss of RNA strands in any natural setting all follow this pattern, at least approximately.

1.3.3. The number e

Exponential change can be written with any base. Doubling uses base 2 and halving uses base 1/2. In science, however, one base is used far more than any other: the number **e**, equal to 2.718 28… (its digits continue without repeating). It arises naturally whenever a quantity changes at a rate proportional to its own size. If a quantity grows continuously at a rate of k per unit time, measured as a fraction of its current value, then after time t it has grown by the factor

$$e^{kt}$$

and if it decays continuously at a fractional rate k, the fraction remaining is e^{-kt}. Here k is called a **rate constant**; it has units of inverse time, for example s^{-1}, so that the product kt has no units. The expression e^{x} is often written exp(x), which is easier to read when x is a long expression; both forms are common.

The connection with half-lives is direct. Exponential decay written as e^{-kt} and as $(1/2)^{t/t½}$ describes the same curve, provided that

$$t_{1/2} = \frac{\ln 2}{k} \approx \frac{0.693}{k}$$

The symbol ln, the natural logarithm, is explained in the next section. For now it is enough to note that a large rate constant means a short half-life, and that the two are inversely proportional: doubling k halves $t_{1/2}$.

1.3.4. Everyday example: compound interest

Compound interest follows the same mathematics. A deposit that earns 5 % interest per year, added to the balance once a year, is multiplied by 1.05 each year. After 14 years it has grown by the factor $1.05^{14} = 1.98$, so it has almost doubled; after 28 years by $1.05^{28} = 3.92$, almost quadrupled. The doubling time is constant, as for any exponential growth, and is 14.2 years. The formula above, with k = 0.05 per year, gives 0.693/0.05 = 13.9 years; the small difference arises because the formula assumes that interest is added continuously rather than once a year.

1.4. Logarithms

1.4.1. The logarithm answers “which power?”

The **logarithm** to base 10 of a number x, written $\log_{10} x$, is the power to which 10 must be raised to give x. Thus $\log_{10} 1000 = 3$, because $10^3 = 1000$; $\log_{10} 0.01 = -2$, because

$10^{-2} = 0.01$; and $\log_{10} 1 = 0$. Logarithms of numbers that are not exact powers of ten lie between whole numbers: $\log_{10} 2 = 0.301$ (to three decimal places), because $10^{0.301} = 2.00$, and $\log_{10} 50 = 1.699$. The logarithm of zero or of a negative number is not defined, since no power of ten is zero or negative.

A logarithm is therefore the exponent of a number written as a power of ten. It discards the leading part and keeps the order of magnitude, refined to a decimal.

1.4.2. Rules for logarithms

The rules for powers translate directly into rules for logarithms:

$$\log_{10}(a \times b) = \log_{10} a + \log_{10} b, \qquad \log_{10} \frac{a}{b} = \log_{10} a - \log_{10} b, \qquad \log_{10} (a^n) = n \log_{10} a$$

The first rule is the reason logarithms were invented: it turns multiplication into addition. It also explains a property of every logarithmic quantity: multiplying a quantity by 10 adds exactly 1 to its logarithm, and multiplying it by 2 adds 0.301.

1.4.3. Logarithmic scales

A **logarithmic scale** reports a quantity by its logarithm instead of by the quantity itself. Equal steps on such a scale correspond to equal *factors*, not equal amounts. Two logarithmic scales will be used in this book and are common in the chemistry of life.

- The **pH scale** (Chapter 8) reports the acidity of a solution. The pH is minus the logarithm to base 10 of the concentration of hydrogen ions, so a solution of pH 4 contains ten times the hydrogen-ion concentration of a solution of pH 5, and a thousand times that of a solution of pH 7.
- The **log epsilon scale** used by astronomers reports the abundance of an element in the Sun or a star. It is defined so that hydrogen has the value 12, and every step of 1 means a factor of ten fewer atoms.

Graphs can also be drawn with logarithmic axes. On a **logarithmic axis**, the marks 1, 10, 100 and 1000 are equally spaced. Such an axis can show, on one graph, values that differ by many orders of magnitude, and it turns exponential change into a straight line. Figure 1.3 shows the same exponential decay plotted on an ordinary (linear) axis and on a logarithmic axis.

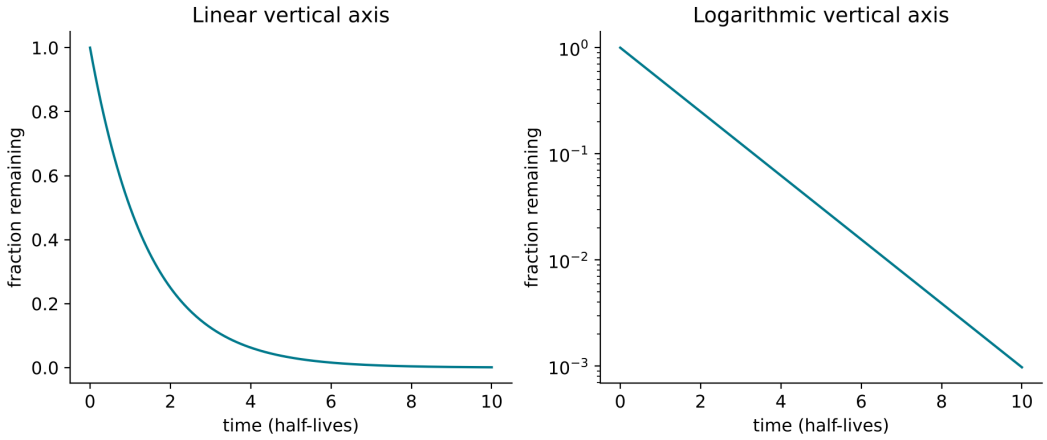


Figure 1.3.: The same exponential decay, with a half-life of 1 time unit, plotted against a linear vertical axis (left) and a logarithmic vertical axis (right). On the logarithmic axis, equal vertical steps are equal factors, and the exponential curve becomes a straight line.

1.4.4. The natural logarithm

Logarithms can be taken to any base. The logarithm to base e is called the **natural logarithm** and is written ln. Thus ln x is the power to which e must be raised to give x: ln e = 1, ln 1 = 0, and ln 2 = 0.693 (to three decimal places). The natural logarithm undoes the exponential function: $\ln(e^x) = x$ and $e^{\ln x} = x$. It obeys the same rules as $\log_{10}$.

Natural and base-10 logarithms differ only by a constant factor:

$$\ln x = 2.303 \times \log_{10} x$$

where 2.303 is ln 10, rounded to three decimal places. This factor appears whenever a natural logarithm, which arises in physical equations, is converted into a pH or another base-10 scale.

The natural logarithm explains the formula for the half-life given in the previous section. After one half-life the fraction remaining is 1/2, so $e^{-kt_{1/2}} = 1/2$. Taking natural logarithms of both sides gives $-k\ t_{1/2} = \ln(1/2) = -\ln 2$, and therefore $t_{1/2} = \ln 2 / k$.

1.5. Probability

1.5.1. Probabilities as numbers between 0 and 1

A **probability** is a number between 0 and 1 that expresses how likely an event is: 0 means that it never happens, and 1 that it always happens. A probability can also be written as a percentage, so that a probability of 0.25 is 25 %. If an event has probability p, the probability that it does *not* happen is $1 - p$.

Probabilities in chemistry usually describe the behaviour of many identical particles. When a nucleotide is added to a growing RNA chain with a probability of 0.99 of being the correct one, it means that, over many additions, 99 % are correct and 1 % are errors.

1.5.2. Independent events multiply

Two events are **independent** if the outcome of one does not affect the probability of the other. The probability that two independent events both happen is the product of their probabilities. If each of n independent events has probability p, the probability that all n happen is p^n.

This rule has a striking consequence for long sequences of steps. Suppose that each nucleotide added to an RNA chain is correct with probability 0.99. The probability that a chain of 100 nucleotides is copied with no error at all is

$$0.99^{100} = 0.366$$

so almost two-thirds of the copies contain at least one error. For a chain of 1000 nucleotides the probability falls to $0.99^{1000} = 4.3 \times 10^{-5}$, about 1 in 23 000. Figure 1.4 plots this probability for several error rates. The same calculation lies behind the idea of an error threshold in the study of the origin of life: there is a maximum length of a genetic molecule that can be copied faithfully enough to keep its information from one generation to the next.

1.5.3. Counting possibilities

When each of n positions can be filled independently with one of m possibilities, the number of different arrangements is m^n. A sequence of RNA uses four different building blocks, so the number of different RNA sequences of length n is 4^n. For $n = 10$ this is 1 048 576, about 10^6; for $n = 40$ it is about 1.2×10^{24}. The set of all possible sequences of a given length is called **sequence space**, and its size explains why no experiment can test more than a tiny fraction of the possible RNA molecules of even moderate length.

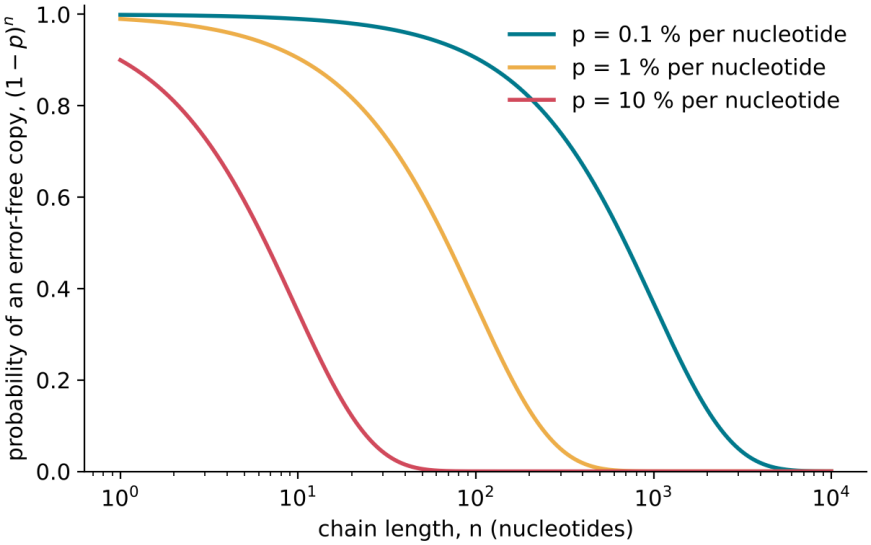


Figure 1.4.: Probability that a chain of n nucleotides is copied without error, when each nucleotide is copied correctly with probability 1 − p, for error rates p of 0.1 %, 1 % and 10 % per nucleotide. The horizontal axis is logarithmic.

Large counts of this kind are easiest to handle with logarithms. Since $\log_{10}(4^n) = n \times \log_{10} 4 = 0.602\ n$, the number of sequences of length 40 has $0.602 \times 40 = 24.1$ as its logarithm, so it is about $10^{24.1}$, or 1.2×10^{24}.

1.6. Reading graphs

1.6.1. Axes, variables and slope

A graph shows how one quantity depends on another. The quantity that is varied or controlled, such as time or temperature, is usually placed on the horizontal axis; the quantity that is measured is placed on the vertical axis. Each axis is labelled with the name of the quantity and its unit, often written in the form “Time (s)” or “Time / s”; the second form means that the numbers on the axis are the time divided by the unit, which are pure numbers.

The **slope** of a straight line is the change in the vertical quantity divided by the corresponding change in the horizontal quantity. Its unit is the vertical unit divided by the horizontal unit. On a graph of the amount of a product against time, the slope is the rate at which the product is formed. When the graph is curved, the slope at a point is the slope of the straight line that just touches the curve there.

1.6.2. Graphs with logarithmic axes

When one or both axes are logarithmic, as in Figure 1.3 and Figure 1.4, the meaning of position changes. Equal distances along a logarithmic axis are equal factors, so a point halfway between the marks 10 and 100 stands for about 32 (since $10^{1.5} = 31.6$), not 55. Before reading values from any scientific graph, check the labels of both axes and whether each is linear or logarithmic.

Two kinds of logarithmic graph are especially useful in chemistry. A graph of the logarithm of a quantity against time is a straight line when the quantity changes exponentially; its slope gives the rate constant. A graph of the logarithm of a rate constant against the reciprocal of the absolute temperature, 1/T, is a straight line for most chemical reactions; it is called an Arrhenius plot, and its slope gives the energy barrier of the reaction (Chapter 10).

Key points

- In scientific notation a number is written as a value between 1 and 10 times a power of ten, such as 5.29×10^{-11} m. Powers multiply by adding exponents and divide by subtracting them.
- A quantity is a number times a unit. SI base units used here are m, kg, s, K and mol; prefixes such as k (10^3), μ (10^{-6}) and n (10^{-9}) multiply them by powers of ten. Units cancel like algebraic factors.
- Absolute temperature T in kelvin equals the Celsius temperature plus 273.15.
- Exponential change multiplies a quantity by a constant factor in each interval. Exponential decay has a constant half-life; after n half-lives a fraction $(1/2)^n$ remains.
- Continuous exponential decay is written e^{-kt}, where e = 2.718 28… and k is the rate constant; the half-life is $t_{1/2} = \ln 2 \,/\, k \approx 0.693/k$.
- The logarithm $\log_{10} x$ is the power of ten that gives x. Multiplying x by 10 adds 1 to its logarithm. The pH scale and the log epsilon scale of abundances are logarithmic.
- The natural logarithm ln x is the power of e that gives x; $\ln x = 2.303 \log_{10} x$.
- The probability that n independent events, each of probability p, all occur is p^n. The number of RNA sequences of length n is 4^n.
- On a logarithmic axis equal distances are equal factors; exponential change appears as a straight line.

Self-test

1. Write in scientific notation: (a) 0.000 45 m; (b) 6 371 000 m. Then calculate $(2 \times 10^{-3}) \times (4 \times 10^{5})$ and $(9 \times 10^{4}) \,/\, (3 \times 10^{-2})$.

2. Convert 37 °C, the normal temperature of the human body, into kelvin, and convert 2.5 nm into metres.
3. A radioactive isotope has a half-life of 8 days. What fraction of a sample remains after 32 days? What is its rate constant k in day^{-1}?
4. Solution A has a hydrogen-ion concentration of 10^{-3} mol per litre and solution B of 10^{-6} mol per litre. By what factor do the concentrations differ, and what are the two pH values?
5. Each nucleotide in a copying reaction is correct with probability 0.999. What is the probability that a chain of 500 nucleotides is copied without any error?

2. What Matter Is Made Of

The question of this chapter

Water, air, rock and living tissue look and behave nothing alike, yet all of them are built from the same small set of ingredients. This chapter asks: *what are those ingredients, and how are they told apart?* It distinguishes elements, compounds and mixtures, describes the three particles of which every atom is made, explains how isotopes and ions arise, and introduces the periodic table and the scale on which the masses of atoms are measured.

2.1. Substances and mixtures

2.1.1. Pure substances

A **pure substance** is a sample of matter with a fixed composition and fixed properties. Pure water, for example, always contains hydrogen and oxygen in the same proportion by mass, and at a pressure of one standard atmosphere it always freezes at 0 °C and boils close to 100 °C. A **mixture** contains two or more pure substances whose proportions can vary. Sea water, air and granite are mixtures: their compositions differ from place to place, and their components can be separated by physical means such as evaporation, distillation or sorting, without any chemical change.

2.1.2. Elements and compounds

Pure substances are of two kinds. An **element** is a pure substance that cannot be broken down into simpler substances by any chemical process. Hydrogen, carbon, oxygen, iron and gold are elements. A **compound** is a pure substance made of two or more elements combined in fixed proportions. Water is a compound of hydrogen and oxygen; table salt is a compound of sodium and chlorine. A compound can be broken down into its elements by chemical reactions, and its properties are usually quite different from those of the elements it contains: sodium is a soft, reactive metal and chlorine is a poisonous green gas, but sodium chloride is the salt sprinkled on food.

Figure 2.1 summarises these categories.

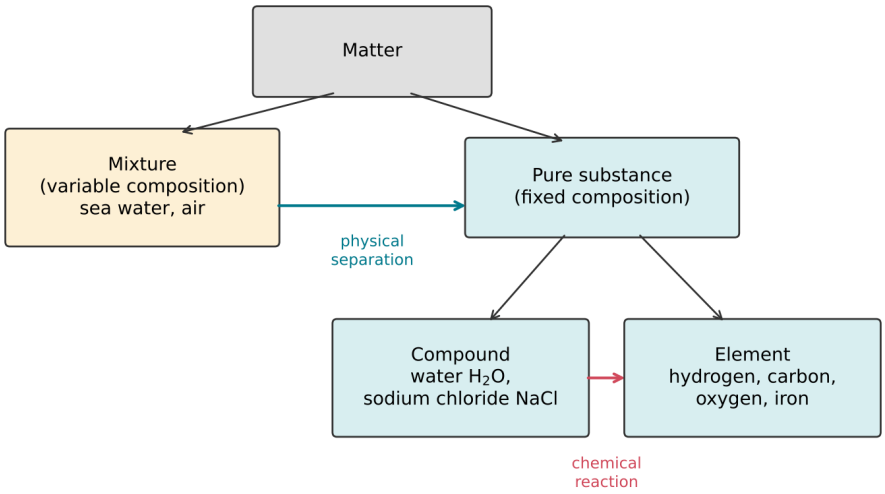


Figure 2.1.: Classification of matter. A mixture can be separated into pure substances by physical means; a compound can be separated into elements only by chemical reactions; an element cannot be broken down further by chemical means.

2.1.3. Atoms and molecules

Every element consists of tiny particles called **atoms**. An atom is the smallest unit of an element that retains its chemical identity. All atoms of one element behave chemically in the same way, and differ from the atoms of every other element.

In most compounds, atoms are held together in groups called **molecules**. A molecule is a group of two or more atoms bonded together, and it is the smallest unit of a compound that has the compound's chemical identity. A molecule of water consists of two hydrogen atoms bonded to one oxygen atom. Some elements also exist as molecules: the oxygen in air consists of molecules containing two oxygen atoms each, and the nitrogen of air likewise of two-atom molecules. Not every compound is made of separate molecules; sodium chloride, for example, is a continuous arrangement of charged particles, described in Section 2.4.

2.1.4. Chemical formulas

A **chemical formula** states which elements a substance contains, using one- or two-letter **element symbols**, and how many atoms of each are present, using subscript numbers. The formula of water is H_2O: two hydrogen atoms (H) and one oxygen atom (O); a subscript of 1 is not written. Oxygen gas is O_2, carbon dioxide is CO_2, and methane, the main component of natural gas, is CH_4. Some symbols come from Latin

names: sodium is Na (*natrium*) and potassium is K (*kalium*). A formula of this kind gives the composition of a molecule but not the arrangement of its atoms; Chapter 4 describes how that arrangement is drawn.

2.1.5. Solid, liquid and gas

The same substance can exist as a **solid**, a **liquid** or a **gas**, depending on temperature and pressure. These are the three common **states of matter**, and they differ in how the particles are arranged and how freely they move.

- In a solid, the particles are packed closely and held in fixed positions, about which they vibrate. A solid keeps its own shape.
- In a liquid, the particles are still close together but can move past one another. A liquid takes the shape of its container but keeps its volume.
- In a gas, the particles are far apart compared with their own size and move freely in all directions. A gas fills any container it is placed in.

A change from one state to another, such as melting or boiling, does not change the molecules themselves: ice, liquid water and steam all consist of H_2O molecules. What changes is the strength of the attraction between molecules compared with the energy of their motion. These attractions between molecules are the subject of Chapter 5.

2.2. Inside the atom

2.2.1. Three particles

An atom consists of a small, dense, positively charged **nucleus** surrounded by a much larger region occupied by **electrons** (Figure 2.2). The nucleus contains two kinds of particle: **protons**, which carry a positive electric charge, and **neutrons**, which carry no charge. Protons and neutrons are known together as **nucleons**.

Electric charge is measured in coulombs (C). The charge of a proton is +e and that of an electron is −e, where e, the **elementary charge**, is exactly $1.602\ 176\ 634 \times 10^{-19}$ C (National Institute of Standards and Technology, n.d.-a). No smaller charge has ever been observed on a free particle, so the charge of any atom or molecule is a whole-number multiple of e.

Table 2.1 compares the three particles. The proton and the neutron have almost the same mass, while the electron is about 1836 times lighter than a proton (National Institute of Standards and Technology, n.d.-a). As a result, more than 99.9 % of the mass of an atom resides in its nucleus.

Table 2.1.: The three particles of which atoms are made (National Institute of Standards and Technology, n.d.-a).

Particle	Location	Charge	Mass (kg)	Mass relative to the proton
Proton	nucleus	+e	1.6726×10^{-27}	1
Neutron	nucleus	0	1.6749×10^{-27}	1.0014
Electron	outside the nucleus	−e	9.1094×10^{-31}	1/1836

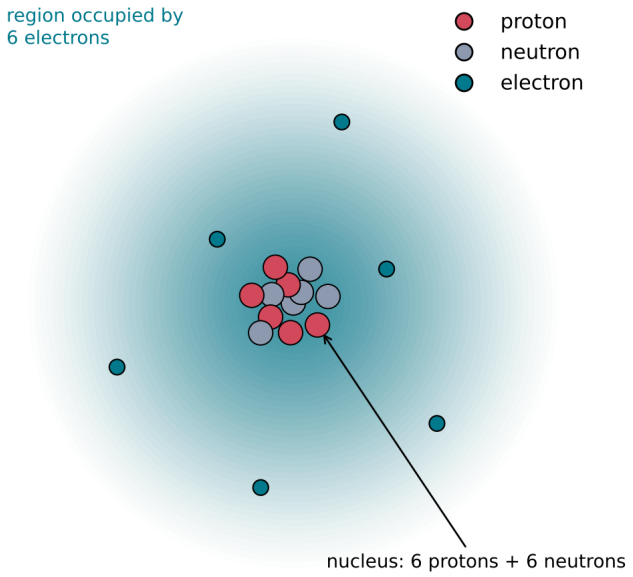


Figure 2.2.: Schematic of a carbon-12 atom: a nucleus of six protons and six neutrons, surrounded by six electrons. The drawing is not to scale. The electrons do not move in fixed orbits; the shading indicates the region in which they are likely to be found.

2.2.2. Mostly empty space

The nucleus is tiny compared with the atom. An atom has a radius of 1 to 2 $\times$ 10^{-10} m, whereas a nucleus ranges from about 4 $\times$ 10^{-15} m across for carbon to about 15 $\times$ 10^{-15} m for lead (Encyclopaedia Britannica, n.d.). The atom is therefore tens of thousands of times wider than its nucleus. If the nucleus of a carbon atom were enlarged to the size of a pea, about 5 mm across, the atom would be several hundred metres across, while still containing almost all of its mass in the pea.

The electrons do not circle the nucleus along fixed paths like planets. Quantum mechanics, the physical theory of matter on the atomic scale, describes each electron by the probability of finding it at each point around the nucleus. The arrangement of these electrons, described in Chapter 3, determines how an atom bonds to other atoms. Chemical reactions rearrange electrons; they never change the nucleus.

2.2.3. Neutral atoms

An atom contains equal numbers of protons and electrons, so its positive and negative charges cancel and it is electrically neutral. Atoms can, however, gain or lose electrons, and the particles that result are the subject of Section 2.4.

2.3. Atomic number, mass number and isotopes

2.3.1. Atomic number defines the element

The number of protons in the nucleus of an atom is its **atomic number**, symbol Z. The atomic number alone determines the element: every atom with one proton is hydrogen, every atom with six is carbon, every atom with eight is oxygen. The definition of an element given in Section 2.1 can therefore be made precise: an element is a substance whose atoms all have the same atomic number. At present 118 elements have been identified and named (International Union of Pure and Applied Chemistry, n.d.-b).

2.3.2. Mass number

The total number of protons and neutrons in a nucleus is its **mass number**, symbol A:

$$A = Z + N$$

where N is the number of neutrons. Because protons and neutrons have nearly the same mass and electrons are so light, the mass number is close to the mass of the atom measured in the unit introduced in Section 2.6.

2.3.3. Isotopes

Atoms of the same element always have the same number of protons, but they can have different numbers of neutrons. Atoms that have the same atomic number but different mass numbers are called **isotopes** of the element. Isotopes of an element have almost identical chemical behaviour, because chemistry depends on the electrons, and the number of electrons in a neutral atom equals the number of protons.

An isotope is written with its mass number as a superscript before the symbol of the element, and sometimes with its atomic number as a subscript below it (Figure 2.3). In words, it is written as the name of the element followed by the mass number: carbon-12, carbon-13.

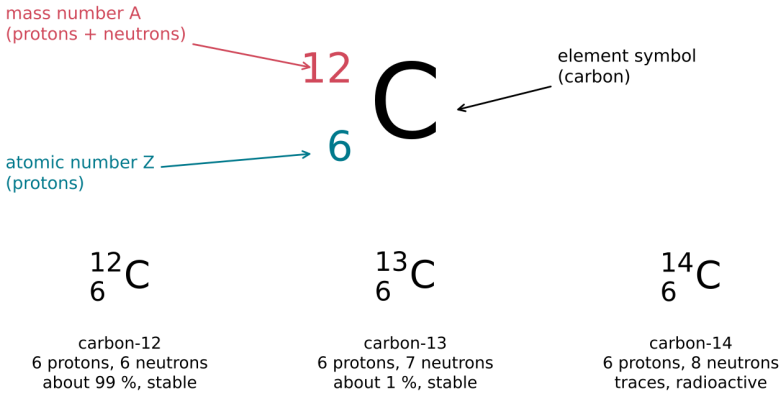


Figure 2.3.: Isotope notation. The mass number A is written at the upper left of the element symbol and the atomic number Z at the lower left; the number of neutrons is A − Z. The three isotopes of carbon found in nature are shown below.

Hydrogen and carbon provide the two examples that matter most in this book.

- Ordinary hydrogen, ^{1}H, has a nucleus consisting of a single proton. Deuterium, ^{2}H, has one proton and one neutron; in the international reference standard for ocean water it accounts for 0.0156 % of hydrogen atoms (IUPAC Commission on Isotopic Abundances and Atomic Weights, n.d.-b). Tritium, ^{3}H, has one proton and two neutrons; it is radioactive, with a half-life of 12.3 years (IUPAC Commission on Isotopic Abundances and Atomic Weights, n.d.-b), and occurs only in traces.
- Carbon-12 (six protons, six neutrons) makes up about 99 % of natural carbon (98.84 to 99.04 %, depending on the source), and carbon-13 (six protons, seven neutrons) about 1 % (IUPAC Commission on Isotopic Abundances and Atomic Weights, n.d.-a). Carbon-14 (six protons, eight neutrons) is radioactive and occurs only in traces.

2.3.4. Stable and radioactive nuclei

Not every combination of protons and neutrons is stable. Protons repel one another electrically; they are held together, with the neutrons, by the **strong nuclear force**, which is much stronger than the electrical repulsion but acts only over distances about the size of a nucleus. A nucleus with too many or too few neutrons for its number of protons is unstable, and sooner or later transforms into a different nucleus, emitting radiation. This is the radioactive decay described in Chapter 1, and the half-lives of such nuclei range from fractions of a second to billions of years. Because decay proceeds at a fixed rate whatever the chemical surroundings, radioactive isotopes serve as clocks: the ages of rocks, and of the Earth itself, are measured from the amounts of radioactive isotopes and their decay products that they contain.

2.4. Ions

2.4.1. Gaining and losing electrons

An atom or group of atoms that has gained or lost electrons carries a net electric charge and is called an **ion**. Losing electrons leaves more protons than electrons, so the ion is positive; such an ion is called a **cation**. Gaining electrons gives a negative ion, called an **anion**. The charge is written as a superscript after the formula, as a multiple of e: a sodium atom that has lost one electron is the cation Na^+, a magnesium atom that has lost two is Mg^{2+}, and a chlorine atom that has gained one is the anion Cl^-.

Only electrons are gained or lost when ions form. The nucleus is unchanged, so an ion of sodium is still sodium, with atomic number 11.

2.4.2. Ionic compounds

Positive and negative ions attract each other strongly. In solid sodium chloride, each Na^+ ion is surrounded by six Cl^- ions and each Cl^- by six Na^+, in a regular three-dimensional arrangement called a crystal lattice. A compound held together in this way is an **ionic compound**. It contains no separate molecules, and its formula, NaCl, gives only the ratio of the ions, which must balance so that the compound as a whole is neutral. Magnesium chloride, for example, has the formula $MgCl_2$, because each Mg^{2+} ion needs two Cl^- ions to balance its charge.

2.4.3. Ions made of several atoms

Some ions consist of several atoms bonded together and carrying a net charge as a whole. Such ions are called **polyatomic ions**. Three are important in the chemistry of life:

- the **hydroxide** ion, OH^-;
- the **ammonium** ion, NH_4^+;
- the **phosphate** ion, PO_4^{3-}, a phosphorus atom surrounded by four oxygen atoms and carrying three negative charges.

Phosphate is a central component of RNA and DNA. Its negative charges, carried by every unit of these molecules, make them strongly charged in water, a property that governs how they dissolve, fold and interact with positive ions such as Mg^{2+}.

2.5. The periodic table

2.5.1. Arranging the elements

The **periodic table** arranges the elements in order of increasing atomic number, in rows of different lengths, so that elements with similar chemical behaviour fall in the same column (Figure 2.4). A row is called a **period**, and a column a **group**. The groups are numbered from 1 to 18 from left to right. The regularity that the table displays was discovered in the nineteenth century from the chemical behaviour of the elements, before anything was known about the structure of atoms; its explanation in terms of electrons is given in Chapter 3.

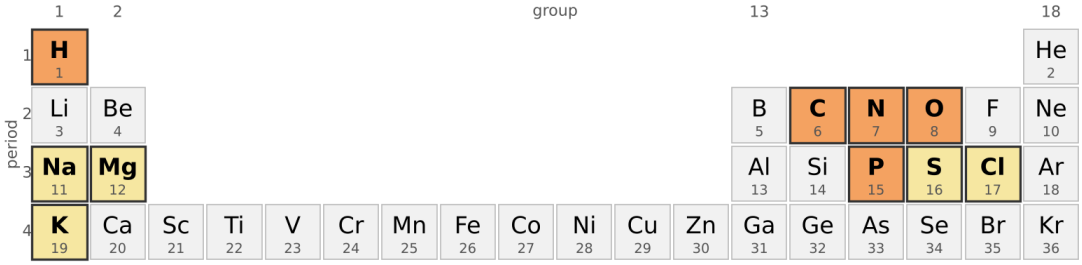


Figure 2.4.: The first four periods of the periodic table (elements 1 to 36). Each box shows the symbol and atomic number. Elements of the chemistry of life discussed in this book are highlighted: the five elements of RNA (orange), and sulfur, sodium, magnesium, potassium and chlorine (yellow).

2.5.2. Metals and non-metals

Most of the elements are **metals**: they conduct electricity and heat, have a characteristic lustre, and in compounds tend to lose electrons and form cations. Sodium, magnesium and iron are metals. The **non-metals**, grouped at the upper right of the table together with hydrogen, include hydrogen, carbon, nitrogen, oxygen, phosphorus, sulfur and chlorine. Non-metals tend to gain electrons or, more often, to share them with other atoms,

as described in Chapter 3. The elements in group 18, helium, neon, argon and their heavier relatives, are the **noble gases**; they hardly react at all.

2.5.3. The elements of life

Living matter is built mainly from a few light non-metals. Every molecule of RNA is made of only five elements: hydrogen, carbon, nitrogen, oxygen and phosphorus. Proteins contain the same elements apart from phosphorus, together with sulfur. For the bacterium *Escherichia coli*, a standard laboratory organism, the dry mass (the mass remaining after all water has been removed) is about 47 % carbon (Heldal et al. 1985), 20 % oxygen, 14 % nitrogen, 8 % hydrogen and 3 % phosphorus (Neidhardt et al., n.d.). A small number of metal ions, such as sodium, potassium and magnesium, are present in smaller amounts but are essential.

2.5.4. Where the elements came from

Elements can be made only by nuclear reactions, which change the number of protons in a nucleus; ordinary chemistry cannot do this. Hydrogen and most of the helium in the Universe were formed in the first minutes after the Big Bang. Carbon, nitrogen, oxygen, phosphorus and the heavier elements were formed later, inside stars, and were returned to space when those stars shed their outer layers or exploded. The carbon in every living cell was therefore made in stars that lived and died before the Sun and the Earth formed.

2.6. Weighing atoms

2.6.1. The atomic mass unit

The masses of atoms are far too small to be expressed conveniently in kilograms. Instead they are measured in the **dalton** (Da), also called the **unified atomic mass unit** (u). It is defined as one-twelfth of the mass of a single atom of carbon-12, and equals 1.660 539 068 92 $\times$ 10^{-27} kg (National Institute of Standards and Technology, n.d.-a). On this scale an atom of carbon-12 has a mass of exactly 12 Da, and the mass of any atom in daltons is close to its mass number. A proton has a mass of 1.007 28 Da and a neutron 1.008 66 Da (National Institute of Standards and Technology, n.d.-a).

2.6.2. Standard atomic weights

Most elements occur in nature as a mixture of isotopes. The average mass of the atoms of an element in a natural sample, expressed relative to one-twelfth of the mass of a carbon-12 atom, is its **atomic weight**, also called its **relative atomic mass**. Because it is a ratio of two masses, it has no unit. The atomic weight of carbon is 12.011, slightly more than 12, because natural carbon contains about 1 % of carbon-13. The atomic weights of the elements are evaluated by the Commission on Isotopic Abundances and Atomic Weights of IUPAC, the International Union of Pure and Applied Chemistry; Table 2.2 lists the values for the elements used most in this book.

Table 2.2.: Abridged standard atomic weights of selected elements (CIAAW, n.d.).

Element	Symbol	Z	Atomic weight
Hydrogen	H	1	1.0080
Carbon	C	6	12.011
Nitrogen	N	7	14.007
Oxygen	O	8	15.999
Sodium	Na	11	22.990
Magnesium	Mg	12	24.305
Phosphorus	P	15	30.974
Sulfur	S	16	32.06
Chlorine	Cl	17	35.45
Potassium	K	19	39.098

2.6.3. Molecular mass

The mass of a molecule is the sum of the masses of its atoms. Using atomic weights, the **relative molecular mass** of water, H_2O, is

$$2 \times 1.0080 + 15.999 = 18.015$$

so a water molecule has an average mass of 18.015 Da. In the same way, carbon dioxide, CO_2, has a relative molecular mass of $12.011 + 2 \times 15.999 = 44.009$. These numbers are the starting point for counting molecules in a sample, which is the subject of Chapter 6.

Key points

- A pure substance has a fixed composition; a mixture does not. Elements cannot be broken down chemically; compounds are made of elements in fixed proportions.

- An atom is the smallest unit of an element; a molecule is a group of bonded atoms, the smallest unit of a molecular compound. A chemical formula such as H_2O gives the number of atoms of each element.
- Solid, liquid and gas differ in how closely and how freely the particles move, not in the particles themselves.
- An atom has a nucleus of protons (+e) and neutrons (no charge), surrounded by electrons (−e). The electron is about 1836 times lighter than a proton; almost all the mass of an atom is in its tiny nucleus.
- The atomic number Z (number of protons) defines the element. The mass number A is the number of protons plus neutrons. Isotopes have the same Z but different A, and almost the same chemistry.
- Ions are atoms or groups of atoms that have lost electrons (cations) or gained them (anions). Phosphate, PO_4^{3-}, is a polyatomic anion found in RNA and DNA.
- The periodic table orders the elements by atomic number, in periods (rows) and groups (columns) of similar behaviour. The elements of life are mainly light non-metals: H, C, N, O, P and S.
- Atomic masses are measured in daltons (1/12 of the mass of a carbon-12 atom). The atomic weight is the average for the natural isotope mixture; molecular masses are sums of atomic weights.

Self-test

1. Classify each of the following as an element, a compound or a mixture: oxygen gas, sea water, carbon dioxide, gold, air.
2. How many protons, neutrons and electrons are there in a neutral atom of oxygen-18 (Z = 8)? How many in the ion $^{16}O^{2-}$?
3. Nitrogen has two stable isotopes, nitrogen-14 and nitrogen-15. What do they have in common, and how do they differ?
4. Write the formula of the ionic compound formed by Mg^{2+} and Cl^-, and of the compound formed by Na^+ and PO_4^{3-}.
5. Using Table 2.2, calculate the relative molecular masses of methane (CH_4) and of ammonia (NH_3).

3. Electrons and Bonds

The question of this chapter

Two hydrogen atoms and one oxygen atom combine into a molecule of water, but helium atoms combine with nothing at all. Carbon forms four bonds, nitrogen three and oxygen two. This chapter asks: *why do atoms join together, and what decides how many bonds each one forms?* The answer lies in the arrangement of the electrons around the nucleus, and in particular in the electrons of the outermost shell.

3.1. Electrons in shells

3.1.1. Energy levels

The electrons of an atom cannot have any energy whatever. They occupy a set of separate, fixed **energy levels**, determined by their attraction to the positive nucleus and their repulsion from one another. This is a result of quantum mechanics, and it can be confirmed directly: atoms absorb and emit light only at particular wavelengths, which correspond to the energy differences between their levels.

Energies on the scale of single atoms are conveniently measured in **electronvolts** (eV). One electronvolt is the energy gained by a particle carrying one elementary charge when it moves through an electric potential difference of one volt; it equals exactly 1.602 176 634 $\times$ 10^{-19} J (National Institute of Standards and Technology, n.d.-a).

3.1.2. Shells, subshells and orbitals

The region around the nucleus in which an electron of a given energy is likely to be found is called an **orbital**. An orbital is not a path. An electron in an atom does not travel along a fixed track; the orbital describes only where it is likely to be.

Orbitals are grouped into **shells**, numbered n = 1, 2, 3, and so on outwards from the nucleus. Electrons in higher shells are, on average, farther from the nucleus and less tightly held. Each shell is divided into **subshells**, labelled s, p, d and f: the first shell contains only an s subshell, the second contains s and p, and the third contains s, p and d. An s subshell has one orbital, which is spherical; a p subshell has three, each with

two lobes on opposite sides of the nucleus, pointing along three perpendicular directions (Figure 3.1).

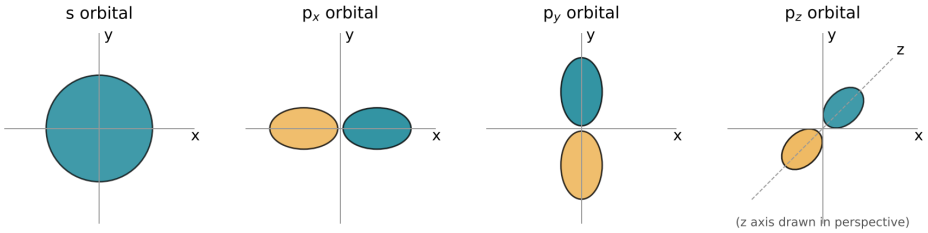


Figure 3.1.: Schematic shapes of an s orbital and the three p orbitals of a subshell, drawn as outlines of the region in which an electron is most likely to be found. The s orbital is spherical; the p orbitals point along three perpendicular axes, x, y and z.

3.1.3. How many electrons fit

Each orbital can hold at most two electrons. This rule, the **Pauli exclusion principle**, follows from a property of electrons called **spin**, which can take one of two values, often written "up" and "down"; two electrons can share an orbital only if their spins are opposite. It follows that an s subshell holds up to 2 electrons and a p subshell up to 6. The first shell, with only an s subshell, is full with 2 electrons. The second shell, with one s and three p orbitals, is full with 8.

3.1.4. Electron configurations

The **electron configuration** of an atom lists how many electrons occupy each subshell, written as the subshell label with the number of electrons as a superscript. In the most stable state of an atom, its **ground state**, electrons fill the lowest available levels first: 1s, then 2s, 2p, 3s and 3p. Table 3.1 gives the configurations of the elements that matter most in this book.

Table 3.1.: Ground-state electron configurations of selected atoms (Kramida et al. 2024).

Element	Z	Electron configuration	Valence electrons
Hydrogen	1	$1s^1$	1
Helium	2	$1s^2$	2 (full shell)
Carbon	6	$1s^2\ 2s^2\ 2p^2$	4
Nitrogen	7	$1s^2\ 2s^2\ 2p^3$	5
Oxygen	8	$1s^2\ 2s^2\ 2p^4$	6
Neon	10	$1s^2\ 2s^2\ 2p^6$	8 (full shell)

Element	Z	Electron configuration	Valence electrons
Sodium	11	$1s^2\ 2s^2\ 2p^6\ 3s^1$	1
Phosphorus	15	$1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^3$	5
Chlorine	17	$1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^5$	7

3.1.5. Valence electrons

The electrons in the outermost occupied shell are called **valence electrons**. They are the electrons farthest from the nucleus and the least tightly held, and they are the ones that take part in chemical bonding. The inner electrons, held close to the nucleus, are hardly affected by chemical reactions.

The periodic table (Chapter 2) is organised by valence electrons. Elements in the same group have the same number of valence electrons and therefore similar chemistry. Nitrogen and phosphorus both have five; oxygen and sulfur both have six. This is the explanation, in terms of atomic structure, of the regularity that nineteenth-century chemists observed.

3.2. Why atoms bond

3.2.1. Measuring how tightly electrons are held

The energy needed to remove one electron from a neutral atom in its ground state is called the **first ionisation energy**. It measures how tightly the atom holds its outermost electron, and it can be measured precisely from the light that atoms absorb. Figure 3.2 shows the values for the first eighteen elements.

Two features stand out. The noble gases helium (24.59 eV), neon (21.56 eV) and argon (15.76 eV) hold their electrons more tightly than any other element in their rows. The elements that follow them, lithium (5.39 eV) and sodium (5.14 eV), hold their single outer electron most weakly (Kramida et al. 2024). An atom with a completely filled outer shell or subshell is unusually stable; an atom with one electron beyond a filled shell loses that electron easily.

3.2.2. The octet rule

This stability of filled shells explains much of chemistry. Atoms tend to react in ways that give each of them the electron arrangement of the nearest noble gas: two electrons in the first shell for hydrogen, and eight in the outermost shell for carbon, nitrogen, oxygen and their heavier relatives. The tendency to reach eight valence electrons is called the **octet rule**. It is a rule of thumb rather than a law, and it has exceptions,

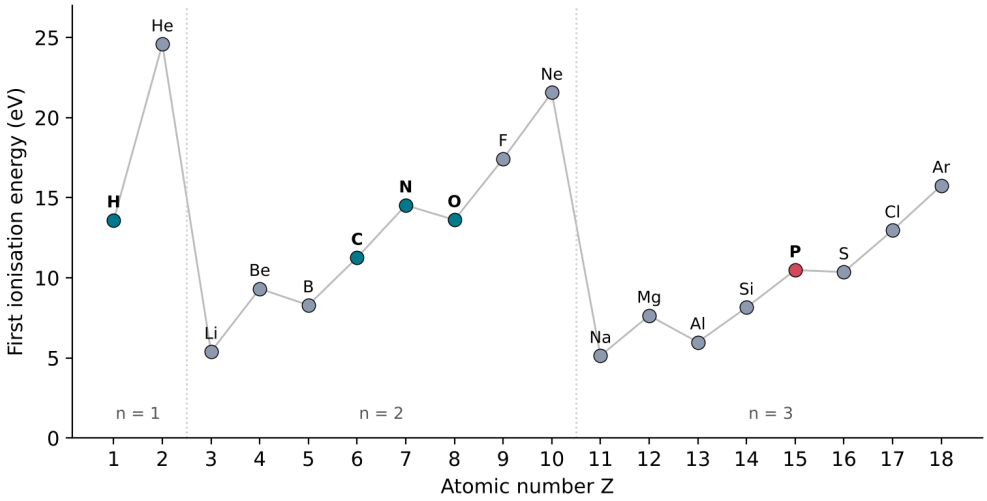


Figure 3.2.: First ionisation energies of the elements from hydrogen (Z = 1) to argon (Z = 18), from the NIST Atomic Spectra Database (Kramida et al. 2024). The elements of RNA are coloured. The peaks at helium, neon and argon correspond to full shells or subshells; the sharp drops at lithium and sodium mark the start of a new shell.

phosphorus among them, but it predicts correctly the bonding of most of the molecules in this book.

An atom can reach a noble-gas arrangement in two ways: by transferring electrons to or from another atom, or by sharing electrons with it. The first gives ionic bonds, the second covalent bonds.

3.3. Ionic bonds

A sodium atom has one valence electron, beyond the filled shells of neon. A chlorine atom has seven valence electrons, one short of the eight of argon. When sodium and chlorine react, each sodium atom transfers its outer electron to a chlorine atom:

$$\mathrm{Na} + \mathrm{Cl} \rightarrow \mathrm{Na}^{+} + \mathrm{Cl}^{-}$$

Both ions now have noble-gas arrangements: Na^{+} that of neon and Cl^{-} that of argon. The oppositely charged ions attract each other and pack together into a crystal. The attraction between oppositely charged ions is called an **ionic bond**. Ionic bonding is typical of compounds between a metal, which loses electrons easily, and a non-metal, which gains them readily.

Ionic bonds are not directional: each ion attracts every neighbour of opposite charge, whatever its direction. Ionic compounds therefore form extended crystals rather than separate molecules, and many dissolve in water, where the ions separate, as Chapter 5 explains.

3.4. Covalent bonds

3.4.1. Sharing a pair of electrons

Two non-metal atoms both tend to gain electrons, so neither gives one up to the other. Instead, they share. A **covalent bond** is a pair of electrons shared between two atoms. The shared electrons are attracted to both nuclei at once, and this attraction holds the atoms together.

The simplest example is the hydrogen molecule, H_2. Each hydrogen atom brings one electron. In the molecule, the two electrons occupy the region between the nuclei and are counted by both atoms, so each atom has two electrons around it, the arrangement of helium. The distance between the two nuclei, the **bond length**, is 0.741 Å (National Institute of Standards and Technology 2022), where the **ångström** (Å) equals 10^{-10} m. The covalent bonds between the atoms of the molecules in this book are all between about 1 and 2 Å long.

3.4.2. How many bonds each atom forms

An atom forms as many covalent bonds as it needs electrons to complete its octet, or its pair in the case of hydrogen. This gives the characteristic number of bonds, or **valence**, of each element:

- hydrogen, with 1 valence electron, needs 1 more and forms **1** bond;
- oxygen, with 6, needs 2 more and forms **2** bonds;
- nitrogen, with 5, needs 3 more and forms **3** bonds;
- carbon, with 4, needs 4 more and forms **4** bonds.

These four numbers, 1, 2, 3 and 4 for H, O, N and C, are worth memorising: they decide the structure of every organic molecule. Methane, CH_4, has one carbon atom bonded to four hydrogen atoms; ammonia, NH_3, one nitrogen bonded to three; and water, H_2O, one oxygen bonded to two.

3.4.3. Lewis structures

A **Lewis structure** is a drawing that shows how the valence electrons of a molecule are arranged. Each shared pair of electrons is drawn as a line between two atoms, and each unshared pair as two dots on the atom that carries it (Figure 3.3). An unshared pair of valence electrons, which belongs to one atom only, is called a **lone pair**. In water, the oxygen atom forms two bonds and carries two lone pairs; in ammonia, the nitrogen forms three bonds and carries one lone pair. Counting the electrons around oxygen in water, two shared pairs and two lone pairs, gives eight, an octet.

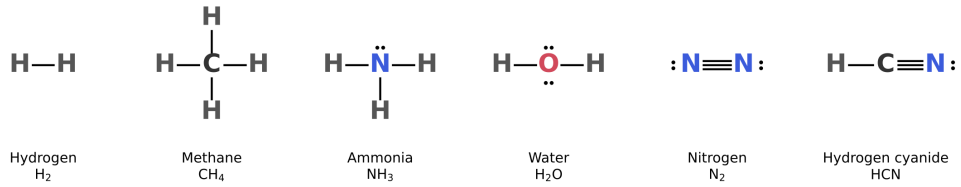


Figure 3.3.: Lewis structures of hydrogen, methane, ammonia, water, nitrogen and hydrogen cyanide. Each line is a shared pair of electrons (a covalent bond); each pair of dots is a lone pair. Three parallel lines mark a triple bond. Every atom except hydrogen is surrounded by eight electrons; hydrogen by two.

Lone pairs are not inert decoration. They give water and ammonia much of their chemical behaviour: a lone pair can accept a hydrogen ion, H^+, as in the formation of the ammonium ion NH_4^+, and it can take part in the weak attractions called hydrogen bonds, described in Chapter 5.

3.4.4. Phosphorus: an exception that matters

Phosphorus has five valence electrons, like nitrogen, and in some compounds it forms three bonds, as the octet rule predicts. In phosphate and its compounds, however, the phosphorus atom is bonded to four oxygen atoms, and the usual Lewis structure draws five bonds to it: one double bond to oxygen and three single bonds. Atoms of the third period, such as phosphorus and sulfur, are commonly drawn with more than eight electrons around them in such structures. Whatever the details of the description, the facts are clear: the phosphorus atom of a phosphate group is bonded to four oxygen atoms arranged at the corners of a tetrahedron, and the group as a whole carries negative charge in water.

3.5. Double and triple bonds

3.5.1. More than one shared pair

Two atoms can share more than one pair of electrons. Two shared pairs form a **double bond**, drawn as two lines, and three shared pairs a **triple bond**, drawn as three lines. The number of shared pairs between two atoms is called the **bond order**.

Multiple bonds allow atoms to complete their octets with fewer partners. In carbon dioxide, O=C=O, the carbon atom forms two double bonds, which count as four bonds, and each oxygen forms one double bond, which counts as two. In the nitrogen molecule, N≡N, the two atoms share three pairs, and each also carries one lone pair. In formaldehyde, $H_2C{=}O$, carbon forms two single bonds to hydrogen and one double bond to oxygen. In hydrogen cyanide, H–C≡N, carbon forms a single bond to hydrogen and a triple bond to nitrogen. Formaldehyde and hydrogen cyanide are two of the simplest compounds of carbon, and both are considered important starting materials in many proposed routes to the building blocks of life.

3.5.2. Shorter and stronger

Each additional shared pair pulls the two atoms closer together. The three simplest compounds of carbon and hydrogen with two carbon atoms show this directly. In ethane, C_2H_6, the carbon atoms are joined by a single bond of length 1.536 Å; in ethylene, C_2H_4, by a double bond of 1.339 Å; and in acetylene, C_2H_2, by a triple bond of 1.203 Å (National Institute of Standards and Technology 2022).

Multiple bonds are also stronger. The strength of a bond is measured by the **bond enthalpy**, the energy needed to break the bond in the gas phase, given for one mole of such bonds (the mole is defined in Chapter 6). Table 3.2 gives some values. The triple bond of N_2 is more than twice as strong as any of the single bonds in the table, which is why the nitrogen gas of the atmosphere is so unreactive.

Table 3.2.: Bond enthalpies at 298.15 K, calculated from standard enthalpies of formation of the gaseous atoms and molecules (National Institute of Standards and Technology, n.d.-b; Cox et al. 1984; Chase 1998).

Bond	Molecule	Bond enthalpy (kJ mol^{-1})
H–H	H_2	436.0
C–H	CH_4 (average of four)	415.9
O–H	H_2O (average of two)	463.5
N≡N	N_2	945.4

3.6. The shapes of molecules

3.6.1. Electron pairs keep apart

Molecules have definite three-dimensional shapes, and shape matters: it decides which molecules can fit against which, and therefore how they react and how they recognise one another. The shape of a small molecule can be predicted from one principle. The electron pairs around a central atom, both bonding pairs and lone pairs, repel each other and arrange themselves as far apart as possible. This is the **valence shell electron pair repulsion** (VSEPR) model.

- Four electron pairs point towards the corners of a **tetrahedron**, a solid with four triangular faces. The angle between any two of them is 109.47°.
- Three groups of electrons lie in a plane, at 120° to each other (**trigonal planar**). A double bond counts as one group.
- Two groups lie on opposite sides of the central atom, at 180° (**linear**).

3.6.2. Methane, ammonia and water

Methane, ammonia and water each have four electron pairs around the central atom, so all three are based on the tetrahedron (Figure 3.4). In methane all four pairs are bonds, and the molecule is a regular tetrahedron with H–C–H angles of 109.47°. In ammonia one pair is a lone pair; the three hydrogen atoms form a low pyramid, with H–N–H angles of 106.67°. In water two pairs are lone pairs, and the molecule is bent, with an H–O–H angle of 104.48° (National Institute of Standards and Technology 2022). Lone pairs spread out more than bonding pairs and push the bonds slightly closer together, which is why the angle shrinks from methane to water.

3.6.3. Flat and straight molecules

When the central atom has a double bond, the number of groups around it falls to three, and the molecule becomes flat. Formaldehyde, $H_2C{=}O$, and ethylene, $H_2C{=}CH_2$, are planar, with angles close to 120°; in ethylene the measured angles are 117.6° for H–C–H and 121.2° for H–C–C (National Institute of Standards and Technology 2022). With two groups, as in carbon dioxide (O=C=O) and hydrogen cyanide (H–C≡N), the molecule is linear.

3.6.4. Hybrid orbitals, briefly

The tetrahedral shape of methane raises a puzzle: carbon's valence electrons occupy one spherical 2s orbital and three 2p orbitals at right angles, none of which points to

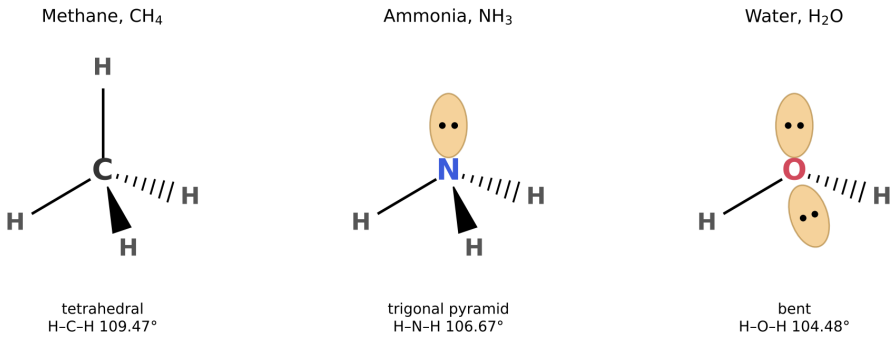


Figure 3.4.: Shapes of methane, ammonia and water, with measured bond angles (National Institute of Standards and Technology 2022). In all three, four electron pairs around the central atom point approximately to the corners of a tetrahedron; lone pairs are shown as shaded lobes. A wedge-shaped bond points towards the viewer and a dashed bond away from the viewer.

the corners of a tetrahedron. Chemists resolve this with the idea of **hybrid orbitals**, mathematical combinations of s and p orbitals that point in the directions of the bonds. A carbon atom forming four single bonds is described by four sp^3 hybrids (tetrahedral); one forming a double bond by three sp^2 hybrids (planar, 120°); and one forming a triple bond by two sp hybrids (linear, 180°). The labels sp^3, sp^2 and sp are used widely in chemistry as a shorthand for these three geometries of carbon.

3.7. Carbon, the element of chains and rings

The bonding rules of this chapter explain why carbon, rather than any other element, forms the backbone of the molecules of life. A carbon atom forms four bonds, more than any other light element, and it bonds strongly to other carbon atoms as well as to hydrogen, oxygen and nitrogen. Carbon atoms can therefore link into chains and rings of almost any length and shape, with other atoms attached at every position. The four bonds of each carbon point in different directions, so these chains and rings are three-dimensional.

Compounds built on frameworks of carbon atoms are called **organic compounds**, and their chemistry is **organic chemistry**. The name dates from a time when such compounds were thought to be made only by living organisms; since the nineteenth century they have been made in the laboratory from inorganic materials, beginning with Friedrich Wöhler's synthesis of urea in 1828 (Wöhler 1828), and the name now refers only to their composition. The sugars, the nucleobases and RNA itself are all organic compounds. Chapter 4 shows how their structures are drawn, and Chapter 11 how their parts behave.

Key points

- Electrons occupy fixed energy levels, grouped into shells (n = 1, 2, 3, …) and subshells (s, p, d). Each orbital holds at most two electrons; the first shell is full with 2 electrons and the second with 8.
- Valence electrons, in the outermost shell, take part in bonding: H 1, C 4, N 5, O 6, P 5. Elements in the same group have the same number.
- Filled shells are especially stable, as shown by the high ionisation energies of the noble gases. Atoms tend to reach a noble-gas arrangement (the octet rule).
- In an ionic bond, electrons are transferred, and oppositely charged ions attract each other. In a covalent bond, a pair of electrons is shared between two atoms.
- Hydrogen forms 1 covalent bond, oxygen 2, nitrogen 3 and carbon 4. Lewis structures show bonds as lines and lone pairs as dots.
- Double and triple bonds share two and three pairs; they are shorter and stronger than single bonds (C–C 1.536 Å, C=C 1.339 Å, C≡C 1.203 Å). The N≡N bond, 945.4 kJ mol^{-1}, makes nitrogen gas unreactive.
- Electron pairs around an atom keep as far apart as possible (VSEPR). Four pairs give tetrahedral directions: methane is tetrahedral (109.47°), ammonia pyramidal (106.67°) and water bent (104.48°). Three groups give planar molecules, two give linear ones.
- Carbon's four bonds and its strong bonds to itself allow chains and rings: the basis of organic chemistry.

Self-test

1. Write the electron configuration of sulfur (Z = 16) and state how many valence electrons it has. Which element of Table 3.1 is in the same group?
2. Explain, using Figure 3.2, why sodium readily forms the ion Na^+ but not Na^{2+}.
3. Draw the Lewis structure of hydrogen sulfide, H_2S, and predict its shape.
4. Draw the Lewis structures of formaldehyde (H_2CO) and hydrogen cyanide (HCN). What is the bond order of the carbon-oxygen bond and of the carbon-nitrogen bond?
5. Predict the number of bonds formed by each atom in methanol, CH_3OH, and sketch its structure.

4. Reading Molecular Drawings

The question of this chapter

Open any book or paper on the chemistry of life and you will find pages covered with hexagons, zigzag lines and letters. These drawings are the language in which chemists describe molecules, and they carry more information than any amount of text. This chapter asks: *how is a molecule drawn, and how can a drawing be read?* It introduces the main types of chemical formula, explains the conventions of the skeletal formula used in almost all modern chemistry, and shows how three-dimensional shape is represented on a flat page. By the end of the chapter, you should be able to look at the drawing of a sugar or a nucleobase and count every atom in it.

4.1. One molecule, many drawings

4.1.1. Why the molecular formula is not enough

Chapter 2 introduced the chemical formula, such as H_2O for water. A formula that gives only the number of atoms of each element in a molecule is called a **molecular formula**. For small molecules it is often all that is needed: there is only one way to join two hydrogen atoms and one oxygen atom so that each forms its usual number of bonds.

For larger molecules, the molecular formula is not enough. The formula C_2H_6O describes two different compounds. In **ethanol**, the alcohol of wine and beer, the two carbon atoms are bonded to each other and the oxygen is bonded to one of them and to a hydrogen atom. In **dimethyl ether**, the oxygen sits between the two carbon atoms. Both arrangements obey the bonding rules of Chapter 3 (carbon 4 bonds, oxygen 2, hydrogen 1), but they are different substances with different properties: at atmospheric pressure ethanol boils at 78 °C and is a liquid at room temperature, whereas dimethyl ether boils at −24 °C and is a gas (National Institute of Standards and Technology, n.d.-b). Compounds that have the same molecular formula but different structures are called **isomers** (Figure 4.1). They are the reason why chemistry needs drawings.

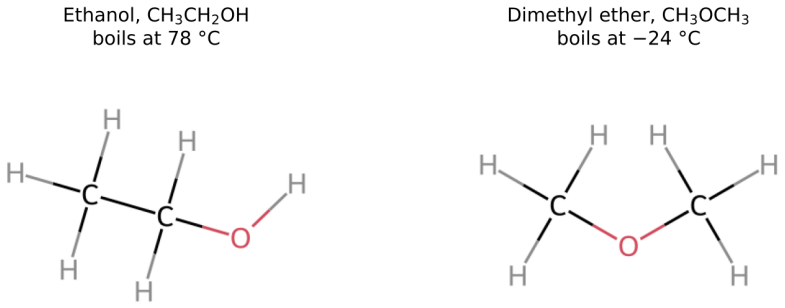


Figure 4.1.: Ethanol and dimethyl ether, two isomers with the molecular formula C_2H_6O, drawn as full structural formulas. All atoms and all bonds are shown. The only difference is the position of the oxygen atom, but it gives two different compounds.

4.1.2. Four ways to draw ethanol

Chemists use several types of formula, which show the same molecule with different amounts of detail. Figure 4.2 shows ethanol in four of them.

1. The **molecular formula**, C_2H_6O, gives the composition only.
2. A **structural formula** shows every atom and every bond as in a Lewis structure, usually omitting the lone pairs.
3. A **condensed formula**, CH_3CH_2OH, writes the atoms in the order in which they are connected, grouping each carbon with the hydrogen atoms attached to it.
4. A **skeletal formula** shows the carbon framework as a zigzag of lines and leaves out the carbon atoms and the hydrogen atoms bonded to them.

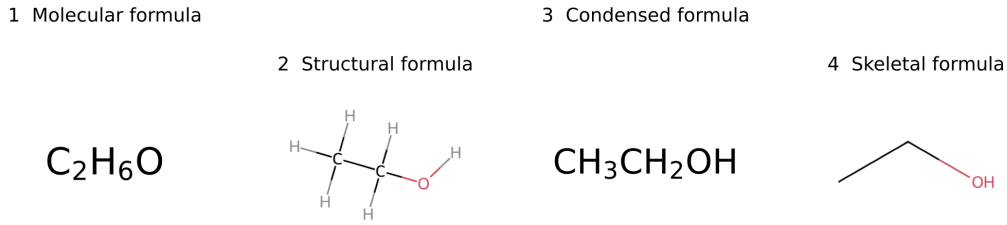


Figure 4.2.: Ethanol in four representations: molecular formula, full structural formula, condensed formula and skeletal formula. All four describe the same molecule; each successive form leaves more to be understood by the reader.

Each type has its uses. The molecular formula is used to calculate masses (Chapter 6).

The structural formula is the clearest for small molecules and for showing exactly which bonds change in a reaction. The condensed formula fits into a line of text. The skeletal formula is the most compact for large molecules, and it is the form in which almost all molecules in modern chemistry and biochemistry are drawn.

4.2. Structural and condensed formulas

4.2.1. Structural formulas

A full structural formula is a Lewis structure in which the lone pairs are usually left out. Every bond appears as a line: one line for a single bond, two for a double bond, three for a triple bond. To check a structural formula, count the lines at each atom. Every carbon must have four, every nitrogen three, every oxygen two and every hydrogen one, unless the atom carries a charge (Section 4.4).

The angles in a structural formula are not the real bond angles. A carbon atom with four single bonds is drawn as a cross with 90° angles for convenience, although in the molecule its bonds point to the corners of a tetrahedron, 109.5° apart (Chapter 3). A structural formula shows which atoms are connected, not the shape of the molecule.

4.2.2. Condensed formulas

A condensed formula lists the atoms along the main chain from one end to the other, writing each carbon followed by the hydrogen atoms attached to it. CH_3CH_2OH is read as: a carbon carrying three hydrogens, bonded to a carbon carrying two hydrogens, bonded to an oxygen carrying one hydrogen. Dimethyl ether is CH_3OCH_3. Bonds within the chain are not drawn, although double and triple bonds are often shown: ethylene is $CH_2{=}CH_2$ and hydrogen cyanide HC≡N.

Groups attached to the chain as branches are written in parentheses after the atom that carries them. Acetone, a common solvent, is $CH_3C(O)CH_3$ or $(CH_3)_2CO$: a central carbon bonded to two CH_3 groups and double-bonded to an oxygen. A subscript after parentheses means that the group appears that many times.

The condensed formula of a group that recurs in many molecules soon becomes familiar. CH_3– is a **methyl group**, –OH a **hydroxy group**, and –COOH a **carboxy group** (a carbon double-bonded to one oxygen and single-bonded to an –OH). Chapter 11 describes these and other functional groups and the chemistry they bring.

4.3. Skeletal formulas

4.3.1. The four rules

Large molecules contain dozens of carbon and hydrogen atoms, and drawing each one would bury the structure in letters. The **skeletal formula** avoids this. It is drawn according to four rules.

1. **Bonds are lines.** Single bonds are drawn as single lines, double bonds as two parallel lines, triple bonds as three.
2. **Every corner and every line end is a carbon atom, unless another atom is written there.** Chains of carbon atoms are drawn as zigzags, so that each carbon atom appears as a corner.
3. **Hydrogen atoms bonded to carbon are not drawn.** Each carbon is understood to carry as many hydrogen atoms as it needs to make four bonds in total.
4. **All other atoms are written with their symbols**, together with any hydrogen atoms attached to them: O, OH, N, NH_2, P and so on.

The zigzag of rule 2 has a physical basis: in a chain of carbon atoms with tetrahedral bonds, the stretched-out shape of the chain really is a zigzag, with angles close to 109.5°. Skeletal formulas are drawn with 120° angles, which makes the zigzag regular on paper.

4.3.2. Reading a skeletal formula

To read a skeletal formula, find every carbon atom and work out its hydrogen atoms. For each corner or line end without a symbol, count the lines that meet there; the number of hydrogen atoms is four minus the number of bonds, counting a double bond as two and a triple bond as three. Figure 4.3 shows the procedure for several molecules, with the hidden hydrogen atoms written in grey.

Three patterns appear so often that they are worth recognising at a glance:

- a line end is a CH_3 group (one bond, so three hydrogen atoms);
- a corner where two single lines meet is a CH_2 group;
- a corner where three single lines meet is a CH group, and one where four meet is a carbon with no hydrogen.

A carbon atom that forms a double bond and one single bond, as in but-2-ene, carries one hydrogen; a carbon atom double-bonded to oxygen and single-bonded to two other atoms, as in acetone or acetic acid, carries none.

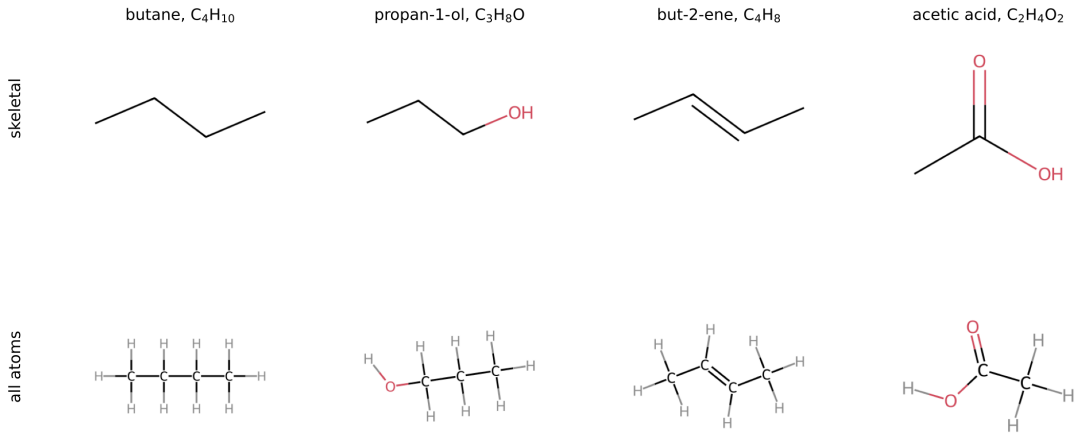


Figure 4.3.: Reading skeletal formulas. Top row: skeletal formulas of butane, propan-1-ol, but-2-ene and acetic acid. Bottom row: the same molecules with every carbon atom labelled and the hydrogen atoms that the skeletal formula leaves out written in grey. A line end is a CH_3 group, a corner with two lines is a CH_2 group, and a carbon with a double bond and one other bond is a CH group.

4.3.3. Rings

Carbon chains can close into **rings**, and skeletal formulas show rings as polygons. A hexagon with no symbols at its corners is cyclohexane, C_6H_{12}: six carbon atoms, each a corner with two lines, and so each carrying two hydrogen atoms. Rings can contain atoms other than carbon, such as oxygen or nitrogen; these are written at the corner they occupy. A ring containing at least one atom other than carbon is called a **heterocycle**. The sugar of RNA and the four nucleobases are all heterocycles.

4.3.4. A worked example: ribose

The sugar in RNA is **ribose**, $C_5H_{10}O_5$. In RNA it is present as a five-membered ring of four carbon atoms and one oxygen atom. Figure 4.4 shows its skeletal formula and the same drawing with every hidden atom made explicit.

The count proceeds corner by corner. The ring has five corners: one is labelled O, so the other four are carbon atoms. A fifth carbon atom is the line end outside the ring, bonded to an OH group. Each of the five carbon atoms then receives hydrogen atoms to make up four bonds. The carbon atom numbered 1 is bonded to the ring oxygen, to carbon 2 and to an OH group, so it carries one hydrogen. Carbons 2, 3 and 4 are each bonded to two ring neighbours and to one other group, and also carry one hydrogen each. Carbon 5 is bonded to carbon 4 and to an OH group, and carries two hydrogens.

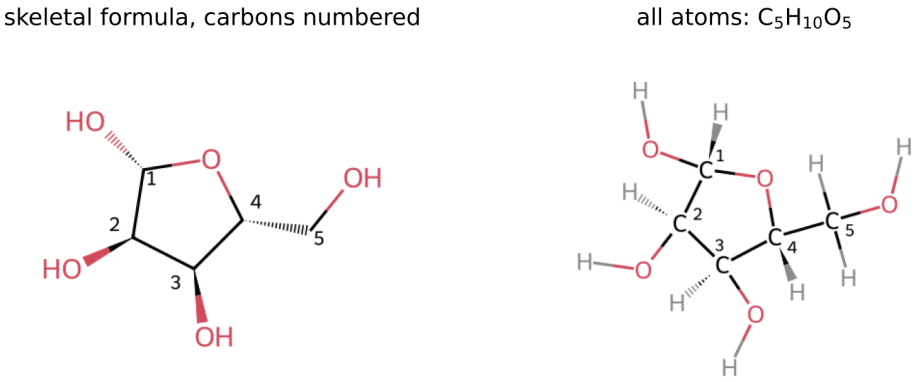


Figure 4.4.: The sugar ribose (in its five-membered ring form), as a skeletal formula (left) and with every carbon and hydrogen atom shown (right). The carbon atoms are numbered 1 to 5 according to the standard convention for sugars. Counting gives 5 carbon atoms, 10 hydrogen atoms and 5 oxygen atoms: $C_5H_{10}O_5$.

Adding the hydrogen atoms of the four OH groups gives $1 + 1 + 1 + 1 + 2 + 4 = 10$ hydrogen atoms, and the oxygen atoms number 4 in OH groups plus 1 in the ring, giving 5. The molecular formula is $C_5H_{10}O_5$.

The numbering of the carbon atoms is part of the language of biochemistry. In a sugar, carbon 1 is the carbon that carries two bonds to oxygen atoms (here the ring oxygen and an OH group), and the numbering continues along the chain. When the sugar is part of a larger molecule such as RNA, its carbon numbers receive a prime, written $1'$, $2'$, $3'$ and so on, to distinguish them from the numbered atoms of the attached nucleobase. Chapter 12 uses this numbering to describe how the parts of RNA are joined.

4.4. Charges and lone pairs in drawings

Atoms in a drawing occasionally carry a charge. An oxygen atom that forms only one bond and carries three lone pairs has one electron more than a neutral oxygen atom in a molecule, and is written O^-. A nitrogen atom that forms four bonds, as in the ammonium ion, has given part of its lone pair to the fourth bond and is written N^+. Such a charge, assigned to an individual atom in a structure, is called a **formal charge**. The rule for bond counts given in Section 4.2 is therefore modified: an O^- forms one bond and an N^+ forms four.

The phosphate groups of RNA are usually drawn with one or two O^- atoms, because at the acidity of most natural waters they have lost hydrogen ions, as Chapter 8 explains.

When a phosphate is drawn as P bonded to four O atoms, one of them by a double line and one or more of them carrying a minus sign, this is the convention being used.

Lone pairs are usually omitted from skeletal formulas, but every oxygen and nitrogen atom still has them. A neutral oxygen with two bonds has two lone pairs, and a neutral nitrogen with three bonds has one. These lone pairs take part in many reactions and in the hydrogen bonds of Chapter 5, so a reader should learn to imagine them even when they are not drawn.

4.5. Three dimensions on paper

4.5.1. Wedges and dashes

Molecules are three-dimensional, and some of their properties depend on the arrangement of atoms in space, not only on which atoms are connected. To show this arrangement, chemists use two kinds of special bond. A solid **wedge**, widening towards one atom, represents a bond pointing out of the page towards the viewer. A **dashed** (hashed) bond represents a bond pointing behind the page, away from the viewer. Ordinary lines lie in the plane of the page. With these three kinds of line, the tetrahedral arrangement around a carbon atom can be drawn as two ordinary bonds in the plane, one wedge and one dash, as in the shapes of methane, ammonia and water in Chapter 3.

The arrangement in space matters most when a carbon atom carries four different groups. Such a carbon can exist in two arrangements that are mirror images of each other and cannot be superimposed, like a left and a right hand. Chapter 11 explains this property, called chirality, and why the sugar of RNA occurs in only one of its mirror-image forms.

4.5.2. Drawing sugar rings: the Haworth projection

Rings of sugars are often drawn in a special perspective called a **Haworth projection** (Figure 4.5). The ring is drawn flat, in a plane roughly perpendicular to the page, as if seen obliquely from above; the lower edge is often drawn thicker to indicate that it is nearer to the viewer. The ring oxygen is placed at the back, behind the plane of the page (International Union of Pure and Applied Chemistry, n.d.-a). Each carbon atom in the ring has two remaining bonds, which are drawn as vertical lines, one pointing up and one pointing down. The groups on these lines, OH or H, are placed above or below the plane of the ring.

The Haworth projection makes it easy to see which groups lie on the same side of the ring. In ribose, the hydroxy groups on carbons 2 and 3 point to the same side, and this arrangement is one of the features that distinguish ribose from other sugars with the same formula, $C_5H_{10}O_5$. The ring is not really flat; five-membered rings are slightly

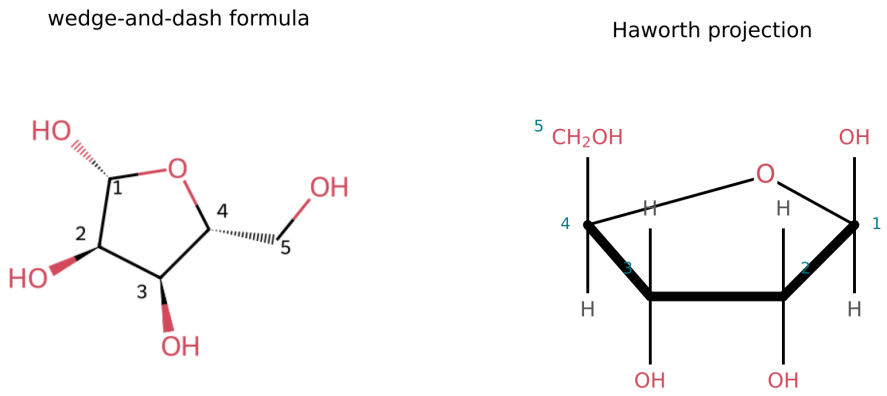


Figure 4.5.: Two representations of ribose in its five-membered ring form. Left: a skeletal formula with wedges (towards the viewer) and dashes (away from the viewer). Right: a Haworth projection, in which the ring is seen obliquely and the thick lower edge is nearest to the viewer; groups are placed above or below the plane of the ring. In both, the OH groups on carbons 2 and 3 lie on the same side of the ring, below it in the Haworth projection, and the CH_2OH group (carbon 5) lies above it.

puckered. The Haworth projection is a convention for showing the relative positions of groups, not the exact shape.

4.5.3. Ball-and-stick and space-filling models

Computer graphics and physical models show molecules in other ways. In a **ball-and-stick model**, atoms are spheres and bonds are rods; the model shows bond angles correctly and makes the three-dimensional arrangement clear. In a **space-filling model**, each atom is drawn as a sphere of roughly its real size, and the spheres overlap; the model shows the overall shape and surface of the molecule, which decides how it fits against other molecules. Models of large biological molecules often use further simplifications, such as ribbons that trace the backbone of an RNA or protein chain.

4.6. Flat rings with alternating double bonds

4.6.1. Benzene

A ring of six carbon atoms with alternating single and double bonds is **benzene**, C_6H_6. The ring structure was proposed by August Kekulé in 1865 (Kekulé 1865), and it raised a puzzle: if the bonds alternated, the double bonds should be shorter than the single

bonds, yet all six carbon-carbon bonds in benzene are found to have the same length, 1.397 Å (National Institute of Standards and Technology 2022), between the lengths of a single and a double bond (Chapter 3).

The explanation is that benzene cannot be described by a single Lewis structure. Two structures can be drawn, differing only in the positions of the double bonds, and the real molecule is a blend of the two, with the six electrons of the three double bonds spread evenly around the whole ring (Figure 4.6). This situation, in which a molecule is described by two or more Lewis structures that differ only in the placement of electrons, is called **resonance**. Benzene is sometimes drawn as a hexagon with a circle inside, to show the evenly spread electrons, but the form with alternating double bonds is more common, and it must be understood as one of two equivalent drawings.

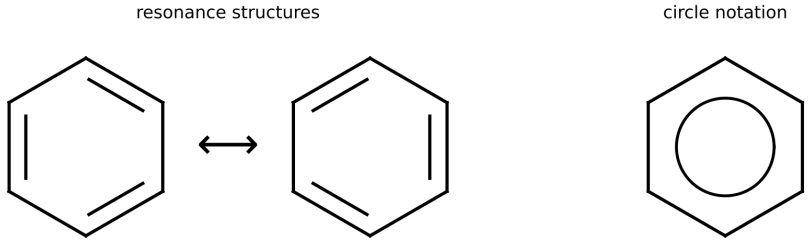


Figure 4.6.: Benzene. Left: the two equivalent structures with alternating double bonds; the double-headed arrow indicates resonance, not a reaction. Right: the representation with a circle, showing the six electrons spread evenly around the ring. All six carbon-carbon bonds have the same length, 1.397 Å.

4.6.2. Aromatic rings

Benzene and other flat rings with electrons spread evenly around them in this way are called **aromatic**. (The name is historical: many early-known compounds of this kind had strong smells.) Aromatic rings are flat and unusually stable. Many contain nitrogen atoms in the ring. The four nucleobases of RNA, adenine, guanine, cytosine and uracil, all contain flat, aromatic heterocycles with nitrogen atoms (Figure 4.7). Their flatness allows them to stack on top of each other in a strand of RNA, and the nitrogen and oxygen atoms on their edges form the hydrogen bonds that pair bases with each other. Both properties are described in Chapters 5 and 12.

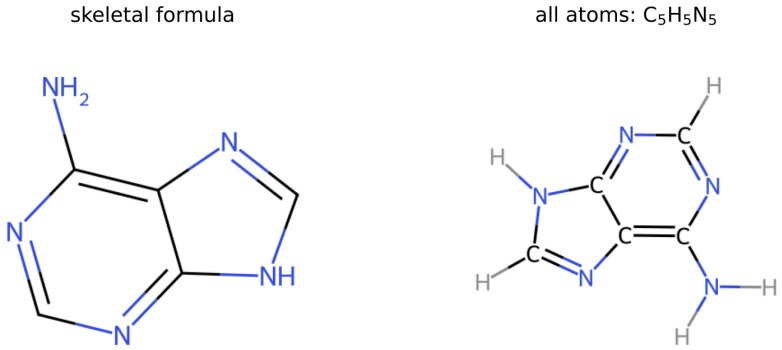


Figure 4.7.: Adenine, one of the four nucleobases of RNA, as a skeletal formula (left) and with all atoms shown (right). It contains two fused aromatic rings, a six-membered and a five-membered ring, with nitrogen atoms at five ring positions. Counting gives $C_5H_5N_5$.

Key points

- A molecular formula gives composition only. Isomers, such as ethanol and dimethyl ether (both C_2H_6O), have the same molecular formula but different structures and properties.
- A structural formula shows every atom and bond; a condensed formula such as CH_3CH_2OH lists atoms in order of connection; a skeletal formula shows the carbon framework only.
- In a skeletal formula, every corner and line end is a carbon atom, hydrogen atoms on carbon are omitted, and each carbon carries enough hydrogen atoms to make four bonds. All other atoms are written with their hydrogen atoms.
- A line end is CH_3, a corner with two single lines is CH_2, and one with three is CH.
- Rings are drawn as polygons; rings containing atoms other than carbon are heterocycles. Ribose, $C_5H_{10}O_5$, has a five-membered ring of four carbons and one oxygen; its carbons are numbered 1 to 5, and $1'$ to $5'$ when the sugar is part of a larger molecule.
- An O^- forms one bond and an N^+ four; such formal charges appear on phosphate groups and ammonium groups.
- Wedges point towards the viewer and dashes away. Haworth projections show sugar rings obliquely, with groups above or below the ring.
- Benzene is a flat ring whose six electrons of the double bonds are spread around the ring (resonance). Such aromatic rings, often containing nitrogen, form the nucleobases of RNA.

Self-test

1. Write the condensed formula and draw the skeletal formula of propane, C_3H_8.
2. In the skeletal formula of acetic acid in Figure 4.3, how many hydrogen atoms does each carbon atom carry? What is its molecular formula?
3. Draw the full structural formula of the compound whose condensed formula is $CH_3CH(OH)CH_3$. Is it an isomer of propan-1-ol?
4. How many hydrogen atoms are carried by the carbon atoms of cyclohexane, and of benzene? Write both molecular formulas.
5. In the Haworth projection of ribose in Figure 4.5, which groups lie above the plane of the ring and which below?

5. Polarity, Hydrogen Bonds and Water

The question of this chapter

Water and methane are molecules of almost the same mass. At room temperature methane is a gas that boils at −162 °C, whereas water is a liquid that boils only at 100 °C. Salt and sugar dissolve in water but not in oil. Ice floats. This chapter asks: *what makes water so different, and what holds molecules together when no chemical bond joins them?* The answer starts with a single property of atoms, electronegativity, and leads to the hydrogen bond, the interaction that holds the two strands of DNA together and gives RNA its shapes.

5.1. Unequal sharing

5.1.1. Electronegativity

In the hydrogen molecule, H_2, the two atoms are identical and share their electron pair equally. When two different atoms are bonded, this is generally not so: one atom attracts the shared electrons more strongly than the other, and the electrons spend more time near it. The tendency of an atom in a bond to attract the shared electrons towards itself is called its **electronegativity**, often written with the Greek letter χ (chi).

Electronegativity is not measured directly, like a mass. It is derived from measured bond energies by a procedure introduced by Linus Pauling in 1932 (Pauling 1932), and the values are pure numbers on what is called the Pauling scale. Table 5.1 gives the values for the elements of this book.

Table 5.1.: Pauling electronegativities (Allred 1961).

Element	χ (Pauling scale)	Element	χ (Pauling scale)
Hydrogen	2.20	Fluorine	3.98
Carbon	2.55	Sodium	0.93
Nitrogen	3.04	Magnesium	1.31
Oxygen	3.44	Sulfur	2.58
Phosphorus	2.19	Chlorine	3.16

Two trends are visible. Across a row of the periodic table, electronegativity increases from left to right, as the nuclear charge increases while the valence electrons stay in the same shell. Down a column it decreases, because the valence electrons lie farther from the nucleus: phosphorus (2.19) is much less electronegative than nitrogen (3.04), although both have five valence electrons. Of the elements of life, oxygen and nitrogen are by far the most electronegative. Carbon, hydrogen and phosphorus lie close together, between 2.19 and 2.55.

5.1.2. Polar bonds

A covalent bond in which the electrons are shared unequally is a **polar bond**. Its more electronegative end carries a small excess of negative charge, and the other end a small deficit. These partial charges are written $\delta-$ and $\delta+$ (δ, the Greek letter delta, is read "a fraction of"), to distinguish them from the whole-number charges of ions. The larger the difference in electronegativity, the more polar the bond. A C–H bond (difference 0.35) is nearly non-polar. An O–H bond (difference 1.24) is strongly polar, with the negative end on oxygen, and so are N–H, C=O and P–O bonds.

When the difference becomes very large, as between sodium (0.93) and chlorine (3.16), the electron is not shared at all but effectively transferred, and the result is a pair of ions (Chapter 3). Covalent and ionic bonds are therefore two ends of a continuous range. Non-polar covalent bonds lie at one end, ionic bonds at the other, and polar covalent bonds in between.

5.1.3. Polar molecules

A molecule as a whole is **polar** if the partial charges of its bonds do not cancel, so that one side of the molecule is slightly negative and the other slightly positive. Whether this happens depends on the shape of the molecule as well as on its bonds. A molecule with separated positive and negative charge is said to have a **dipole**, and the size of the separation is measured by its **dipole moment**, given in a unit called the debye (D).

Water is bent (Chapter 3), so the two O–H bond dipoles point partly in the same direction and add up: water has a dipole moment of 1.857 D, with its negative end at the oxygen atom (Figure 5.1). Ammonia, also not symmetric, has 1.476 D. Carbon dioxide, O=C=O, has two strongly polar C=O bonds, but the molecule is straight, so they point in exactly opposite directions and cancel; carbon dioxide has no dipole moment. Methane is also non-polar, because its four slightly polar C–H bonds point to the corners of a regular tetrahedron and cancel exactly (National Institute of Standards and Technology 2022).

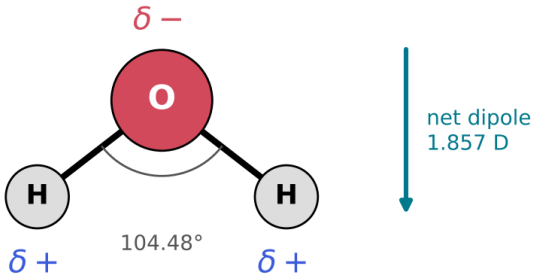


Figure 5.1.: Polarity of the water molecule. Each O–H bond is polar, with partial negative charge ($\delta-$) on oxygen and partial positive charge ($\delta+$) on hydrogen. Because the molecule is bent, the two bond dipoles add to a net dipole moment of 1.857 D (National Institute of Standards and Technology 2022).

5.2. Forces between molecules

5.2.1. Bonds within molecules and forces between them

Covalent bonds hold the atoms of a molecule together. Molecules also attract one another, but by much weaker **intermolecular forces**. These forces do not create new molecules, and at ordinary temperatures they are constantly broken and re-formed as the molecules move. They nevertheless decide whether a substance is a gas, a liquid or a solid at a given temperature, and whether it dissolves in water. All intermolecular forces are electrical in origin: they arise from the attraction between regions of positive and negative charge on neighbouring molecules.

5.2.2. Dipole-dipole and dispersion forces

Polar molecules attract each other because the positive end of one is drawn towards the negative end of another. This is a **dipole-dipole interaction**.

Non-polar molecules attract each other too, more weakly. The electron cloud of any molecule fluctuates, so that at any instant it may have a small, temporary dipole. That dipole distorts the electron cloud of a neighbouring molecule, creating a dipole in it, and the two temporary dipoles attract. This is the **dispersion force**, also called the London force. It acts between all molecules, and it is larger for larger molecules, which have more electrons. Dipole-dipole and dispersion forces, with related interactions, are together called **van der Waals forces**.

5.2.3. The hydrogen bond

One intermolecular interaction is of special importance for water and for the molecules of life. IUPAC defines it as follows: "The hydrogen bond is an attractive interaction between a hydrogen atom from a molecule or a molecular fragment X–H in which X is more electronegative than H, and an atom or a group of atoms in the same or a different molecule, in which there is evidence of bond formation" (Arunan et al. 2011).

In practice, a **hydrogen bond** forms when a hydrogen atom bonded to oxygen or nitrogen, and so carrying a $\delta+$ charge, points towards a lone pair on another oxygen or nitrogen atom (Figure 5.2). It is written X–H $\cdots$ Y, with the dotted line for the hydrogen bond. The group X–H is the **hydrogen-bond donor**, and the atom Y with its lone pair is the **hydrogen-bond acceptor**. Hydrogen bonds are stronger than other intermolecular forces, but much weaker than covalent bonds, so they are made and broken easily at ordinary temperatures. They are also directional: they are strongest when X, H and Y lie on a straight line (Arunan et al. 2011).

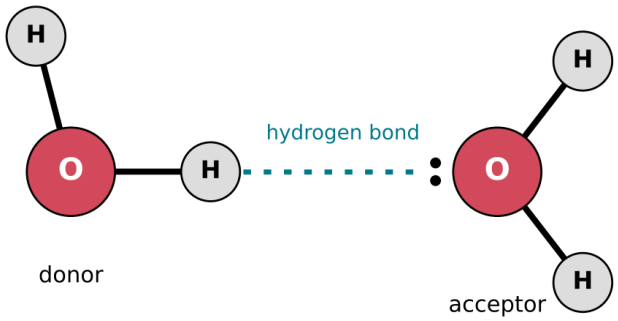


Figure 5.2.: A hydrogen bond between two water molecules. The O–H group of the left-hand molecule (the donor) points towards a lone pair on the oxygen atom of the right-hand molecule (the acceptor). Covalent bonds are drawn as solid lines and the hydrogen bond as a dotted line. Schematic; distances not to scale.

Hydrogen bonds are not limited to water. They form wherever O–H or N–H groups meet oxygen or nitrogen atoms with lone pairs: between molecules of alcohols, sugars and nucleobases, and between different parts of the same large molecule. The two strands of a DNA double helix are held together by hydrogen bonds between pairs of bases, and the same kind of pairing folds a single strand of RNA into a shape. Chapter 12 shows which atoms form these bonds.

5.2.4. Evidence from boiling points

When a liquid boils, its molecules separate from one another, so a higher boiling point indicates stronger intermolecular forces. Table 5.2 compares four small molecules containing hydrogen.

Table 5.2.: Normal boiling points, at a pressure of one standard atmosphere, from the NIST Chemistry WebBook (National Institute of Standards and Technology, n.d.-b). Molar mass is introduced in Chapter 6.

Substance	Formula	Molar mass (g mol^{-1})	Normal boiling point (°C)
Methane	CH_4	16.04	−162
Water	H_2O	18.02	100.02
Hydrogen fluoride	HF	20.01	19.6
Hydrogen sulfide	H_2S	34.08	−60.28

Methane and water have almost the same mass, yet water boils 262 °C higher. Methane is non-polar and forms no hydrogen bonds; water forms many, through its two O–H groups and its two lone pairs. Hydrogen sulfide, H_2S, is nearly twice as heavy as water, but sulfur (2.58) is much less electronegative than oxygen (3.44), and H_2S forms only weak hydrogen bonds; it boils 160 °C lower than water. Hydrogen fluoride, which contains the most electronegative element of all, also boils unusually high for its size.

5.3. Water, the solvent of life

5.3.1. A small molecule with four hydrogen-bonding sites

A water molecule is small, bent and polar. Its oxygen atom carries two lone pairs and is bonded to two hydrogen atoms, so each molecule can donate two hydrogen bonds and accept two. In liquid water, each molecule is linked to its neighbours by a network of hydrogen bonds that are constantly broken and re-formed. This network is responsible for the unusually high boiling point of water, and for most of its other unusual properties.

5.3.2. Solutions

A **solution** is a mixture in which one substance is spread evenly through another at the level of individual molecules or ions. The substance present in the larger amount, usually a liquid, is the **solvent**; the substance dissolved in it is the **solute**. A solution in which water is the solvent is an **aqueous solution**. The maximum amount of a solute that dissolves in a given amount of solvent at a given temperature is its **solubility**.

A useful rule of thumb is "like dissolves like". Polar solutes dissolve in polar solvents, such as water, and non-polar solutes dissolve in non-polar solvents, such as oils. A molecule that can form hydrogen bonds with water, such as an alcohol, a sugar or a nucleobase, can take a place in the hydrogen-bond network of water and dissolves well. Table sugar dissolves in water in large amounts for this reason: each of its molecules carries eight O–H groups.

5.3.3. Dissolving ions

Water also dissolves many ionic compounds, such as sodium chloride, which do not dissolve in oils. Two properties of water are responsible.

First, water greatly weakens the electrical force between charges. The force between two charges in a medium is smaller than in a vacuum by a factor called the **relative permittivity** of the medium. For liquid water at 25 °C, the relative permittivity is 78.4 (International Association for the Properties of Water and Steam 2011): the attraction between a sodium ion and a chloride ion in water is almost eighty times weaker than at the same distance in a vacuum, and the ions can drift apart.

Second, water molecules arrange themselves around each dissolved ion, forming a **hydration shell** (Figure 5.3). Around a positive ion, such as Na^+, the water molecules turn their $\delta-$ oxygen atoms towards the ion; around a negative ion, such as Cl^-, they turn their $\delta+$ hydrogen atoms towards it. The attraction between the ion and these oriented water molecules holds the ion in solution.

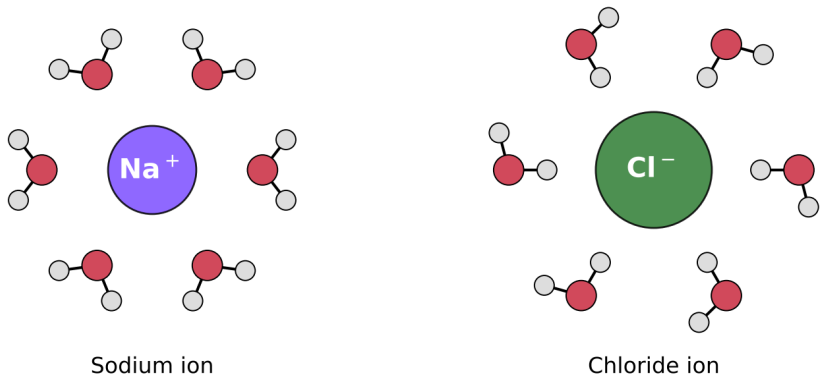


Figure 5.3.: Hydration of ions (schematic). Water molecules around a sodium ion (left) point their partially negative oxygen atoms towards the positive ion; those around a chloride ion (right) point their partially positive hydrogen atoms towards the negative ion. Only one layer of water molecules is shown, in two dimensions.

5.3.4. The hydrophobic effect

Non-polar molecules, such as the hydrocarbons of oil, dissolve poorly in water and tend to cluster together when placed in it; oil and water separate. This is the **hydrophobic effect** (from the Greek for "water-fearing"). It is not caused by a special attraction between the non-polar molecules. A non-polar molecule cannot form hydrogen bonds, and the water molecules around it are restricted in the hydrogen bonds they can form with one another. When the non-polar molecules cluster together, less of their surface touches water, and fewer water molecules are restricted. Polar and charged groups, which interact favourably with water, are called **hydrophilic** ("water-loving"); non-polar groups are **hydrophobic**.

Some molecules have a hydrophilic part and a hydrophobic part. Such molecules are called **amphiphilic**. Soap is an example: a long hydrocarbon tail attached to a charged head. In water, amphiphilic molecules arrange themselves with their tails together, away from the water, and their heads facing outwards. Depending on their shape, they form small spherical clusters or double layers that close into hollow spheres. The membranes that surround every living cell are such double layers, and similar structures formed from simple amphiphilic molecules are studied as possible compartments for the first cells.

5.3.5. Some unusual properties of water

The hydrogen-bond network gives water several properties that are unusual for so small a molecule.

- **Ice floats.** In ice, every water molecule forms four hydrogen bonds in a regular, open, tetrahedral arrangement, which occupies more space than the disordered liquid. At 0 °C and atmospheric pressure, ice has a density of 916.7 kg m^{-3} and liquid water 999.8 kg m^{-3} (Feistel and Wagner 2006; Wagner and Pruß 2002). Because ice is less dense, it floats, and bodies of water freeze from the top down, leaving liquid water beneath.
- **Liquid water is densest at about 4 °C.** As it cools below this temperature, the liquid begins to form open, ice-like arrangements and expands slightly. The maximum density, 999.97 kg m^{-3}, is reached at 3.98 °C (Wagner and Pruß 2002).
- **Water stores a lot of heat.** Heating water requires energy to break hydrogen bonds as well as to make the molecules move faster. The energy needed to raise the temperature of 1 g of liquid water by 1 °C at 25 °C is 4.18 J (International Association for the Properties of Water and Steam 2011), several times more than for most common liquids and solids. Oceans and lakes therefore change temperature slowly.

Key points

- Electronegativity measures how strongly an atom in a bond attracts the shared electrons. On the Pauling scale: H 2.20, C 2.55, N 3.04, O 3.44, P 2.19. It increases across a period and decreases down a group.
- Unequal sharing makes a bond polar, with partial charges $\delta+$ and $\delta-$. O–H and N–H bonds are strongly polar; C–H bonds nearly non-polar.
- A molecule is polar if its bond dipoles do not cancel. Bent water (1.857 D) is polar; straight CO_2 and symmetric CH_4 are not.
- Intermolecular forces are much weaker than covalent bonds: dipole-dipole forces, dispersion forces and, strongest, hydrogen bonds.
- A hydrogen bond X–H $\cdots$ Y links a hydrogen on oxygen or nitrogen to a lone pair on another oxygen or nitrogen. It is directional and easily broken.
- Hydrogen bonding explains why water boils at 100 °C while methane, of similar mass, boils at −162 °C.
- Water dissolves polar molecules by hydrogen bonding and ions by weakening the force between them (relative permittivity 78.4) and forming hydration shells. Like dissolves like.
- Non-polar molecules cluster in water (the hydrophobic effect). Amphiphilic molecules form layers and membranes.
- Ice is less dense than liquid water and floats; liquid water is densest at 3.98 °C and has a high heat capacity.

Self-test

1. Using Table 5.1, rank the bonds C–H, N–H, O–H and C–O from least to most polar, and mark the $\delta-$ end of each.
2. Carbon dioxide contains two polar bonds but is non-polar. Explain why, and say whether sulfur dioxide, which is bent, should be polar.
3. Identify the hydrogen-bond donors and acceptors in a molecule of ammonia, NH_3. Can two ammonia molecules form a hydrogen bond with each other?
4. Explain why ethanol (CH_3CH_2OH) mixes with water in all proportions, whereas hexane (C_6H_{14}) does not dissolve in water.
5. What would happen to lakes in winter if ice were denser than liquid water?

6. Counting Molecules: Moles and Concentration

The question of this chapter

A chemist who wants to make water from hydrogen and oxygen needs two molecules of hydrogen for every molecule of oxygen. Molecules, however, cannot be counted one by one; even a speck of matter contains far too many. This chapter asks: *how do chemists count molecules?* The answer is the mole, a unit that links the number of particles to a mass that can be weighed. From it follow the molar mass, the concentration of a solution, and the behaviour of gases.

6.1. The mole

6.1.1. Counting by weighing

A bank that receives a large bag of identical coins does not count them one by one; it weighs them. If one coin has a mass of 5.00 g and the bag contains 2.50 kg of coins, the bag holds 500 coins. Chemists count atoms and molecules in the same way: the mass of a sample, divided by the mass of one particle, gives the number of particles. The only difficulty is that the numbers are enormous, and a special unit is used to handle them.

6.1.2. Definition of the mole

The **mole** (symbol mol) is the SI unit of **amount of substance**. Since 2019 it has been defined by fixing the number of particles it contains: one mole contains exactly 6.022 140 76 $\times$ 10^{23} specified elementary entities (Bureau International des Poids et Mesures 2019). The entities may be atoms, molecules, ions or electrons, and they must always be stated: one mole of water molecules, one mole of sodium ions. The fixed number, expressed per mole, is the **Avogadro constant**:

$$N_{\mathrm{A}} = 6.022\,140\,76 \times 10^{23}\ \mathrm{mol}^{-1}$$

It is exact by definition (National Institute of Standards and Technology, n.d.-a). The amount of substance n and the number of particles N are related by

$$N = n \times N_{\mathrm{A}}$$

Two moles of hydrogen molecules therefore contain $2 \times 6.022 \times 10^{23} = 1.204 \times 10^{24}$ molecules.

The mole is a counting unit, like a dozen, but for a number with 24 digits. Its size was chosen for convenience. Before 2019 the mole was defined by requiring the molar mass of carbon-12 to be exactly 12 g per mole, so that the Avogadro constant was the number of atoms in 12 g of carbon-12 (Bureau International des Poids et Mesures, n.d.). The new definition was chosen to agree with the old one within the precision of measurement. As a result, one mole of any substance has a mass in grams equal to the relative mass of its molecules, introduced in Chapter 2. This is what makes the mole so useful.

6.2. Molar mass

6.2.1. From daltons to grams per mole

The **molar mass** of a substance, symbol M, is the mass of one mole of it, usually given in grams per mole (g mol^{-1}). Numerically, the molar mass in g mol^{-1} equals the relative molecular mass (or the atomic weight, for an element). A water molecule has a relative molecular mass of 18.015 (Chapter 2), so the molar mass of water is 18.015 g mol^{-1}: one mole of water, 6.022×10^{23} molecules, has a mass of 18.015 g, a little more than a tablespoon of water.

The molar mass of a compound is calculated by adding the atomic weights of its atoms. For carbon dioxide, $\mathrm{CO_2}$:

$$M(\mathrm{CO_2}) = 12.011 + 2 \times 15.999 = 44.009 \text{ g mol}^{-1}$$

For ribose, $\mathrm{C_5H_{10}O_5}$ (Chapter 4):

$$M = 5 \times 12.011 + 10 \times 1.0080 + 5 \times 15.999 = 150.13 \text{ g mol}^{-1}$$

6.2.2. Converting between mass, amount and number

The amount of substance in a sample of mass m is

$$n = \frac{m}{M}$$

and the number of particles follows by multiplying by N_A. Chapter 1 stated that a drop of water of about 0.05 mL contains about 1.7×10^{21} water molecules. The calculation runs as follows. Water at room temperature has a density of 997 kg m^{-3} (International Association for the Properties of Water and Steam 2011), very nearly 1 g per millilitre, so the drop has a mass of about 0.050 g. Its amount of substance is

$$n = \frac{0.050 \text{ g}}{18.015 \text{ g mol}^{-1}} = 2.8 \times 10^{-3} \text{ mol}$$

and the number of molecules is 2.8×10^{-3} mol $\times$ 6.022×10^{23} mol^{-1} $= 1.7 \times 10^{21}$.

The same steps run in reverse. To prepare 0.10 mol of sodium chloride (molar mass $22.990 + 35.45 = 58.44$ g mol^{-1}), weigh out $0.10 \times 58.44 = 5.8$ g.

6.2.3. How small a molecule is

The Avogadro constant also gives the mass of a single molecule. One water molecule has a mass of 18.015 g mol^{-1} divided by 6.022×10^{23} mol^{-1}, which is 2.99×10^{-23} g. Since each molecule in liquid water occupies a volume equal to the molar volume divided by N_A, and one mole of liquid water occupies about 18 mL, each molecule has about 3.0×10^{-23} mL, or 3.0×10^{-29} m^3, to itself: the volume of a cube about 0.31 nm on each side. A water molecule is therefore a few tenths of a nanometre across.

6.3. Concentration

6.3.1. Amount concentration

Most of the chemistry of life takes place in water. To describe a solution, it is not enough to know how much solute it contains; what matters is how much solute there is in a given volume. The **amount concentration**, often simply called the **concentration**, is the amount of solute divided by the volume of the solution:

$$c = \frac{n}{V}$$

It is usually expressed in moles per litre (mol L^{-1}), a unit written **M** and called "molar". A solution containing 0.10 mol of sodium chloride in a total volume of 1.0 L has a concentration of 0.10 mol L^{-1}, or 0.10 M. The concentration of a substance X is often written [X], so that this statement can be written [NaCl] = 0.10 M.

Concentrations in the chemistry of life span many orders of magnitude, and the SI prefixes are used: millimolar (mM, 10^{-3} M), micromolar (μM, 10^{-6} M) and nanomolar (nM, 10^{-9} M).

6.3.2. Some concentrations for orientation

- **Pure water.** One litre of water at 25 °C has a mass of 997 g, which is 997/18.015 = 55.3 mol. The "concentration" of water in water is therefore about 55 M. This large number reappears whenever water takes part in a reaction.
- **Sea water.** In the standard composition of present-day sea water, the main dissolved ions are chloride (0.546 mol per kilogram of sea water), sodium (0.469 mol kg^{-1}), magnesium (0.0528 mol kg^{-1}) and sulfate (0.0282 mol kg^{-1}) (Millero et al. 2008). Sea water is therefore roughly half a mole per litre in sodium chloride.
- **Dilute building blocks.** Many experiments on the origin of life use the starting materials at millimolar concentrations or higher. Natural waters on the early Earth may have contained some of these molecules at much lower concentrations, and how they could have been concentrated is one of the central questions of the field.

Figure 6.1 places these concentrations on a logarithmic scale.

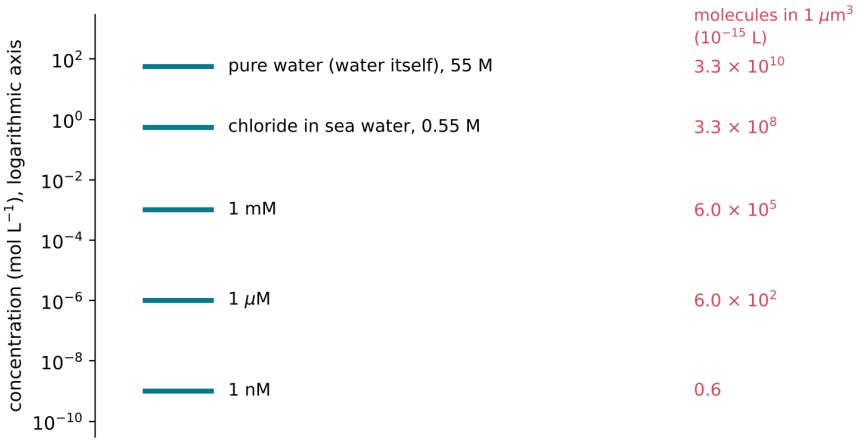


Figure 6.1.: Concentrations on a logarithmic scale, from pure water (about 55 M) to 1 nM, with the corresponding average number of molecules in a volume of 1 μm^3 (10^{-15} L), calculated as $c \times V \times N_A$. The sea-water value is for chloride (Millero et al. 2008).

Concentrations are sometimes given in units of mass instead of amount, such as grams per litre (g L^{-1}) or milligrams per litre. To convert a mass concentration into an amount concentration, divide by the molar mass: 9.0 g L^{-1} of sodium chloride is 9.0/58.44 = 0.15 M.

6.3.3. Concentration and the number of molecules in a small volume

At low concentrations and in small volumes, the number of molecules can become very small. A volume of 1 μm^3, a cube one micrometre on each side, is 10^{-15} L. At a concentration of 1 nM it contains

$$10^{-9}\ \text{mol L}^{-1} \times 10^{-15}\ \text{L} \times 6.022 \times 10^{23}\ \text{mol}^{-1} = 0.6$$

molecules on average, that is, sometimes none and sometimes one or two. A bacterial cell is about this size. Inside it, “1 nanomolar” therefore means roughly one molecule per cell, and any notion of a smoothly varying concentration breaks down.

6.4. Dilution and concentration

6.4.1. Dilution

Adding solvent to a solution does not change the amount of solute it contains; it only spreads the solute through a larger volume. Since n = cV is the same before and after,

$$c_1 V_1 = c_2 V_2$$

where c_1 and V_1 are the concentration and volume before dilution, and c_2 and V_2 after. Diluting 10 mL of a 1.0 M solution to 100 mL gives a concentration of 1.0 M × 10 mL / 100 mL = 0.10 M, a tenfold dilution.

6.4.2. Concentration by evaporation

The same equation, read in the other direction, describes concentration. If water evaporates from a solution and the solute stays behind, the volume falls and the concentration rises in proportion. A pond that loses 99 % of its water by evaporation concentrates every non-volatile solute a hundredfold, provided that nothing is lost by other routes, such as crystallisation, seepage or reaction. Repeated drying and wetting of shallow pools is one of the main mechanisms proposed for concentrating the building blocks of life on the early Earth, and the arithmetic of concentration is simply the dilution equation run backwards.

6.5. Gases

6.5.1. The ideal gas law

In a gas, the molecules are far apart and move independently. For most gases near ordinary temperatures and pressures, pressure p, volume V, amount n and absolute temperature T are related by the **ideal gas law**:

$$pV = nRT$$

where R is the **molar gas constant**, 8.314 J mol^{-1} K^{-1} (National Institute of Standards and Technology, n.d.-a). Its value is exact, since it is the product of two constants fixed by definition, the Avogadro constant and the Boltzmann constant. Pressure is the force per unit area exerted by the gas on the walls of its container, measured in pascals (Pa); the **bar**, equal to 10^5 Pa, is close to atmospheric pressure at sea level, and the standard atmosphere is exactly 101 325 Pa.

The ideal gas law contains a remarkable fact: the volume of a gas depends on the number of its molecules, not on what they are. At 25 °C (298.15 K) and 1 bar, one mole of any ideal gas occupies

$$V = \frac{nRT}{p} = \frac{1\ \text{mol} \times 8.314\ \text{J mol}^{-1}\ \text{K}^{-1} \times 298.15\ \text{K}}{10^5\ \text{Pa}} = 0.0248\ \text{m}^3 = 24.8\ \text{L}$$

6.5.2. Mixtures of gases and partial pressure

In a mixture of gases, each gas contributes to the total pressure in proportion to its amount. The pressure that a gas in a mixture would exert if it alone filled the container is its **partial pressure**, and the total pressure is the sum of the partial pressures. The partial pressure of each gas equals its fraction of the molecules multiplied by the total pressure.

Dry air today consists, by volume and therefore by number of molecules, of 78.08 % nitrogen, 20.95 % oxygen and 0.93 % argon, together with smaller amounts of carbon dioxide and other gases (NASA Space Science Data Coordinated Archive 2024). At a total pressure of 1 bar, the partial pressure of oxygen is therefore about 0.21 bar. The atmosphere of the early Earth was very different: it contained essentially no free oxygen, and its composition is one of the main uncertainties in the study of the origin of life.

Key points

- The mole is the SI unit of amount of substance. One mole contains exactly 6.022 140 76 × 10^{23} entities; this number per mole is the Avogadro constant N_A. N = n N_A.
- The molar mass M, in g mol^{-1}, equals the relative molecular mass: water 18.015 g mol^{-1}, CO_2 44.009 g mol^{-1}, ribose 150.13 g mol^{-1}. The amount of substance is n = m / M.
- A drop of water (0.05 mL) contains about 1.7 × 10^{21} molecules; a water molecule is a few tenths of a nanometre across.
- Concentration c = n / V is usually given in mol L^{-1} (M), with mM, μM and nM for dilute solutions. Pure water is about 55 M; sea water about 0.5 M in sodium chloride.
- In a volume of 1 μm^3, a concentration of 1 nM corresponds to less than one molecule on average.
- In dilution and concentration, $c_1V_1 = c_2V_2$. Evaporating 99 % of the water concentrates a non-volatile solute a hundredfold.
- For gases, pV = nRT, with R = 8.314 J mol^{-1} K^{-1}. One mole of gas occupies 24.8 L at 25 °C and 1 bar. Each gas in a mixture contributes its partial pressure.

Self-test

1. Calculate the molar mass of glucose, $C_6H_{12}O_6$, using Table 2.2. How many moles are there in 90 g of glucose?
2. How many molecules are there in 1.0 g of water?
3. 5.85 g of sodium chloride is dissolved in water to make 250 mL of solution. What is its concentration in mol L^{-1}?
4. A pool contains a solute at 2 μM. Evaporation reduces its volume to one-fiftieth, with no loss of solute. What is the new concentration?
5. What volume does 2.0 mol of nitrogen gas occupy at 25 °C and 1 bar, assuming ideal behaviour?

7. Chemical Reactions and Equations

The question of this chapter

When methane burns, it disappears, and carbon dioxide and water appear in its place. When iron rusts, a grey metal becomes a red-brown powder. In every such change, atoms are neither created nor destroyed; they are only rearranged into new combinations. This chapter asks: *how is a chemical reaction described, and what kinds of reaction are there?* It shows how reactions are written as balanced equations, how the amounts of substances in a reaction are calculated, and how most reactions in the chemistry of life fall into a few types: transfers of protons, transfers of electrons, and the joining and splitting of molecules with water.

7.1. What happens in a reaction

7.1.1. Reactants and products

A **chemical reaction** is a process in which substances are converted into different substances by the breaking and forming of chemical bonds. The substances present at the start are the **reactants**, and those formed are the **products**. In the burning of methane, the reactants are methane and oxygen and the products are carbon dioxide and water. The C–H bonds of methane and the O=O bonds of oxygen are broken, and new C=O and O–H bonds are formed.

A chemical reaction changes only the arrangement of electrons and the way atoms are bonded together. The nuclei are untouched, so every atom present among the reactants is still present among the products, as an atom of the same element. This distinguishes chemical reactions from the nuclear reactions of stars and radioactive decay, which change one element into another.

7.1.2. Conservation of mass

Because atoms are only rearranged, the total mass of the products equals the total mass of the reactants. This **law of conservation of mass** was established by careful weighing in the late eighteenth century, most influentially by Antoine Lavoisier (Wikipedia contributors, n.d.-a). It holds to the precision of any chemical balance. (According to

relativity, the energy released in a reaction corresponds to a tiny loss of mass, but for chemical reactions it is far too small to measure by weighing.)

7.2. Chemical equations

7.2.1. Writing an equation

A reaction is written as a **chemical equation**, with the formulas of the reactants on the left, the formulas of the products on the right, and an arrow between them pointing from reactants to products. The formation of water from hydrogen and oxygen is written

$$2\,H_2 + O_2 \rightarrow 2\,H_2O$$

The numbers in front of the formulas are called **coefficients**. They give the number of molecules (or moles) of each substance that take part. A coefficient of 1 is not written. The equation reads: two molecules of hydrogen react with one molecule of oxygen to give two molecules of water (Figure 7.1).

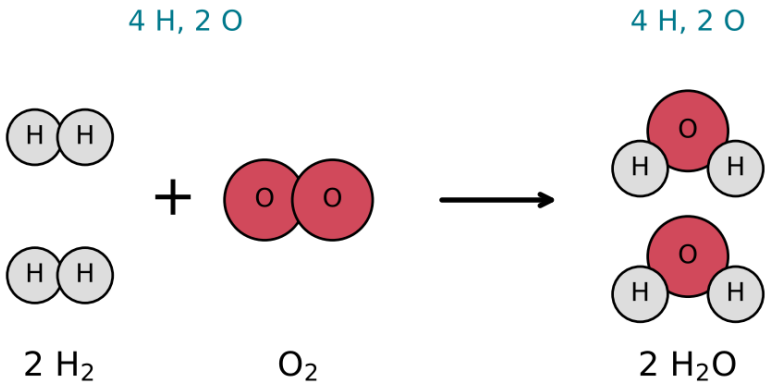


Figure 7.1.: The reaction $2\,H_2 + O_2 \rightarrow 2\,H_2O$ drawn as molecules. Hydrogen atoms are grey and oxygen atoms red. The left side contains four hydrogen atoms and two oxygen atoms, and so does the right side: the equation is balanced.

7.2.2. Balancing an equation

An equation is **balanced** when each side contains the same number of atoms of every element, and the same total charge. To balance an equation, change only the coefficients, never the subscripts in the formulas: changing a subscript would describe a different

substance. The usual procedure is to balance first the elements that appear in only one substance on each side, and to leave hydrogen and oxygen, which often appear in several, until last.

As an example, consider the burning of methane. The unbalanced equation is

$$\mathrm{CH_4 + O_2 \rightarrow CO_2 + H_2O}$$

Carbon is already balanced, with one atom on each side. Methane supplies four hydrogen atoms, so two molecules of water are needed: $\mathrm{CH_4 + O_2 \rightarrow CO_2 + 2\ H_2O}$. The right side now contains $2 + 2 = 4$ oxygen atoms, so two molecules of oxygen are needed on the left:

$$\mathrm{CH_4 + 2\,O_2 \rightarrow CO_2 + 2\,H_2O}$$

A final count confirms one carbon, four hydrogen and four oxygen atoms on each side.

Equations involving ions must also balance charge. In the reaction of sodium with chlorine to give ions, $\mathrm{Na + Cl \rightarrow Na^+ + Cl^-}$, the total charge is zero on both sides.

7.2.3. State symbols

The physical state of each substance is often added in parentheses after its formula: (s) for solid, (l) for liquid, (g) for gas and (aq) for a substance dissolved in water (from the Latin *aqua*). Dissolving table salt is written

$$\mathrm{NaCl(s) \rightarrow Na^+(aq) + Cl^-(aq)}$$

7.2.4. Arrows

Different arrows have different meanings, and they must not be confused.

- A single arrow, $\rightarrow$, shows the direction of a reaction from reactants to products.
- A pair of half-arrows, $\rightleftharpoons$, shows a **reversible reaction**: one that proceeds in both directions at once, and in which reactants and products coexist. Such a reaction reaches a state of balance, called equilibrium, described in Chapter 8. Strictly, every reaction is reversible; the single arrow is used when the reverse reaction is negligible.
- A double-headed arrow, $\leftrightarrow$, does not denote a reaction at all. It connects two resonance structures of the same molecule (Chapter 4).
- Curved arrows, drawn on structures, show the movement of pairs of electrons during a reaction. They are used to describe reaction mechanisms, the step-by-step paths by which reactions occur (Chapter 11).

7.3. Amounts in reactions

7.3.1. Coefficients are ratios of moles

The coefficients of a balanced equation give the ratios in which molecules react. Since a mole is a fixed number of molecules (Chapter 6), they also give the ratios in moles. The equation $CH_4 + 2\ O_2 \rightarrow CO_2 + 2\ H_2O$ states that one mole of methane reacts with two moles of oxygen to give one mole of carbon dioxide and two moles of water. The calculation of amounts of reactants and products from a balanced equation is called **stoichiometry** (from the Greek *stoicheion*, element, and *metron*, measure).

7.3.2. A worked example

How much carbon dioxide and water are produced when 16.0 g of methane burns completely? The molar mass of methane is $12.011 + 4 \times 1.0080 = 16.04$ g mol^{-1}, so 16.0 g is 0.998 mol, very nearly one mole. By the equation, this gives 0.998 mol of CO_2 and $2 \times 0.998 = 2.00$ mol of H_2O. With molar masses of 44.01 and 18.015 g mol^{-1}, these are 43.9 g of carbon dioxide and 36.0 g of water.

The masses confirm conservation of mass. The reactants are 16.0 g of methane and 2×0.998 mol $\times$ 32.00 g mol^{-1} = 63.9 g of oxygen, a total of 79.9 g. The products are 43.9 g + 36.0 g = 79.9 g.

7.3.3. The limiting reactant

Reactants are rarely mixed in exactly the ratio of the equation. The reactant that runs out first is the **limiting reactant**; it determines how much product is formed, and the others are left over in excess. If 1 mol of methane is burned with only 1 mol of oxygen, oxygen is limiting: it can react with only 0.5 mol of methane, giving 0.5 mol of carbon dioxide and 1 mol of water, and 0.5 mol of methane remains. In the chemistry of the origin of life, the scarcest reactant, whether phosphate, a particular sugar or an energy-rich compound, often limits how much of a product can form, however abundant the others may be.

7.4. Transferring protons: acids and bases

Many reactions in water involve the transfer of a single **proton**, that is, a hydrogen ion H^+, which is a hydrogen atom without its electron. An **acid** is a substance that donates a proton, and a **base** is a substance that accepts one. When hydrogen chloride dissolves in water, it donates its proton to a water molecule:

$$HCl + H_2O \rightarrow H_3O^+ + Cl^-$$

Here HCl is the acid and water is the base. The ion H_3O^+, a water molecule carrying an extra proton, is the **hydronium ion**. When ammonia dissolves in water, water acts as the acid and ammonia as the base:

$$NH_3 + H_2O \rightleftharpoons NH_4^+ + OH^-$$

The products are the ammonium ion and the hydroxide ion. Water can act either as an acid or as a base, depending on its partner. Proton transfers are among the fastest of all reactions, and the balance between acids and bases in a solution is measured by its pH. Both are the subject of Chapter 8.

7.5. Transferring electrons: oxidation and reduction

7.5.1. Definitions

A second large class of reactions involves the transfer of electrons. **Oxidation** is the loss of electrons by an atom, ion or molecule, and **reduction** is the gain of electrons. The two always occur together, because electrons lost by one substance must be gained by another, and the combined process is called a **redox reaction** (from *red*uction and *ox*idation). The substance that gains electrons, and so causes the oxidation of its partner, is the **oxidising agent**; the substance that loses electrons is the **reducing agent**.

The reaction of sodium with chlorine is a redox reaction: each sodium atom loses an electron (it is oxidised) and each chlorine atom gains one (it is reduced). Another example, important in the geochemistry of the early Earth, is the oxidation of dissolved iron(II) ions, Fe^{2+}, to iron(III), Fe^{3+}, each by loss of one electron. Oxygen is a strong oxidising agent, and the word oxidation originally meant combination with oxygen; but many oxidations involve no oxygen at all.

7.5.2. Oxidation numbers

In reactions between molecules, electrons are often not transferred completely but shifted partly from one atom towards another. To keep track of such shifts, each atom is assigned an **oxidation number** (or oxidation state): the charge it would have if all its bonds were fully ionic, with the shared electrons of each bond given to the more electronegative atom (Chapter 5). For most purposes a few rules are enough.

1. An atom in an element (H_2, O_2, N_2, C as graphite) has oxidation number 0.

2. Hydrogen bonded to a more electronegative atom, as in all compounds in this book, has +1. Oxygen usually has −2.
3. The oxidation numbers of the atoms in a molecule add up to its charge: zero for a neutral molecule, and the charge of the ion for an ion.

An increase in oxidation number is oxidation, and a decrease is reduction.

7.5.3. A ladder of carbon compounds

The rules show that carbon can exist at many levels of oxidation. In methane, CH_4, each hydrogen is +1, so carbon is −4. In methanol, CH_3OH, three hydrogens on carbon (+3 in total), one oxygen (−2) and one hydrogen on oxygen (+1) leave carbon at −2. Continuing in this way gives the ladder of Figure 7.2: methane (−4), methanol (−2), formaldehyde, H_2CO (0), formic acid, HCOOH (+2), and carbon dioxide, CO_2 (+4). Each step up the ladder is an oxidation by two electrons, and each step down a reduction.

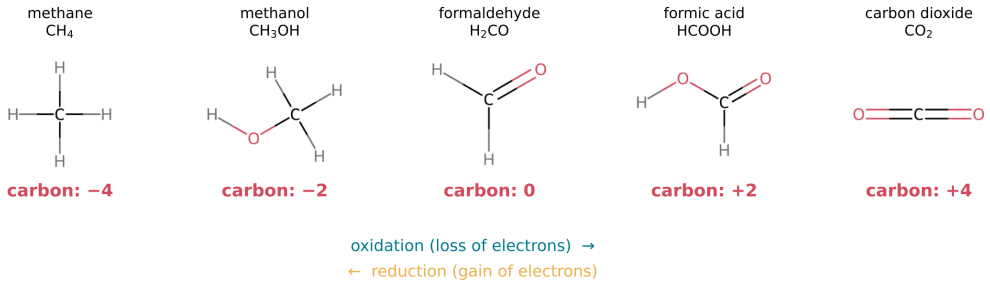


Figure 7.2.: Oxidation numbers of carbon in five one-carbon compounds. Moving to the right, each step removes two electrons from carbon (oxidation); moving to the left adds two electrons (reduction). Carbon dioxide is the most oxidised form of carbon, methane the most reduced.

This ladder matters for the origin of life. The carbon of living matter lies mostly in the middle of the ladder, at oxidation numbers between −2 and +2, as in sugars, whose carbon atoms average 0. On the early Earth, much of the carbon was present as carbon dioxide, at the top. Making the building blocks of life from carbon dioxide therefore requires reduction: a supply of electrons from a reducing agent, such as hydrogen gas, iron(II) or the energy of sunlight. Whether the early atmosphere and oceans were reducing (rich in electron donors) or more oxidised is one of the long-standing questions of the field.

7.6. Joining and splitting with water

7.6.1. Condensation

A third type of reaction is central to the chemistry of life. In a **condensation** reaction, two molecules join to form a larger molecule, and a small molecule, usually water, is released. The reaction of an acid with an alcohol to form a compound called an ester is an example (Figure 7.3):

$$CH_3COOH + CH_3CH_2OH \rightleftharpoons CH_3COOCH_2CH_3 + H_2O$$

An OH group from the acid and an H atom from the alcohol leave together as water, and a new bond joins the remaining parts. The product, ethyl acetate, is an ester; Chapter 11 describes this and other groups formed by condensation.

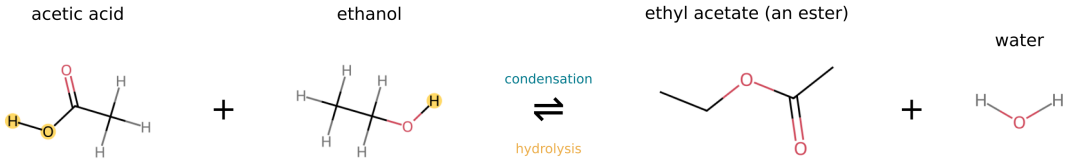


Figure 7.3.: Condensation and hydrolysis. Acetic acid and ethanol combine to form the ester ethyl acetate, releasing a molecule of water (condensation, forward direction). Adding water splits the ester back into the acid and the alcohol (hydrolysis, reverse direction). The atoms that form the water are highlighted.

The large molecules of life are all built by condensation. Proteins are chains of amino acids, each joined to the next by a condensation that releases water. RNA and DNA are chains of nucleotides, joined by condensations between phosphate groups and the hydroxy groups of sugars. Even the attachment of a nucleobase to a sugar is a condensation.

7.6.2. Hydrolysis

The reverse of condensation is **hydrolysis** (from the Greek for "splitting by water"): a reaction in which water breaks a bond, its H going to one fragment and its OH to the other. Ethyl acetate is hydrolysed back to acetic acid and ethanol; proteins are hydrolysed to amino acids, and RNA to its nucleotides.

Condensation and hydrolysis are a reversible pair, and in water, where water molecules are present at about 55 mol L^{-1} (Chapter 6), the balance usually lies on the side of

hydrolysis. A chain of RNA dissolved in water is therefore always at risk of being cut by hydrolysis, and its building blocks do not join of their own accord. How chains could nevertheless have formed in water on the early Earth is one of the central problems of the origin of life. Chapters 8 and 9 explain why the large amount of water favours hydrolysis, and Chapter 12 returns to the question.

7.6.3. Addition

In an **addition** reaction, two molecules combine to form a single product, and no small molecule is released. Additions to double bonds are common. Water can add to the C=O double bond of formaldehyde, and hydrogen cyanide can add to it to form a larger molecule containing two carbon atoms:

$$\mathrm{HCN} + \mathrm{H_2CO} \rightarrow \mathrm{HOCH_2CN}$$

Additions of this kind, which build new carbon-carbon bonds from small molecules such as hydrogen cyanide and formaldehyde, are thought to be among the first steps by which more complex organic molecules formed on the early Earth.

Key points

- A chemical reaction rearranges atoms by breaking and forming bonds; atoms of each element are conserved, and so is mass.
- A chemical equation lists reactants, an arrow, and products, with coefficients giving the number of molecules or moles. A balanced equation has the same number of atoms of each element, and the same charge, on both sides.
- $\rightarrow$ marks a reaction, $\rightleftharpoons$ a reversible reaction, $\leftrightarrow$ resonance (not a reaction). State symbols are (s), (l), (g), (aq).
- Coefficients are mole ratios. Stoichiometry converts between masses and amounts: 16 g of methane burns to 44 g of CO_2 and 36 g of water. The limiting reactant determines the yield.
- Acid-base reactions transfer protons (H^+). An acid donates a proton and a base accepts one; water can act as either.
- Redox reactions transfer electrons: oxidation is loss, reduction gain. Oxidation numbers track the shifts; carbon ranges from -4 in CH_4 to $+4$ in CO_2. Making biological molecules from CO_2 requires reduction.
- Condensation joins two molecules and releases water; hydrolysis is the reverse. Proteins, RNA and DNA are built by condensation and broken by hydrolysis, which water favours.
- In an addition reaction, two molecules combine without losing any atoms.

Self-test

1. Balance the equation for the burning of ethanol: $C_2H_5OH + O_2 \rightarrow CO_2 + H_2O$.
2. How many grams of water are formed when 4.0 g of hydrogen gas reacts completely with oxygen?
3. In the reaction $NH_3 + HCl \rightarrow NH_4^+ + Cl^-$, which reactant is the acid and which the base?
4. Find the oxidation number of carbon in carbon monoxide (CO) and of nitrogen in ammonia (NH_3) and in nitrogen gas (N_2).
5. Glycine (H_2NCH_2COOH) and alanine ($H_2NCH(CH_3)COOH$) join by condensation, the amino group of one reacting with the carboxy group of the other. What small molecule is released, and what is the molecular formula of the product?

8. Equilibrium, Acids and pH

The question of this chapter

Many reactions do not go to completion. They stop partway, with reactants and products both present, and nothing further seems to happen however long one waits. This chapter asks: *where does a reaction stop, and why?* The answer is chemical equilibrium, a balance between a reaction and its reverse that can be described by a single number, the equilibrium constant. The same idea explains what acids and bases do in water, what pH measures, and why the joining of building blocks into chains is so difficult in dilute solution.

8.1. Dynamic equilibrium

8.1.1. Two reactions at once

Consider a reversible reaction, A $\rightleftharpoons$ B. When A is first dissolved, only the forward reaction, A $\rightarrow$ B, can occur, and B begins to accumulate. As soon as some B is present, the reverse reaction, B $\rightarrow$ A, also begins. As A is used up, the forward reaction slows; as B accumulates, the reverse reaction speeds up. Eventually the two rates become equal. From then on, A is converted into B exactly as fast as B is converted back into A, and the amounts of both stay constant (Figure 8.1).

This state is called **chemical equilibrium**. It is a **dynamic equilibrium**: at the level of molecules, both reactions continue without pause, but their effects cancel, and nothing changes on the scale of the whole sample. The same final state is reached whether one starts with pure A, pure B, or any mixture of the two, provided that the temperature and the total amounts are the same.

8.1.2. Equilibrium is not equality

Equilibrium does not mean that reactants and products are present in equal amounts. It means that their amounts no longer change. In some reactions, the equilibrium mixture contains almost nothing but products; in others, almost nothing but reactants; and in many, appreciable amounts of both. Which of these applies is expressed by the equilibrium constant.

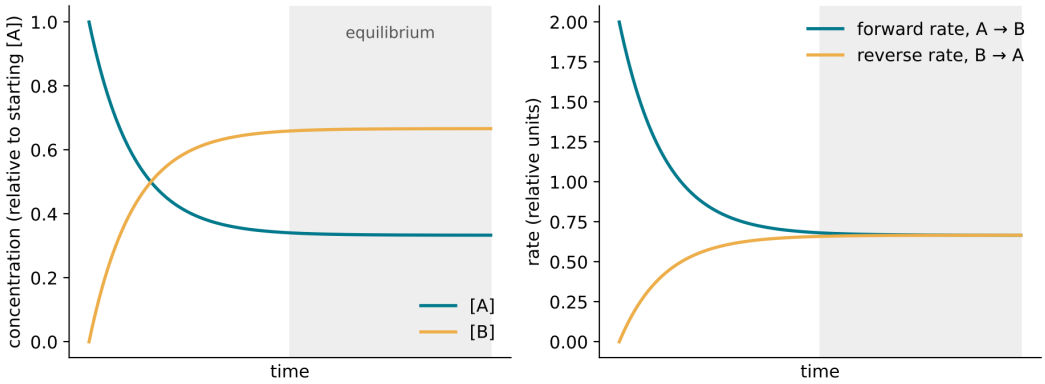


Figure 8.1.: Approach to equilibrium for a reaction A $\rightleftharpoons$ B, starting from pure A. Left: concentrations of A (teal) and B (orange) against time; both level off at their equilibrium values. Right: rates of the forward (A $\rightarrow$ B) and reverse (B $\rightarrow$ A) reactions, which become equal at equilibrium. The curves are calculated for a forward rate constant twice as large as the reverse one, so that at equilibrium [B] = 2[A].

8.2. The equilibrium constant

8.2.1. Definition

For a reaction written in general form as

$$a\,\mathrm{A} + b\,\mathrm{B} \rightleftharpoons c\,\mathrm{C} + d\,\mathrm{D}$$

where the lower-case letters are the coefficients, the **equilibrium constant** K is

$$K = \frac{[\mathrm{C}]^c\,[\mathrm{D}]^d}{[\mathrm{A}]^a\,[\mathrm{B}]^b}$$

where the square brackets denote the concentrations at equilibrium. The products appear in the numerator and the reactants in the denominator, each raised to the power of its coefficient. At a given temperature, K has a fixed value for each reaction, whatever the starting amounts.

Two conventions make K a pure number and keep it simple. First, each concentration is divided by a standard concentration of 1 mol L^{-1}, so that [A] stands for the number of moles per litre without the unit. Strictly, K is defined with **activities**, effective concentrations that allow for the interactions between dissolved particles, but in dilute solutions activities are close to concentrations divided by 1 mol L^{-1}. Second, pure solids and pure liquids, and the solvent of a dilute solution, have activities of 1 and are left

out of the expression. In reactions in dilute aqueous solution, water therefore does not appear in K, even when it is a reactant or a product.

8.2.2. What the size of K means

- If K is much larger than 1, the numerator dominates at equilibrium: the reaction lies far to the right, and the equilibrium mixture consists mostly of products.
- If K is much smaller than 1, the reaction lies far to the left, and little product forms.
- If K is close to 1, appreciable amounts of both reactants and products are present.

Because equilibrium constants range over many orders of magnitude, from 10^{-30} or less to 10^{30} or more, they are often given as logarithms, in the form $\mathrm{pK} = -\log_{10} \mathrm{K}$, which turns the range into a manageable span of numbers. This is the origin of the "p" in pH and $\mathrm{pK_a}$, described below.

8.2.3. An important consequence: dilution and joining

The concentrations in K are raised to powers, and this has a consequence that runs through the whole chemistry of the origin of life. Consider a condensation in water, in which two building blocks A and B join to form AB and release water:

$$\mathrm{A} + \mathrm{B} \rightleftharpoons \mathrm{AB} + \mathrm{H_2O}$$

Since water is the solvent, it does not appear in K, and

$$K = \frac{[\mathrm{AB}]}{[\mathrm{A}]\,[\mathrm{B}]}, \qquad \text{so} \qquad \frac{[\mathrm{AB}]}{[\mathrm{A}]} = K \times [\mathrm{B}]$$

The fraction of A that is joined to B depends not only on K but also on the concentration of B. Suppose, for example, that K = 10. If B is present at 1 mol $\mathrm{L^{-1}}$, then [AB]/[A] = 10: most of A is joined. If B is present at 1 mM, then [AB]/[A] = 10 × 0.001 = 0.01: only about 1 % of A is joined. At 1 μM, only about one A in 100 000. The more dilute the solution, the more the equilibrium favours the separate building blocks and the less it favours the joined product. Dilute solutions favour hydrolysis.

This simple relation explains why the chains of life do not form spontaneously from dilute building blocks in water, and why so much attention is paid to ways of concentrating them, by evaporation, freezing or adsorption on mineral surfaces, and of removing water, by drying.

8.3. Shifting an equilibrium

8.3.1. Le Chatelier's principle

An equilibrium can be disturbed by changing the conditions: adding or removing a reactant or product, changing the temperature, or, for gases, changing the pressure. The response follows a general rule known as **Le Chatelier's principle**: when a system at equilibrium is disturbed, the equilibrium shifts in the direction that partly counteracts the disturbance.

- **Adding a reactant** shifts the equilibrium to the right, consuming some of the added reactant and forming more product. **Removing a product** has the same effect. A reaction whose product is continually removed, for example because it crystallises out of solution or escapes as a gas, can be driven almost to completion even if its K is small.
- **Removing water** from a condensation, A + B $\rightleftharpoons$ AB + H_2O, shifts it towards AB. In dilute solution the water is the solvent and cannot be removed without concentrating everything else; but when a solution is dried, water is no longer present in excess, its activity falls below 1, and condensation becomes more favourable. This is one reason why wet-dry cycles are thought to have helped build chains on the early Earth.
- **Changing the temperature** changes the value of K itself. Raising the temperature favours the direction that absorbs heat. Chapter 9 explains why.

Adding a catalyst, a substance that speeds up a reaction without being consumed (Chapter 10), does not shift an equilibrium at all. It speeds up the forward and reverse reactions by the same factor, so that equilibrium is reached sooner, but the equilibrium mixture is the same.

8.4. Water, acids and bases

8.4.1. Water ionises itself

Water itself takes part in an equilibrium. A small fraction of water molecules transfer a proton to a neighbouring water molecule:

$$2\,H_2O \rightleftharpoons H_3O^+ + OH^-$$

Because water is the solvent, its activity is 1, and the equilibrium constant reduces to the product of the activities of the two ions. This constant is the **ion product of water**, K_w. At 25 °C, $pK_w = -\log_{10} K_w = 13.99$ (International Association for the Properties of Water and Steam 2024), so $K_w = 1.0 \times 10^{-14}$. In pure water the two ions are present

in equal amounts, each at a concentration of about 10^{-7} mol L^{-1}. Since pure water contains about 55 mol L^{-1} of water molecules (Chapter 6), only about two molecules in a thousand million are ionised at any moment.

8.4.2. pH

The concentration of hydronium ions in a solution determines its acidity. Because this concentration varies over many orders of magnitude, it is expressed on a logarithmic scale (Chapter 1). The **pH** of a solution is defined as

$$\mathrm{pH} = -\log_{10} a(\mathrm{H_3O^+})$$

where $a(H_3O^+)$ is the activity of the hydronium ion, which in dilute solution is close to its concentration in mol L^{-1}. A solution with $[H_3O^+] = 10^{-7}$ M has pH 7; one with 10^{-3} M has pH 3. Each unit of pH is a factor of ten, so a solution of pH 3 contains ten thousand times as many hydronium ions as a solution of pH 7.

Since the activities of H_3O^+ and OH^- multiply to K_w, an increase in one means a decrease in the other. A solution is **neutral** when the two are equal, which at 25 °C is at pH 7.00. A solution with a lower pH is **acidic**, and one with a higher pH is **basic** (or alkaline). Figure 8.2 shows the scale.

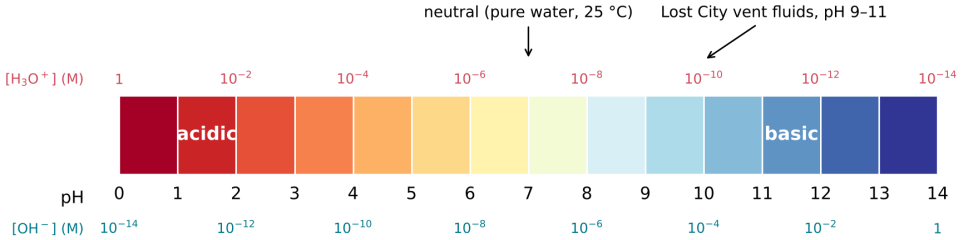


Figure 8.2.: The pH scale at 25 °C. Each step of one unit is a factor of ten in the concentration of hydronium ions (top) and, in the opposite direction, of hydroxide ions (bottom). Pure water is neutral at pH 7; the fluids of the Lost City hydrothermal field have pH 9 to 11 (Kelley et al. 2005).

8.4.3. Neutral is not always pH 7

The value of K_w depends strongly on temperature, and so does the pH of neutral water (Table 8.1). At 0 °C neutral water has pH 7.48, and at 100 °C it has pH 6.13. A hot spring at 100 °C with pH 6.5 is therefore slightly basic, not acidic. This matters when

comparing the waters of cold pools, hot springs and hydrothermal vents, all of which have been proposed as settings for the origin of life.

Table 8.1.: Ion product of water and the pH of neutral water at three temperatures (International Association for the Properties of Water and Steam 2024). Neutral pH is $pK_w/2$, rounded.

Temperature (°C)	pK_w	pH of neutral water
0	14.95	7.48
25	13.99	7.00
100	12.25	6.13

8.5. Strong and weak acids

8.5.1. Strong acids

An acid is a proton donor (Chapter 7). A **strong acid**, such as hydrochloric acid, gives up its proton to water essentially completely; in its solution, no undissociated HCl molecules remain. A 0.01 M solution of hydrochloric acid therefore contains 0.01 M H_3O^+ and has pH 2.

8.5.2. Weak acids and K_a

Most acids in the chemistry of life are **weak acids**: they donate their protons only partly, and an equilibrium is established between the acid and its deprotonated form. For a weak acid HA,

$$HA + H_2O \rightleftharpoons H_3O^+ + A^-$$

The species A^- that remains after HA has lost its proton is called the **conjugate base** of HA. The equilibrium constant of this reaction, with water omitted, is the **acid dissociation constant**:

$$K_a = \frac{[H_3O^+]\,[A^-]}{[HA]}$$

and it is usually given as $pK_a = -\log_{10} K_a$. The smaller the pK_a, the stronger the acid. Acetic acid, the acid of vinegar, has $pK_a = 4.76$ (Goldberg et al. 2002).

8.5.3. pH and pK_a: which form is present?

Taking logarithms of the expression for K_a gives a simple and very useful relation between the pH and the ratio of the two forms of an acid:

$$\mathrm{pH} = \mathrm{p}K_\mathrm{a} + \log_{10} \frac{[\mathrm{A}^-]}{[\mathrm{HA}]}$$

It is known as the Henderson-Hasselbalch equation, and its consequences can be read off directly.

- When $\mathrm{pH} = \mathrm{p}K_\mathrm{a}$, the logarithm is zero, so $[\mathrm{A}^-] = [\mathrm{HA}]$: the acid is half dissociated.
- For each unit of pH above the $\mathrm{p}K_\mathrm{a}$, the ratio $[\mathrm{A}^-]/[\mathrm{HA}]$ increases tenfold. Two units above the $\mathrm{p}K_\mathrm{a}$, the acid is 99 % in its deprotonated form, A^-.
- For each unit below, the ratio decreases tenfold. Two units below, the acid is 99 % in its protonated form, HA.

At pH 7, acetic acid, with $\mathrm{p}K_\mathrm{a} = 4.76$, is more than 99 % in the form of the acetate ion, CH_3COO^-. The carboxy groups of amino acids and other biological acids are likewise almost entirely deprotonated, and negatively charged, at neutral pH.

8.5.4. Phosphoric acid and phosphate

Phosphoric acid, H_3PO_4, has three O–H groups, each of which can lose a proton, with three successive $\mathrm{p}K_\mathrm{a}$ values of about 2.1, 7.2 and 12.4 (Wikipedia contributors, n.d.-b). Depending on the pH, it is present as H_3PO_4, $H_2PO_4^-$, HPO_4^{2-} or PO_4^{3-} (Figure 8.3). Near pH 7, $H_2PO_4^-$ and HPO_4^{2-} are present in similar amounts. The general word **phosphate** covers all these forms.

In RNA, each phosphate group is joined to two sugars and keeps only one O–H group of its own. This remaining group is a strong acid, much stronger than the second O–H group of phosphoric acid, and at neutral pH it has lost its proton: each phosphate linkage of an RNA chain carries one negative charge (Westheimer 1987). An RNA chain of 100 nucleotides carries about 100 negative charges along its backbone.

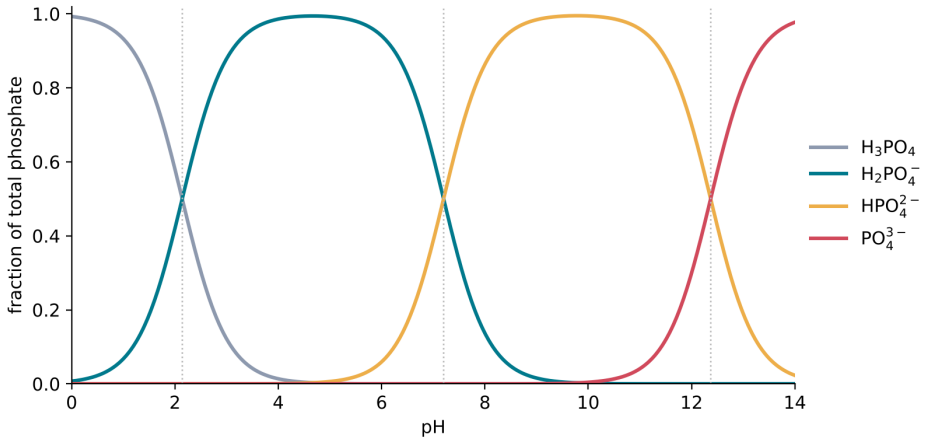


Figure 8.3.: Fraction of phosphate present in each of its four forms as a function of pH, calculated from the three pK_a values 2.14, 7.20 and 12.37 (Wikipedia contributors, n.d.-b). Each crossing point, where two forms are present in equal amounts, lies at a pK_a.

8.6. Buffers

A solution that contains a weak acid together with its conjugate base, in comparable amounts, resists changes in pH. Such a solution is called a **buffer**. If a strong acid is added, the added H_3O^+ ions are largely taken up by A^-, forming HA; if a strong base is added, the OH^- ions are largely neutralised by HA, forming A^- and water. In either case the ratio $[A^-]/[HA]$ changes only a little, and by the Henderson-Hasselbalch equation the pH changes only a little.

A buffer works best when the pH is close to the pK_a of its acid, where the two forms are present in similar amounts. A mixture of $H_2PO_4^-$ and HPO_4^{2-}, with p$K_a = 7.2$, buffers near neutral pH, and phosphate buffers are among the most widely used in laboratories. Natural waters are also buffered, mainly by dissolved carbon dioxide and its ions, and by reactions with minerals. Laboratory experiments on the origin of life are usually run in buffered solutions, so that the pH stays where the experimenter has set it; on the early Earth, the pH of each setting would have been set by its own buffers.

Key points

- A reversible reaction reaches dynamic equilibrium when its forward and reverse rates are equal; the amounts then stay constant, though both reactions continue.
- For $aA + bB \rightleftharpoons cC + dD$, $K = [C]^c[D]^d / ([A]^a[B]^b)$, with concentrations in mol L^{-1} (strictly, activities). Pure solids, pure liquids and the solvent are left out. $K \gg 1$ favours products; $K \ll 1$ favours reactants.

- For a condensation A + B $\rightleftharpoons$ AB + H_2O in water, [AB]/[A] = K[B]: dilution strongly favours the separate building blocks.
- Le Chatelier's principle: a disturbed equilibrium shifts to counteract the disturbance. Removing a product or water drives a reaction forward. A catalyst does not shift an equilibrium.
- Water ionises: $K_w = 1.0 \times 10^{-14}$ at 25 °C. pH = $-\log_{10}$ a(H_3O^+). Neutral water has pH 7.00 at 25 °C, 7.48 at 0 °C and 6.13 at 100 °C.
- Strong acids dissociate completely; weak acids partly, with K_a and pK_a. pH = $pK_a + \log_{10}([A^-]/[HA])$. At pH = pK_a the acid is half dissociated.
- Phosphoric acid has pK_a values of about 2.1, 7.2 and 12.4. Each phosphate linkage of RNA carries one negative charge at neutral pH.
- A buffer, a weak acid with its conjugate base, resists changes in pH, best near its pK_a.

Self-test

1. Write the expression for K for the reaction $N_2(g) + 3\ H_2(g) \rightleftharpoons 2\ NH_3(g)$, using concentrations.
2. A condensation A + B $\rightleftharpoons$ AB + H_2O has K = 100. What fraction of A is joined when [B] is held at 0.1 M, and when it is held at 10 μM?
3. What is the pH of a solution with $[H_3O^+] = 2.0 \times 10^{-5}$ M? Is it acidic or basic at 25 °C?
4. Formic acid has $pK_a = 3.75$. What is the ratio $[HCOO^-]/[HCOOH]$ at pH 5.75?
5. Why does adding a little hydrochloric acid to a phosphate buffer at pH 7.2 change the pH much less than adding the same amount to pure water?

9. Energy and the Direction of Change

The question of this chapter

Some changes happen of their own accord and others never do. Water left in an open dish evaporates; a mixture of hydrogen and oxygen, once ignited, turns into water; but water never separates by itself into hydrogen and oxygen, and the evaporated water does not return to the dish. This chapter asks: *what decides the direction in which a change can go?* The answer involves two quantities, enthalpy and entropy, which combine into a third, the Gibbs energy. The Gibbs energy also explains the size of the equilibrium constant met in Chapter 8, and why joining building blocks into chains in water needs a source of energy.

9.1. Energy, heat and temperature

9.1.1. Energy and its units

Energy is the capacity to do work or to supply heat. It takes several forms. **Kinetic energy** is the energy of motion. **Potential energy** is energy stored in the positions of objects that attract or repel each other, such as a raised weight in the field of the Earth, or electrons and nuclei in the electric field of their charges. The **chemical energy** of a substance is the potential and kinetic energy of its electrons and nuclei, and it changes when bonds are broken and formed.

The SI unit of energy is the **joule** (J), equal to 1 kg m^2 s^{-2} (Bureau International des Poids et Mesures 2019). Chemical energies are usually given per mole of reaction, in kilojoules per mole (kJ mol^{-1}). Older literature and food labels use the **calorie**; the thermochemical calorie is defined as exactly 4.184 J (Thompson and Taylor 2008), and the "Calorie" of nutrition labels is a kilocalorie.

The **first law of thermodynamics** states that energy is conserved: it can be transferred from one body to another and converted from one form into another, but it is neither created nor destroyed. When a reaction releases energy, that energy is not new; it was stored in the reactants and is passed to the surroundings, usually as heat.

9.1.2. Heat and temperature are not the same

Heat is energy transferred from one body to another because of a difference in temperature. **Temperature** is a measure of how vigorously the particles of a body move. In a gas, the average kinetic energy of the molecules moving from place to place is proportional to the absolute temperature. A bathtub of warm water and a cup of water at the same temperature have the same temperature but very different amounts of thermal energy, because the bathtub contains far more molecules.

The energy of thermal motion per particle is of the order of k_BT, where k_B is the **Boltzmann constant**, exactly 1.380 649 $\times$ 10^{-23} J K^{-1} (Bureau International des Poids et Mesures 2019). Per mole of particles the corresponding quantity is RT, where R is the molar gas constant, 8.314 462 618 J K^{-1} mol^{-1} (National Institute of Standards and Technology, n.d.-a). At 25 °C (298.15 K), RT = 2.479 kJ mol^{-1}. This number is a useful yardstick. Energies of a few kJ mol^{-1} are comparable with thermal motion and are easily supplied or overcome by collisions; energies of hundreds of kJ mol^{-1}, typical of covalent bonds, are not.

9.2. Enthalpy: the heat of a reaction

9.2.1. Exothermic and endothermic

Most chemical reactions are carried out at constant pressure, open to the atmosphere. The heat exchanged with the surroundings under these conditions is equal to the change in a quantity called **enthalpy**, H. Only changes in enthalpy, ΔH, can be measured, not absolute values. A reaction that releases heat to its surroundings has $\Delta H < 0$ and is called **exothermic**; one that absorbs heat has $\Delta H > 0$ and is called **endothermic** (Figure 9.1).

9.2.2. Bonds and enthalpy

Breaking a chemical bond always requires energy, and forming one always releases energy. The energy needed to break one mole of a given bond in the gas phase is its bond enthalpy (Chapter 3, Table 3.2). A reaction is exothermic when the bonds formed in the products are, in total, stronger than the bonds broken in the reactants, and endothermic when they are weaker. The very strong triple bond of N_2, 945.4 kJ mol^{-1}, explains why nitrogen gas, which makes up most of the air, reacts so reluctantly: any reaction of N_2 must begin by breaking it.

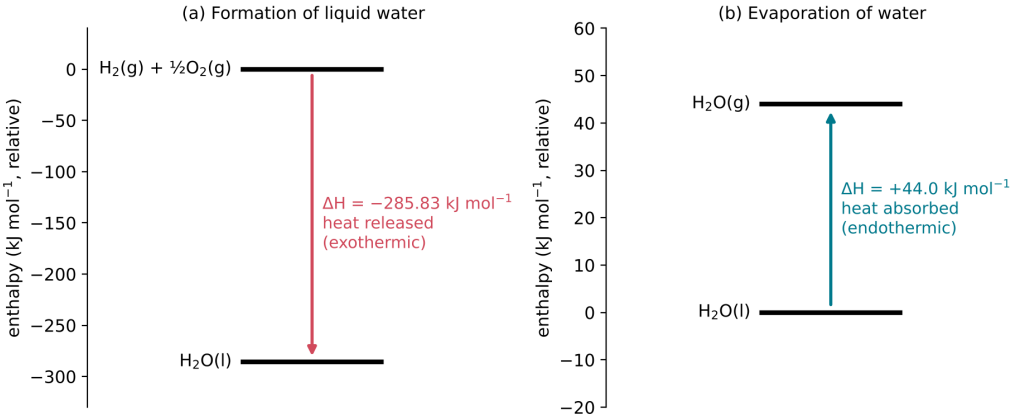


Figure 9.1.: Enthalpy changes drawn as levels. (a) The formation of liquid water from hydrogen and oxygen releases 285.83 kJ per mole of water: the products lie lower than the reactants. (b) The evaporation of water absorbs 44.0 kJ per mole: the vapour lies higher than the liquid. Values for 25 °C and 1 bar (National Institute of Standards and Technology, n.d.-b; Cox et al. 1984).

9.2.3. Enthalpies of formation and Hess's law

Enthalpy changes are tabulated as **standard enthalpies of formation**, $\Delta_f H^\circ$: the enthalpy change when one mole of a compound is formed from its elements in their most stable forms, with all substances in their **standard states**. The degree sign ° marks a standard value. For gases, the standard state is a pressure of 1 bar; for dissolved substances, a concentration of 1 mol L^{-1}; tables usually refer to 298.15 K. By definition, $\Delta_f H^\circ$ of an element in its most stable form, such as $H_2(g)$ or $O_2(g)$, is zero.

Enthalpy is a **state function**: its value depends only on the present state of a system, not on how that state was reached. It follows that the enthalpy change of a reaction is the same whether the reaction occurs in one step or in several, and that it can be calculated from enthalpies of formation:

$$\Delta H^\circ = \sum \Delta_f H^\circ(\text{products}) - \sum \Delta_f H^\circ(\text{reactants})$$

each multiplied by its coefficient in the equation. This rule is **Hess's law**. As an example, consider the burning of methane, the main component of natural gas:

$$CH_4(g) + 2\,O_2(g) \rightarrow CO_2(g) + 2\,H_2O(l)$$

With $\Delta_f H^\circ = -74.87$ kJ mol^{-1} for methane (Chase 1998), -393.51 kJ mol^{-1} for carbon dioxide and -285.830 kJ mol^{-1} for liquid water (Cox et al. 1984),

$$\Delta H^\circ = [-393.51 + 2(-285.830)] - [-74.87 + 0] = -890.30 \text{ kJ mol}^{-1}$$

The reaction releases about 890 kJ of heat for each mole (16 g) of methane burned.

9.3. Entropy: the number of arrangements

9.3.1. Heat alone does not decide

If the release of heat were the only thing that mattered, endothermic changes would never happen of their own accord. Yet they do. Water evaporates from a dish at room temperature, absorbing heat from its surroundings; ice melts in a warm room, absorbing heat. A second quantity is needed to explain the direction of change.

9.3.2. What entropy measures

Entropy, S, is a measure of the number of microscopic arrangements of a system's particles and energy that are consistent with what is observed on the large scale. A state that can be realised in many ways has a high entropy; one that can be realised in few ways has a low entropy. Some general trends follow:

- A gas has a much higher entropy than the liquid from which it forms, because its molecules move freely through a much larger volume. The standard molar entropy of water vapour is 188.835 J K^{-1} mol^{-1}, and that of liquid water 69.95 J K^{-1} mol^{-1} (National Institute of Standards and Technology, n.d.-b; Cox et al. 1984).
- A liquid has a higher entropy than its solid, in which the particles are held in fixed positions.
- Entropy increases with temperature, because the particles can share a larger amount of energy in more ways.
- A reaction that increases the number of independent particles, especially of gas molecules, usually increases entropy; one that joins particles together, such as the joining of two building blocks into one, usually decreases it.

Entropy has units of energy divided by temperature, J K^{-1}, or J K^{-1} mol^{-1} for one mole.

9.3.3. The second law

The **second law of thermodynamics** states that in any spontaneous change, the total entropy of a system and its surroundings increases. The word **spontaneous** here has a technical meaning: a spontaneous change is one that can happen without being driven from outside. It says nothing about speed. A spontaneous change may be fast, like the neutralisation of an acid by a base, or so slow that it is never observed, as Section 9.7 shows.

The second law refers to the *total* entropy. The entropy of a system can decrease, as when water freezes into ordered ice, provided that the entropy of the surroundings increases by more. Freezing releases heat, and heat flowing into the surroundings increases their entropy. When an amount of heat q flows into surroundings at absolute temperature T, their entropy increases by q/T. The same amount of heat therefore increases entropy more when the surroundings are cold than when they are hot, and this is why water freezes below 0 °C but not above.

9.4. Gibbs energy: one quantity for both

9.4.1. Definition

Keeping track of the entropy of the surroundings is inconvenient. For a change at constant temperature and pressure, it can be replaced by a quantity that refers only to the system: the **Gibbs energy** (also called the Gibbs free energy), G, named after the American physicist Josiah Willard Gibbs. Its change is

$$\Delta G = \Delta H - T\Delta S$$

where T is the absolute temperature. The link to the second law is as follows. At constant pressure, the heat released into the surroundings is $-\Delta$H, so their entropy changes by $-\Delta$H/T, while the entropy of the system changes by ΔS. The total entropy change is therefore ΔS $-$ ΔH/T, and multiplying by $-$T gives exactly ΔH $-$ TΔS $=$ ΔG. A decrease in the Gibbs energy of the system thus corresponds to an increase in the total entropy, and the second law takes a simple form:

- If ΔG < 0, the change is spontaneous: it can occur of its own accord.
- If ΔG > 0, the change is not spontaneous; the reverse change is.
- If ΔG $= 0$, the system is at equilibrium.

A change with ΔG < 0 is also described as **favourable** or **exergonic**, and one with ΔG > 0 as **unfavourable** or **endergonic**.

9.4.2. Four combinations

Since ΔH and ΔS can each be positive or negative, there are four cases (Table 9.1).

Table 9.1.: The sign of ΔG for the four combinations of ΔH and ΔS.

ΔH	ΔS	ΔG = ΔH − TΔS	Example
− (releases heat)	+ (more arrangements)	negative at all temperatures	decomposition of hydrogen peroxide into water and oxygen gas
+ (absorbs heat)	− (fewer arrangements)	positive at all temperatures	the reverse: water and oxygen forming hydrogen peroxide
−	−	negative at low temperature, positive at high	freezing of water; many reactions that join two molecules into one
+	+	positive at low temperature, negative at high	melting of ice; evaporation of water

9.4.3. Two worked examples: water changing state

For the evaporation of one mole of liquid water at 298.15 K and 1 bar, the standard values give

$$\Delta H = -241.826 - (-285.830) = +44.004\,\mathrm{kJ\,mol^{-1}}, \qquad \Delta S = 188.835 - 69.95 = +118.89\,\mathrm{J\,K^{-1}\,mol^{-1}}$$

using $\Delta_f H^\circ = -241.826$ kJ mol^{-1} for water vapour (National Institute of Standards and Technology, n.d.-b; Cox et al. 1984). Before the two terms are combined, ΔS must be converted to kJ K^{-1} mol^{-1} (118.89 J = 0.11889 kJ):

$$\Delta G = 44.004 - 298.15 \times 0.11889 = +8.56\ \mathrm{kJ\,mol^{-1}}$$

ΔG is positive: at 25 °C, liquid water does not turn into vapour at a pressure of 1 bar. (Water in an open dish evaporates nonetheless, because the pressure of water vapour in ordinary air is far below 1 bar; Section 9.5 shows how ΔG depends on the amounts present.) The enthalpy term favours the liquid, the entropy term favours the vapour, and the entropy term grows with temperature. The two balance, ΔG = 0, when T =

$\Delta H/\Delta S$ = 44 004 / 118.89 ≈ 370 K (Figure 9.2 a). This estimate assumes that ΔH and ΔS do not change with temperature, and it comes within 3 K of the boiling point of water at 1 atmosphere, 373.12 K (99.97 °C on the present international temperature scale, calculated from the international reference formulation for water (Wagner and Pruß 2002)).

The melting of ice follows the same pattern on a smaller scale. At 0 °C and 1 atmosphere, melting absorbs 6.01 kJ mol^{-1} and increases the entropy by 22.0 J K^{-1} mol^{-1} (values calculated from the international reference formulations for ice and liquid water (Feistel and Wagner 2006; Wagner and Pruß 2002)). Below 273.15 K the enthalpy term wins and ice is stable; above it the entropy term wins and the ice melts (Figure 9.2 b). At the melting point itself, $\Delta G = 0$ and ice and water coexist at equilibrium.

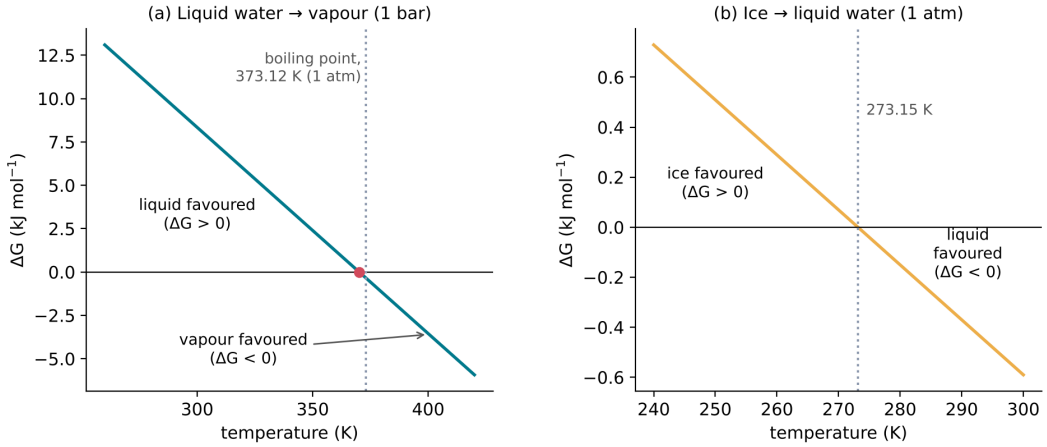


Figure 9.2.: Gibbs energy change as a function of temperature, $\Delta G = \Delta H - T\Delta S$, assuming that ΔH and ΔS are constant. (a) Evaporation of water at 1 bar, with ΔH = 44.004 kJ mol^{-1} and ΔS = 118.89 J K^{-1} mol^{-1} (National Institute of Standards and Technology, n.d.-b; Cox et al. 1984). ΔG crosses zero near 370 K; the dotted line marks the boiling point at 1 atm, 373.12 K (Wagner and Pruß 2002). (b) Melting of ice at 1 atm, with ΔH = 6.01 kJ mol^{-1} (Feistel and Wagner 2006; Wagner and Pruß 2002); ΔG crosses zero at the melting point.

9.5. Gibbs energy and equilibrium

9.5.1. The standard Gibbs energy and K

The Gibbs energy change of a reaction depends on the amounts of reactants and products present. The **standard Gibbs energy change**, $\Delta G°$, refers to the reaction with every substance in its standard state, for example every dissolved substance at 1 mol L^{-1}.

It is a fixed number for each reaction at a given temperature, and it is related to the equilibrium constant of Chapter 8 by

$$\Delta G^\circ = -RT \ln K \tag{9.1}$$

where ln is the natural logarithm (Chapter 1). Rearranged, $K = e^{-\Delta G^\circ/RT}$. A negative ΔG° gives $K > 1$, so the products are favoured at equilibrium; a positive ΔG° gives $K < 1$.

Because K depends exponentially on ΔG°, modest differences in energy produce large differences in K. At 298.15 K, RT ln 10 = 5.71 kJ mol^{-1}, so every 5.71 kJ mol^{-1} of ΔG° changes K by a factor of ten (Figure 9.3). Table 9.2 gives some values.

Table 9.2.: Equilibrium constants for some values of the standard Gibbs energy change, from Equation 9.1.

ΔG° (kJ mol^{-1})	K at 298.15 K
−40	1.0×10^{7}
−20	3.2×10^{3}
−10	56
0	1
+10	0.018
+20	3.1×10^{-4}
+40	9.8×10^{-8}

9.5.2. Away from equilibrium: the reaction quotient

When a reaction mixture is not at equilibrium, the ratio of products to reactants, written in the same form as K but with the actual concentrations, is called the **reaction quotient**, Q. The Gibbs energy change under these actual conditions is

$$\Delta G = \Delta G^\circ + RT \ln Q$$

This equation states three things. If $Q < K$, ΔG is negative and the reaction proceeds forwards, making more product. If $Q > K$, ΔG is positive and the reaction proceeds backwards. If $Q = K$, $\Delta G = 0$ and the mixture is at equilibrium. A reaction with a positive ΔG° can therefore still proceed, if the products are kept at low concentration or the reactants at high concentration, because a small Q makes RT ln Q large and negative. This is the quantitative form of Le Chatelier's principle (Chapter 8): adding reactant or removing product lowers Q and drives the reaction forwards.

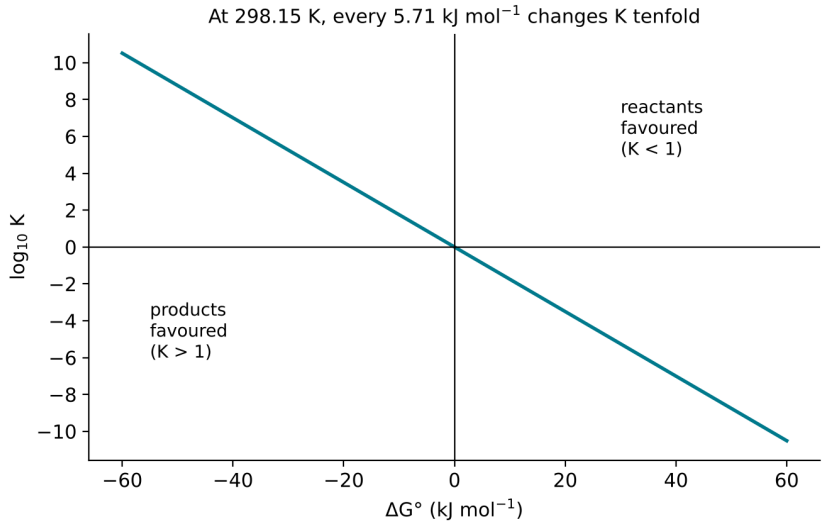


Figure 9.3.: The relation between ΔG° and $\log_{10}$ K at 298.15 K, from Equation 9.1. The line falls by one unit of $\log_{10}$ K for every 5.71 kJ mol^{-1} increase in ΔG°.

9.5.3. Why water favours hydrolysis

Chapters 7 and 8 showed that a condensation in water, A + B $\rightleftharpoons$ AB + H_2O, is held back by the large excess of water. In the language of this chapter, the standard Gibbs energy change of such a condensation, with water as the solvent, is positive for many of the links found in living molecules; the reverse reaction, hydrolysis, is favourable. The equilibrium mixture therefore contains mostly separate building blocks, and dilution makes the situation worse, because the amount of joined product at equilibrium is proportional to the concentration of the partners ([AB]/[A] = K[B], Chapter 8). To make chains, either the water must be removed, so that the solvent no longer pushes the equilibrium back, or the joining must be paid for with Gibbs energy from another source, as described in Section 9.6.

9.5.4. How temperature changes K

Combining Equation 9.1 with $\Delta G^\circ = \Delta H^\circ - T\Delta S^\circ$ and dividing by $-RT$ gives

$$\ln K = -\frac{\Delta H^\circ}{RT} + \frac{\Delta S^\circ}{R}$$

If ΔH° and ΔS° change little with temperature, the second term is constant, and the effect of temperature is carried by the first term. For an exothermic reaction ($\Delta H^\circ < 0$), the first term is positive and becomes smaller as T increases, so K decreases on heating: the equilibrium shifts towards the reactants. For an endothermic reaction ($\Delta H^\circ > 0$),

K increases on heating. This is the explanation promised in Chapter 8 for the rule that raising the temperature favours the direction that absorbs heat. The relation is known as the **van 't Hoff equation**, after the Dutch chemist Jacobus Henricus van 't Hoff.

9.6. Driving unfavourable reactions

9.6.1. Coupling

An unfavourable reaction can still take place if it is linked to a favourable one in such a way that the two occur together as a single overall process. Because Gibbs energy is a state function, the Gibbs energy changes of the two steps add. If a reaction A + B → AB has $\Delta G = +15$ kJ mol^{-1} and another reaction X → Y has $\Delta G = -30$ kJ mol^{-1}, a combined process A + B + X → AB + Y has $\Delta G = -15$ kJ mol^{-1} and is favourable (Figure 9.4). This is called **coupling**.

Coupling is not achieved by simply mixing the two reactions in one vessel. The two must share a common step or intermediate. Typically, X reacts first with A to form a high-energy intermediate, and this intermediate then reacts with B to give AB, releasing Y. The favourable reaction pays for the unfavourable one only through this shared chemistry.

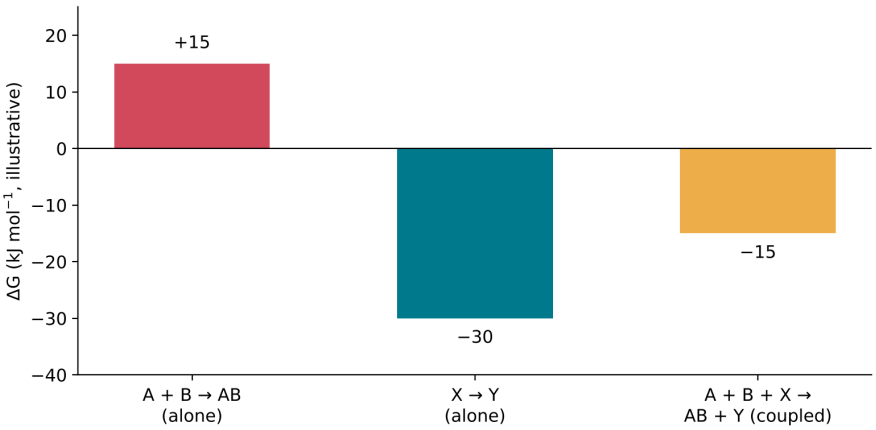


Figure 9.4.: Coupling of reactions (illustrative values). A reaction with a positive ΔG (left) can be driven by a reaction with a larger negative ΔG (centre) if the two share an intermediate, because the Gibbs energy changes add (right).

9.6.2. Activated building blocks

A molecule that carries a built-in source of Gibbs energy, ready to be released in a coupled reaction, is called **activated**. In living cells, the most common example is adenosine

triphosphate (ATP). Its chain of three phosphate groups is linked by P–O–P bonds, called phosphoanhydride bonds, whose hydrolysis is strongly favourable. Cells use the transfer of a phosphate group from ATP to drive many reactions that would otherwise be unfavourable, including the joining of building blocks into the chains of RNA and proteins. The building blocks that cells join into RNA are themselves activated in this way: each arrives carrying a chain of three phosphates, and the release of two of them pays for the new link.

On the early Earth, before cells existed, any joining of building blocks in water needed a comparable source of Gibbs energy. Possible sources include chemically activated molecules formed in the environment, the removal of water by drying, and the input of energy from sunlight, heat or chemical gradients. Which of these could have operated is one of the central questions in research on the origin of life.

9.7. Favourable is not the same as fast

The formation of liquid water from its elements,

$$\mathrm{H_2(g)} + \tfrac{1}{2}\,\mathrm{O_2(g)} \rightarrow \mathrm{H_2O(l)}$$

has $\Delta\mathrm{H}^\circ = -285.830$ kJ mol^{-1} and $\Delta\mathrm{S}^\circ = 69.95 - 130.680 - \tfrac{1}{2}(205.152) = -163.31$ J K^{-1} mol^{-1} (National Institute of Standards and Technology, n.d.-b; Cox et al. 1984). At 298.15 K,

$$\Delta G^\circ = -285.830 + 298.15 \times 0.16331 = -237.14\ \mathrm{kJ\,mol^{-1}}$$

The reaction is strongly favourable. Yet a mixture of hydrogen and oxygen can be kept at room temperature for a very long time without reacting appreciably, until a spark ignites it. Thermodynamics says whether a change *can* happen and how far it will go; it says nothing about how fast. The hydrolysis of RNA in water is another example: it is favourable, yet a chain of RNA survives in pure water for long enough to be studied, because the reaction is slow. Speed is the subject of the next chapter.

Key points

- Energy is conserved (first law). Chemical energies are given in kJ mol^{-1}; at 25 °C the thermal energy per mole, RT, is 2.479 kJ mol^{-1}.
- Enthalpy change, ΔH, is the heat exchanged at constant pressure: negative for exothermic, positive for endothermic reactions. Hess's law gives $\Delta\mathrm{H}^\circ$ of a reaction from enthalpies of formation.

- Entropy, S, measures the number of microscopic arrangements. The total entropy increases in every spontaneous change (second law).
- The Gibbs energy change, $\Delta G = \Delta H - T\Delta S$, decides direction at constant temperature and pressure: $\Delta G < 0$ spontaneous, $\Delta G > 0$ not, $\Delta G = 0$ equilibrium.
- $\Delta G^\circ = -RT \ln K$: at 25 °C, every 5.71 kJ mol^{-1} changes K tenfold. Under actual conditions $\Delta G = \Delta G^\circ + RT \ln Q$.
- Heating lowers K of an exothermic reaction and raises K of an endothermic one.
- Unfavourable reactions can be driven by coupling to favourable ones through a shared intermediate; activated molecules such as ATP serve this purpose in cells.
- A favourable reaction can be extremely slow: thermodynamics gives direction, not speed.

Self-test

1. Use the standard enthalpies of formation in this chapter to calculate ΔH° for $H_2O(g) \rightarrow H_2O(l)$. Is condensation of water vapour exothermic or endothermic?
2. A reaction has $\Delta H = +20$ kJ mol^{-1} and $\Delta S = +100$ J K^{-1} mol^{-1}. Is it spontaneous at 25 °C (298.15 K)? Above what temperature does it become spontaneous, assuming ΔH and ΔS are constant?
3. A reaction has $\Delta G^\circ = -11.4$ kJ mol^{-1} at 298.15 K. Without a calculator, estimate K. (Hint: use the 5.71 kJ mol^{-1} rule.)
4. A reaction has $K = 10$ and is started with $Q = 1000$. In which direction does it proceed? What is the sign of ΔG?
5. Explain, using the four cases of Table 9.1, why joining two building blocks into one tends to become less favourable as the temperature rises, if the joining releases heat.

10. How Fast: Reaction Rates

The question of this chapter

Chapter 9 ended with a puzzle. A mixture of hydrogen and oxygen can turn into water with a large release of Gibbs energy, yet it can be kept for a long time without reacting. Some favourable reactions are over in a fraction of a second; others take years, or millions of years. This chapter asks: *what determines how fast a reaction goes?* The answer involves the concentrations of the reactants, an energy barrier that every reaction must cross, the temperature, and catalysts that lower the barrier. Together they explain why a molecule can be thermodynamically doomed and still survive, and why the amount of a molecule that accumulates depends on a race between its formation and its destruction.

10.1. The rate of a reaction

10.1.1. Definition

The **rate of a reaction** is the change in the concentration of a reactant or product per unit time. For a reaction A $\rightarrow$ B, the rate is the decrease in [A] per second, or equivalently the increase in [B] per second. Its usual unit is mol L^{-1} s^{-1}. When the coefficients in the balanced equation are not all 1, the change in each concentration is divided by its coefficient, so that the rate has the same value whichever substance is followed (IUPAC, n.d.-m). For 2 H_2 + O_2 $\rightarrow$ 2 H_2O, for example, hydrogen is used up twice as fast as oxygen, but the rate of the reaction is one value.

The rate of most reactions changes as the reaction proceeds. It is usually greatest at the start, when the concentrations of the reactants are highest, and falls as they are used up.

10.1.2. Rate laws and rate constants

The dependence of the rate on concentration is described by a **rate law**, found by experiment. Two simple forms cover many reactions:

- In a **first-order reaction**, the rate is proportional to the concentration of one reactant: rate = k[A]. Doubling [A] doubles the rate. Reactions in which a single molecule falls apart or rearranges are often first order, and so are reactions with the solvent, such as hydrolysis in water, because the concentration of water hardly changes.
- In a **second-order reaction**, the rate is proportional to the product of two concentrations: rate = k[A][B], or k[A]^2 when two molecules of A react with each other. Doubling both [A] and [B] quadruples the rate. Reactions that require two molecules to meet are often second order.

The proportionality factor k is the **rate constant**. It does not depend on concentration, but it depends strongly on temperature and is changed by catalysts. Its unit depends on the rate law: s^{-1} for a first-order reaction, and L mol^{-1} s^{-1} for a second-order reaction, so that the rate always comes out in mol L^{-1} s^{-1}.

The **order** of a reaction is the sum of the powers of the concentrations in its rate law. It cannot, in general, be read from the balanced equation, which shows only the overall change and not the steps by which it happens. For the same reason, a rate law is measured, not deduced.

The rate law of a second-order joining reaction contains a consequence that matters for the origin of life. If [A] and [B] are both diluted tenfold, the rate of A + B → AB falls a hundredfold, while a first-order destruction of AB, such as its hydrolysis, is not slowed at all. Dilution therefore slows the joining of molecules far more than their breakdown, in addition to the effect on equilibrium described in Chapter 8.

10.2. First-order reactions and half-life

10.2.1. Exponential decay

For a first-order reaction A → products, the rate at which A disappears is proportional to how much A is left. A quantity that decreases in proportion to its own size decreases exponentially (Chapter 1):

$$[\mathrm{A}] = [\mathrm{A}]_0 \, \mathrm{e}^{-kt}$$

where $[\mathrm{A}]_0$ is the concentration at time t = 0 and e = 2.718… is the base of natural logarithms.

10.2.2. Half-life

The **half-life**, $t_{1/2}$, is the time in which the concentration of a reactant falls to half its initial value. Setting $[A] = [A]_0/2$ in the equation above and taking natural logarithms gives

$$t_{1/2} = \frac{\ln 2}{k} = \frac{0.693}{k}$$

For a first-order reaction, the half-life does not depend on the starting concentration: whatever amount is present, half of it is gone after one half-life, three-quarters after two, seven-eighths after three, and a fraction $(1/2)^n$ remains after n half-lives (Figure 10.1). After ten half-lives, less than one part in a thousand remains, since $(1/2)^{10} = 1/1024$.

Radioactive decay follows the same law, which is why half-lives are used both for radioactive nuclei and for chemical reactions. Carbon-14, for example, has a half-life of 5700 years (Chechev and Bé 2012) (Chapter 1). The difference is that radioactive decay is a property of the nucleus and is practically unaffected by temperature or chemical environment, whereas the half-life of a chemical reaction can change by many orders of magnitude with temperature, pH or the presence of a catalyst.

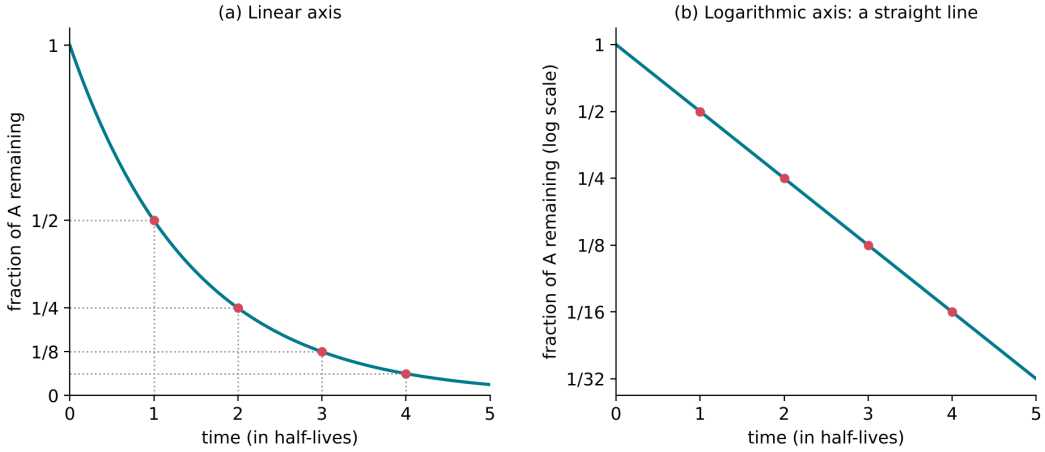


Figure 10.1.: First-order decay. (a) The fraction of A remaining halves in each half-life. (b) On a logarithmic vertical axis, the same curve is a straight line, whose slope gives the rate constant (Chapter 1).

10.3. The energy barrier

10.3.1. Collisions

For two molecules to react, they must meet. In a gas or a solution, molecules collide with one another billions of times per second, yet in a slow reaction only a tiny fraction of these collisions lead to products. Two conditions must be met. The molecules must meet in a suitable orientation, with the atoms that will form the new bond close to each other. And they must bring enough energy into the collision to begin breaking existing bonds.

10.3.2. Activation energy and the transition state

As two molecules react, their atoms pass through arrangements in which the old bonds are partly broken and the new ones partly formed. Such arrangements have a higher energy than either the reactants or the products. The arrangement of highest energy along the path is the **transition state**, and the energy that the reactants must acquire to reach it is the **activation energy**, E_a (Figure 10.2). The activation energy is a barrier between reactants and products. It is separate from the overall Gibbs energy change: a reaction may be strongly favourable (products far lower than reactants) and still have a high barrier, as the mixture of hydrogen and oxygen shows.

10.3.3. Only a few molecules have enough energy

At any temperature, the molecules of a substance do not all have the same energy. Collisions continually redistribute energy among them, so that most molecules have energies near the average, of the order of RT per mole (Chapter 9), while a few have much more (Figure 10.3). Only those with at least the activation energy can react. The fraction of molecules with an energy of at least E_a is, to a good approximation, proportional to the **Boltzmann factor**

$$e^{-E_a/RT}$$

For an activation energy of 50 kJ mol^{-1} at 25 °C (298.15 K), $E_a/RT = 20.2$, and the Boltzmann factor is 1.7×10^{-9}: fewer than two molecules in a billion have enough energy at any moment. At 35 °C the factor is 3.3×10^{-9}, nearly twice as large. A small rise in temperature greatly increases the small number of molecules in the high-energy tail of the distribution, and this is why rates are so sensitive to temperature.

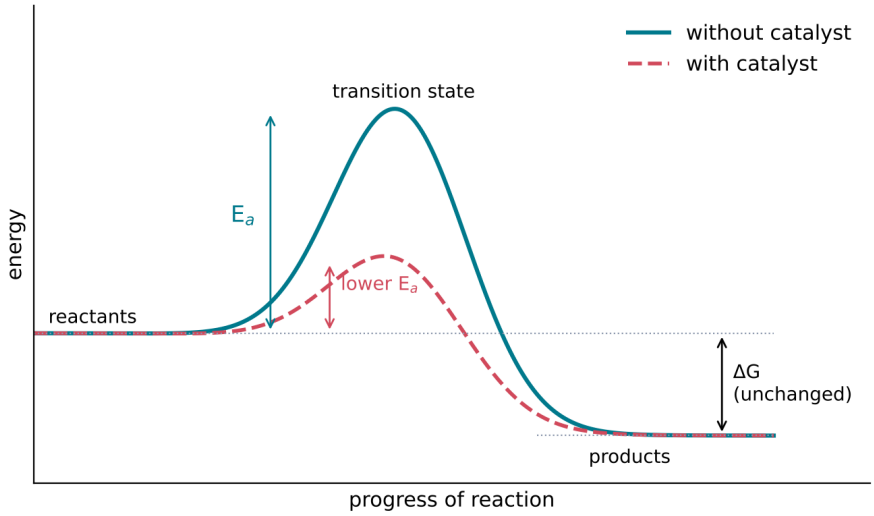


Figure 10.2.: Energy profile of a reaction (schematic). Reactants must pass over the barrier at the transition state, whose height above the reactants is the activation energy E_a. A catalyst (dashed curve) provides a different path with a lower barrier. The energy difference between reactants and products, and therefore the position of equilibrium, is the same with or without the catalyst.

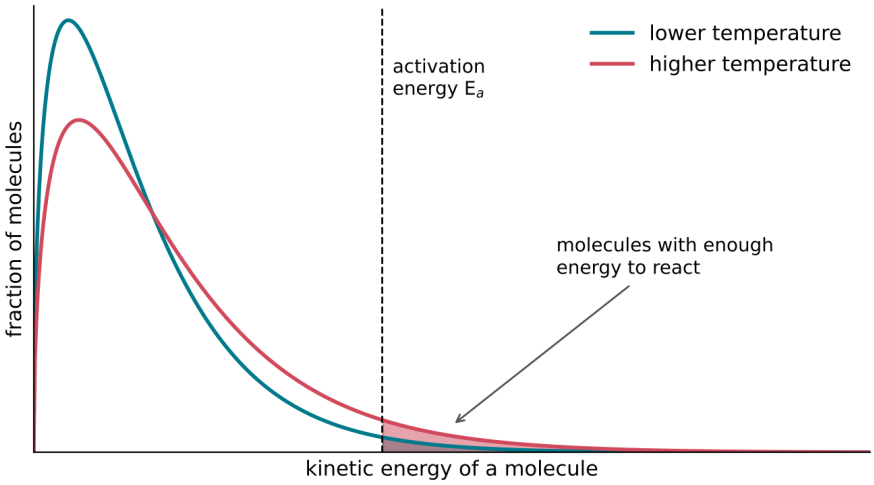


Figure 10.3.: Distribution of kinetic energies among the molecules of a gas at two temperatures (schematic). The shaded areas show the fraction of molecules with an energy greater than the activation energy E_a. At the higher temperature, the distribution is broader and this fraction is much larger, although the average energy is only somewhat higher.

10.4. Temperature and the Arrhenius equation

10.4.1. The equation

The dependence of a rate constant on temperature is described by the **Arrhenius equation**, proposed by the Swedish chemist Svante Arrhenius in 1889 (IUPAC, n.d.-d):

$$k = A\,\mathrm{e}^{-E_\mathrm{a}/RT} \tag{10.1}$$

Here A, the **pre-exponential factor**, has the same unit as k and reflects how often molecules collide in a suitable orientation; the exponential factor is the Boltzmann factor of the previous section; T is the absolute temperature; and R is the molar gas constant, 8.314 462 618 J K^{-1} mol^{-1} (National Institute of Standards and Technology, n.d.-a). E_a must be expressed in J mol^{-1}, so that E_a/RT is a pure number.

10.4.2. Comparing two temperatures

Taking the ratio of the rate constants at two temperatures T_1 and T_2, the factor A cancels:

$$\frac{k(T_2)}{k(T_1)} = \exp\left[\frac{E_\mathrm{a}}{R}\left(\frac{1}{T_1} - \frac{1}{T_2}\right)\right] \tag{10.2}$$

where exp(x) means e^x. For a reaction with E_a = 50 kJ mol^{-1}, warming from 25 °C to 35 °C increases the rate constant by a factor of 1.92, and warming to 100 °C by a factor of about 58. For E_a = 100 kJ mol^{-1}, the corresponding factors are 3.70 and about 3300 (Figure 10.4). The often-quoted rule that a rate doubles for every 10 °C rise is therefore true only for activation energies near 50 kJ mol^{-1} close to room temperature; for higher barriers the effect of temperature is larger. In the other direction, a reaction at 0 °C is slower than at 25 °C by a factor of 6.3 (for 50 kJ mol^{-1}) or 40 (for 100 kJ mol^{-1}).

10.4.3. The Arrhenius plot

Taking the natural logarithm of Equation 10.1 gives

$$\ln k = \ln A - \frac{E_\mathrm{a}}{R} \cdot \frac{1}{T}$$

A graph of ln k against 1/T is therefore a straight line with slope $-E_a/R$. This graph, the **Arrhenius plot** mentioned in Chapter 1, is the standard way of measuring an activation energy: the rate constant is measured at several temperatures, the points are plotted,

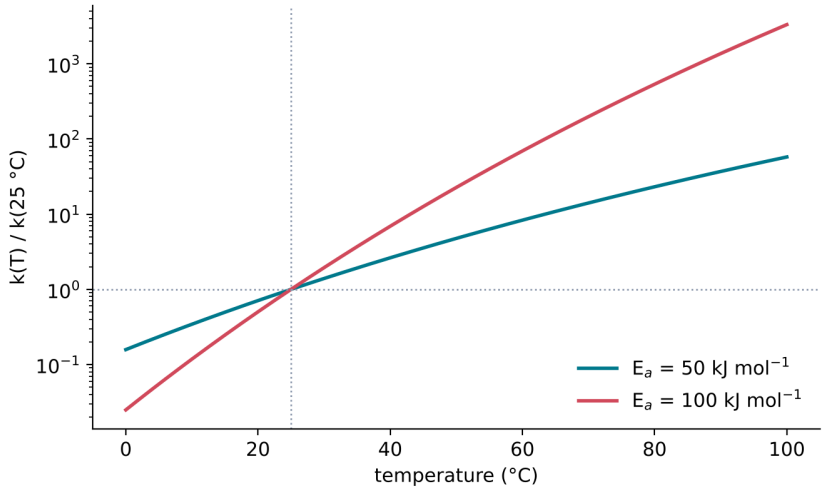


Figure 10.4.: Rate constant relative to its value at 25 °C, calculated from the Arrhenius equation for activation energies of 50 and 100 kJ mol^{-1}. The vertical axis is logarithmic. The higher the activation energy, the more strongly the rate depends on temperature.

and E_a is obtained from the slope. If A and E_a themselves change with temperature, the plot is curved; IUPAC therefore defines the activation energy more generally through the slope at each temperature (IUPAC, n.d.-h).

10.4.4. An example from the origin of life: the decomposition of ribose

Ribose, the sugar of RNA, is not stable in water. Larralde, Robertson and Miller measured how fast it decomposes in neutral solution (pH 7.0) at temperatures between 40 °C and 120 °C (Larralde et al. 1995). The half-life was 73 minutes at 100 °C, and the value extrapolated to 0 °C was 44 years. The two half-lives differ by a factor of about 3.2×10^5. Since the half-life of a first-order reaction is inversely proportional to its rate constant, Equation 10.2 gives an activation energy of about 107 kJ mol^{-1}, and, at 25 °C, a half-life of roughly 300 days. This estimate assumes that the reaction obeys the Arrhenius equation with constant E_a over the whole range. Even at 0 °C, a few decades is a short time on the scale of the early Earth: ribose formed in warm water would have had to be used, or protected, soon after it was made.

10.5. Catalysts

10.5.1. What a catalyst does

A **catalyst** is a substance that increases the rate of a reaction without changing the overall standard Gibbs energy change of the reaction (IUPAC, n.d.-e). It provides an alternative path with a lower activation energy (Figure 10.2) and is regenerated at the end of each cycle, so that a small amount of catalyst can convert a large amount of reactant.

The effect of lowering the barrier follows from the exponential in Equation 10.1. Lowering E_a by 20 kJ mol^{-1} at 25 °C, with A unchanged, multiplies the rate constant by e^20 000/(8.314 × 298.15)^, a factor of about 3200. A reaction that would take nine years would then take about a day.

10.5.2. What a catalyst does not do

Because a catalyst does not change $\Delta G°$, it does not change the equilibrium constant (Chapter 9). It speeds up the forward and reverse reactions by the same factor and only allows equilibrium to be reached sooner. A catalyst cannot make an unfavourable reaction favourable; it can only make a favourable reaction fast. A catalyst for joining building blocks in water is therefore equally a catalyst for cutting them apart.

10.5.3. Catalysts in chemistry and in life

Catalysts take many forms: metal surfaces, minerals, dissolved ions, acids and bases. The catalysts of living cells are mainly proteins called **enzymes**. Their effects can be enormous: the enzyme orotidine 5′-phosphate decarboxylase, for example, makes its reaction about 10^{17} times faster than the same reaction without a catalyst (Radzicka and Wolfenden 1995). Some are RNA molecules, called **ribozymes**. The discovery, in the early 1980s, that RNA can act as a catalyst as well as carry genetic information (Kruger et al. 1982; Guerrier-Takada et al. 1983) is one of the foundations of the idea that RNA preceded proteins in the history of life.

A reaction catalysed by one of its own products is called **autocatalytic** (IUPAC, n.d.-e). In an autocatalytic reaction, the more product there is, the faster more of it is made, and the product can increase exponentially for a while. A molecule that catalyses the formation of copies of itself is the chemical core of replication.

10.6. Competition between rates

10.6.1. Formation against destruction

On the early Earth, as in a laboratory flask, a molecule is usually being made and destroyed at the same time. Suppose that a molecule X is produced at a constant rate P (in mol L^{-1} s^{-1}) and destroyed by a first-order reaction with rate constant k, for example hydrolysis. The concentration of X then changes as

$$\text{rate of change of [X]} = P - k[\text{X}]$$

At first, when little X is present, production wins and [X] rises. As [X] rises, so does its rate of destruction, until the two are equal. From then on [X] stays constant at

$$[\text{X}]_{\text{ss}} = \frac{P}{k}$$

This state is called a **steady state** (Figure 10.5). It resembles an equilibrium in that nothing appears to change, but it is different: a steady state is maintained only as long as X is supplied, and it is not determined by ΔG° but by the ratio of two rates. If the supply stops, X disappears with the half-life of its destruction reaction. Doubling the rate constant of destruction halves the amount of X that can accumulate.

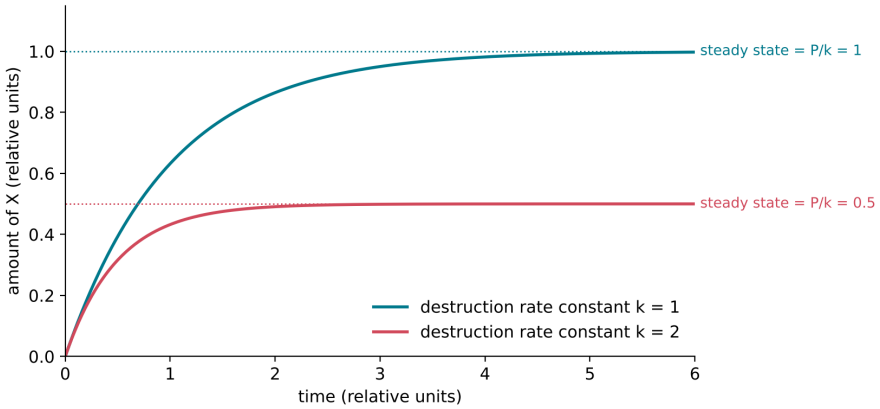


Figure 10.5.: Approach to a steady state for a molecule X produced at a constant rate P and destroyed by a first-order reaction with rate constant k (relative units). [X] levels off where production and destruction balance, at P/k. A faster destruction (red) gives a lower steady state, reached sooner.

10.6.2. Competing reactions

A molecule can often react in more than one way. If a reactant A can be converted into a product B with rate constant k_1 or into a different product C with rate constant k_2, both by first-order reactions, the two products form in the ratio of the rate constants:

$$\frac{\text{amount of B formed}}{\text{amount of C formed}} = \frac{k_1}{k_2}$$

The product that forms fastest dominates, even if another product would be more stable. A reaction whose outcome is decided by the relative rates of competing paths is said to be under **kinetic control**; one whose outcome is decided by the relative stabilities of the products, because the reactions are reversible and equilibrium is reached, is under **thermodynamic control**. Much of the difficulty of prebiotic chemistry lies here: the reactions that produce useful building blocks usually compete with many others that produce useless or harmful products, and a useful product is obtained in quantity only if its path is faster than the alternatives.

Key points

- The rate of a reaction is the change in concentration per unit time. Rate laws, such as rate = k[A] (first order) or k[A][B] (second order), are found by experiment.
- For a first-order reaction, $[A] = [A]_0 e^{-kt}$ and the half-life, $t_{1/2} = 0.693/k$, does not depend on concentration.
- Reactants must cross an energy barrier, the activation energy, at the transition state. Only a small fraction of molecules, given approximately by $e^{-E_a/RT}$, has enough energy.
- The Arrhenius equation, $k = A\ e^{-E_a/RT}$, describes the effect of temperature. For $E_a = 50$ kJ mol^{-1}, a 10 °C rise near room temperature nearly doubles k; higher barriers give larger effects.
- A catalyst lowers the activation energy, speeds up forward and reverse reactions equally, and does not change K. Enzymes and ribozymes are biological catalysts.
- The amount of a molecule that accumulates under continuous production and destruction is a steady state, P/k, set by rates rather than by equilibrium. Competing reactions give products in the ratio of their rate constants.

Self-test

1. A first-order reaction has a rate constant of 0.01 s^{-1}. What is its half-life? What fraction of the reactant remains after 5 minutes?

2. The rate of a reaction A + B $\rightarrow$ C is found to be proportional to [A][B]. By what factor does the rate change if [A] is doubled and [B] is halved? If both are reduced to one tenth?
3. A reaction has $E_a = 100$ kJ mol^{-1}. Using Figure 10.4 or Equation 10.2, by roughly what factor is it faster at 35 °C than at 25 °C? Compare with the "doubling per 10 °C" rule.
4. A catalyst increases the rate of the forward reaction of A $\rightleftharpoons$ B by a factor of 1000. By what factor does it increase the rate of the reverse reaction? What happens to K?
5. A molecule is produced at a constant rate and destroyed by hydrolysis with a half-life of 10 days. If warming the solution shortens the half-life to 1 day, without changing the rate of production, how does the steady-state concentration change?

11. The Carbon Toolkit

The question of this chapter

Millions of carbon compounds are known, and no one could learn the properties of each. Fortunately, there is no need to. The behaviour of an organic molecule is decided largely by a few small groups of atoms attached to its carbon framework, and the same group behaves in much the same way in whatever molecule it occurs. This chapter asks: *how can a small set of patterns be used to read the chemistry of a large molecule?* It introduces the functional groups found in the molecules of life, the ways in which molecules with the same formula can differ, including the left- and right-handed forms called enantiomers, and the electron-rich and electron-poor sites at which reactions begin.

11.1. Skeletons and functional groups

11.1.1. Carbon frameworks

The carbon atoms of an organic molecule form its **skeleton**: a chain, a branched chain, a ring, or a combination of these (Chapter 3). The simplest organic compounds contain only carbon and hydrogen and are called **hydrocarbons**. Three families of hydrocarbons are distinguished by their carbon-carbon bonds:

- **Alkanes** contain only single bonds. Methane, CH_4, ethane, C_2H_6, and propane, C_3H_8, are the first members.
- **Alkenes** contain at least one carbon-carbon double bond, as in ethene, $H_2C{=}CH_2$.
- **Alkynes** contain at least one carbon-carbon triple bond, as in ethyne (acetylene), $HC{\equiv}CH$.

A compound with only single bonds between its carbon atoms is called **saturated**, because each carbon carries as many hydrogen atoms as it can; a compound with double or triple bonds is **unsaturated**.

Systematic names indicate the number of carbon atoms in the main chain by a stem: meth- (1), eth- (2), prop- (3), but- (4), pent- (5), hex- (6). The ending indicates the family: -ane for alkanes, -ene for alkenes, -ol for alcohols, -al for aldehydes, and so on. Thus ethanol is a two-carbon alcohol and pentanal a five-carbon aldehyde. Many familiar compounds also keep older names, such as acetic acid and acetone.

11.1.2. Functional groups

The C–C and C–H bonds of a hydrocarbon skeleton are strong and only slightly polar (Chapter 5), and they react with few reagents. Most of the chemistry of an organic molecule happens elsewhere, at atoms or groups of atoms that contain oxygen, nitrogen, phosphorus or other elements, or at multiple bonds. A **functional group** is a group of atoms that gives a molecule characteristic properties and reactions, which it shows more or less independently of the rest of the molecule (IUPAC, n.d.-j). Organic compounds are therefore classified into families by their functional groups. In formulas, the rest of the molecule is often written as R (for "residue"), so that R–OH stands for any alcohol.

11.2. The functional groups of life

Table 11.1 lists the functional groups that occur most often in the molecules of life and in the small molecules from which they may have formed. Examples are drawn in Figure 11.1.

Table 11.1.: Common functional groups in the molecules of life and their precursors. R and R′ stand for the rest of the molecule.

Family	Functional group	General formula	Example
Alcohol	hydroxy group	R–OH	ethanol; the OH groups of sugars
Ether	oxygen between two carbons	R–O–R′	dimethyl ether; the ring oxygen of ribose
Aldehyde	carbonyl group with one H	R–CHO	formaldehyde, acetaldehyde
Ketone	carbonyl group between two carbons	R–CO–R′	acetone
Carboxylic acid	carboxy group	R–COOH	acetic acid; amino acids
Ester	acid joined to an alcohol	R–CO–O–R′	ethyl acetate; fats
Amine	amino group	R–NH_2	methylamine; amino acids; nucleobases
Amide	carbonyl bonded to nitrogen	R–CO–NH–R′	acetamide; the links of proteins
Phosphate ester	phosphate joined to an alcohol	R–O–PO_3H_2	nucleotides; the backbone of RNA

Family	Functional group	General formula	Example
Nitrile	carbon triple-bonded to nitrogen	R–C≡N	acetonitrile; hydrogen cyanide (R = H)

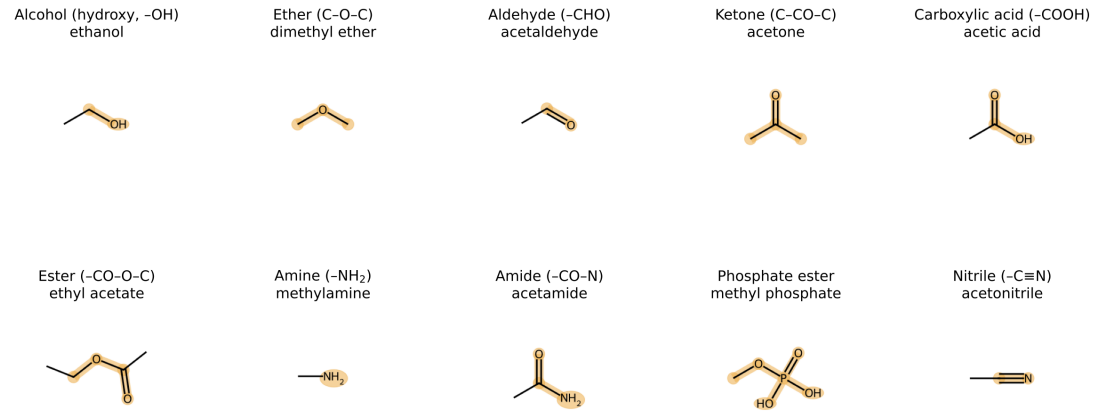


Figure 11.1.: Examples of the functional groups of Table 11.1, drawn as skeletal formulas (Chapter 4). The atoms of each functional group are highlighted.

11.2.1. Groups containing oxygen

- **Alcohols** contain a hydroxy group, –OH, attached to a carbon atom that forms only single bonds (IUPAC, n.d.-a). The polar O–H bond takes part in hydrogen bonds with water and with other hydroxy groups (Chapter 5). This is why small alcohols mix with water, and why sugars, which carry several hydroxy groups, dissolve readily in it.
- **Ethers** contain an oxygen atom bonded to two carbon atoms, C–O–C. They cannot donate hydrogen bonds, because they have no O–H, and they are much less reactive than alcohols. The oxygen atom in the ring of ribose is an ether-like oxygen.
- The **carbonyl group** is a carbon atom double-bonded to an oxygen atom, C=O. Oxygen is more electronegative than carbon, so the carbon atom carries a partial positive charge and the oxygen a partial negative charge. This polarity makes the carbonyl carbon the reactive site of many molecules (Section 11.6). In an **aldehyde**, the carbonyl carbon is bonded to at least one hydrogen atom, R–CHO (IUPAC, n.d.-b); formaldehyde, H_2CO, is the simplest. In a **ketone**, the carbonyl carbon is bonded to two carbon atoms.
- A **carboxylic acid** contains the carboxy group, –COOH, a carbonyl group and a hydroxy group on the same carbon atom. Its hydrogen can be donated as a proton

(Chapter 7). Acetic acid, for example, has $pK_a = 4.76$ (Goldberg et al. 2002), so at pH 7 it, like other carboxylic acids, is present almost entirely as the carboxylate ion, $–COO^-$ (Chapter 8).
- An **ester** is formed from an acid and an alcohol with the loss of water (Chapter 7) (IUPAC, n.d.-i). In a carboxylic ester, the carbonyl carbon is bonded to an oxygen that is in turn bonded to carbon, R–CO–O–R′.

11.2.2. Groups containing nitrogen

- An **amine** contains a nitrogen atom bonded to one or more carbon atoms; the simplest form is the amino group, $–NH_2$. The lone pair on nitrogen can accept a proton, so amines are bases, and many are present mostly in their protonated form, $–NH_3^+$, at pH 7.
- An **amide** is derived from an acid by replacing its –OH with a nitrogen group (IUPAC, n.d.-c). In a carboxylic amide the carbonyl carbon is bonded directly to nitrogen, –CO–N. Unlike an amine, the nitrogen of an amide is hardly basic, because its lone pair is drawn towards the neighbouring carbonyl group. Amide bonds link the amino acids of proteins (Chapter 12).
- A **nitrile** contains a carbon atom triple-bonded to a nitrogen atom, –C≡N. Hydrogen cyanide, HCN, can be regarded as the simplest nitrile, and it is a starting material in many proposed routes to the building blocks of life.

11.2.3. The phosphate group

Phosphoric acid, H_3PO_4 or $(HO)_3P{=}O$, is an acid with three OH groups (Chapter 8). Like a carboxylic acid, it can form esters with alcohols. When one of its OH groups is joined to an alcohol, the product is a **phosphate monoester**; when two are joined to two alcohols, it is a **phosphate diester**. A phosphate diester bridges two alcohol groups through a single phosphorus atom, R–O–P–O–R′, and this is how the units of RNA and DNA are linked into chains (Chapter 12). The remaining OH group of a phosphate diester is strongly acidic, so in water the linkage carries a negative charge.

11.2.4. Molecules with several groups

Many molecules of life carry more than one functional group, and their behaviour combines the behaviour of each. An amino acid carries an amino group and a carboxy group on the same carbon atom. At pH 7, both are charged: the amino group is protonated and the carboxy group deprotonated, so the molecule carries a positive and a negative charge at once. A sugar such as ribose carries an aldehyde group and four hydroxy groups (Figure 11.2). Recognising the groups is the first step in reading any structure.

11.3. Sugars: groups that react with each other

11.3.1. Aldoses and ketoses

Simple sugars are compounds with the general formula $C_n(H_2O)_n$, usually with n from 3 to 7, in which every carbon atom but one carries a hydroxy group and the remaining carbon is part of a carbonyl group. A sugar whose carbonyl group is an aldehyde, at the end of the chain, is an **aldose**; one whose carbonyl group is a ketone, within the chain, is a **ketose**. The name also indicates the number of carbon atoms: a triose has three, a pentose five, a hexose six. Ribose, $C_5H_{10}O_5$, is an aldopentose; glucose, $C_6H_{12}O_6$, is an aldohexose.

11.3.2. Ring formation

A hydroxy group can add to a carbonyl group (Chapter 7). In a sugar, both groups are in the same molecule, and if the chain can bend so that a hydroxy group reaches the carbonyl carbon, the addition closes a ring. In ribose, the hydroxy group on carbon 4 adds to the aldehyde carbon, carbon 1, to form a five-membered ring of four carbon atoms and one oxygen atom (Figure 11.2). The former carbonyl oxygen becomes a new hydroxy group on carbon 1. The reaction is reversible, and in water a sugar exists as an equilibrium mixture of ring forms, with only a small fraction in the open-chain form at any moment. Ribose can also form a six-membered ring, through the hydroxy group on carbon 5. In water, about three-quarters of ribose is present as six-membered rings, about one-quarter as five-membered rings, and only about 0.1 percent as the open chain (Drew et al. 1998). In RNA, however, ribose is always present as the five-membered ring, called a **furanose**.

Ring closure creates a new stereocentre (Section 11.5) at carbon 1: the new hydroxy group can point to one side of the ring or the other. The two forms are called **anomers** and are labelled α and β. In RNA, the bond from carbon 1 to the nucleobase always has the β arrangement, the one drawn in Figure 11.2.

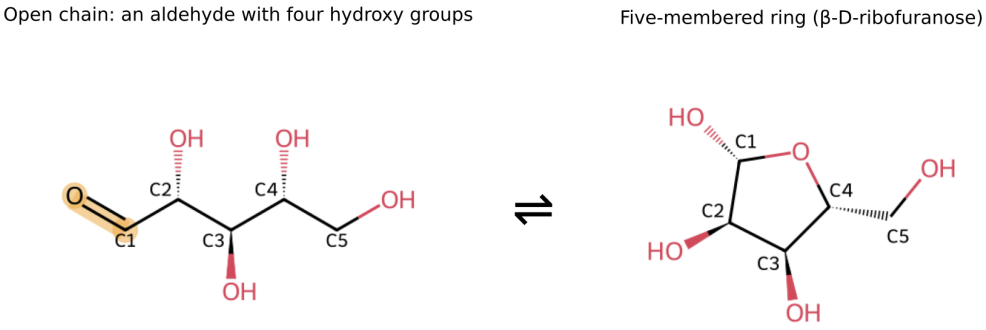


Figure 11.2.: Ribose in its open-chain form (left), an aldehyde (highlighted) with four hydroxy groups, and in its five-membered ring form (right), produced when the hydroxy group on carbon 4 adds to carbon 1. Wedged bonds point towards the viewer and hashed bonds away. The two forms interconvert in water.

11.4. Isomers

Chapter 4 introduced **isomers**: compounds that have the same molecular formula but different structures, and hence different properties (IUPAC, n.d.-k). Isomers are of two broad kinds.

- In **constitutional isomers** (also called structural isomers), the atoms are bonded in a different order. Ethanol and dimethyl ether (C_2H_6O) are one example (Chapter 4). Another is the pair glyceraldehyde and dihydroxyacetone, both $C_3H_6O_3$: in glyceraldehyde the carbonyl group is at the end of the chain, making it an aldose, and in dihydroxyacetone it is in the middle, making it a ketose. Ribose has many constitutional isomers among the pentoses, which is one reason why making ribose selectively is difficult.
- In **stereoisomers**, the atoms are bonded in the same order but arranged differently in space. Stereoisomers include the mirror-image forms described next.

11.5. Chirality

11.5.1. Left and right hands

An object is **chiral** if it cannot be superimposed on its mirror image (IUPAC, n.d.-f). A left hand and a right hand are mirror images, and no rotation will make one fit exactly onto the other. A plain sphere, by contrast, is identical to its mirror image and is **achiral**.

In molecules, chirality most often arises from a carbon atom bonded to four different groups. Because the four bonds point to the corners of a tetrahedron (Chapter 3), the four groups can be arranged in two ways that are mirror images of each other and cannot be interconverted without breaking bonds. Such a carbon atom is called a **stereocentre**. A carbon atom that carries two identical groups is not a stereocentre: its mirror image can always be turned to coincide with the original.

11.5.2. Enantiomers

Two molecules that are non-superimposable mirror images of each other are called **enantiomers**. Enantiomers have identical physical properties, such as melting point, boiling point and solubility, and identical chemical properties towards achiral reagents. They differ only in their interactions with other chiral things: they rotate the plane of polarised light in opposite directions, and they react at different rates with other chiral molecules. A mixture of equal amounts of two enantiomers is called a **racemic mixture** (or racemate). A chemical synthesis from achiral starting materials, without any chiral influence, always produces a racemic mixture.

11.5.3. D and L

Glyceraldehyde, the simplest aldose, has one stereocentre, its central carbon atom, which carries –CHO, $–CH_2OH$, –OH and –H. It therefore exists as two enantiomers, labelled D and L (Figure 11.3). The labels refer to a drawing convention called the **Fischer projection**, in which the carbon chain is drawn vertically with the carbonyl group at the top; at each stereocentre, the vertical bonds point away from the viewer and the horizontal bonds towards the viewer. In D-glyceraldehyde the OH group on the stereocentre is on the right; in L-glyceraldehyde it is on the left.

For longer sugars, the label D or L is determined by the stereocentre farthest from the carbonyl group: a sugar is D if the OH group on that carbon points to the right in the Fischer projection, as in D-glyceraldehyde. In ribose this is carbon 4.

11.5.4. How many stereoisomers?

A molecule with n stereocentres can have up to 2^n stereoisomers. The open chain of an aldopentose has three stereocentres, carbons 2, 3 and 4, and therefore $2^3 = 8$ stereoisomers: four pairs of enantiomers, named D- and L-ribose, arabinose, xylose and lyxose. Only one of the eight, D-ribose, occurs in RNA.

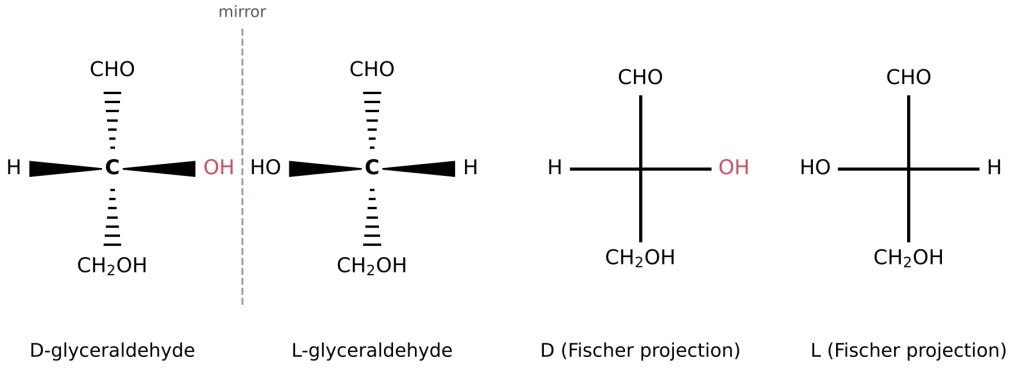


Figure 11.3.: The enantiomers of glyceraldehyde. Left: the two molecules drawn as mirror images, with horizontal bonds (wedges) pointing towards the viewer and vertical bonds (hashed) away from the viewer. Right: the same molecules in Fischer projection, in which this orientation is understood and the bonds are drawn as plain lines. In D-glyceraldehyde the OH group lies on the right.

11.5.5. One hand in life

The molecules of living organisms are not racemic. The sugars of RNA and DNA are D-sugars, and the amino acids of proteins are, apart from glycine, which has no stereocentre, L-amino acids. This uniform handedness, called **homochirality**, is essential for the regular structures of RNA and proteins, and its origin is one of the open questions of the origin of life: a chemical synthesis from achiral starting materials gives equal amounts of both forms, so some process must have favoured one.

11.6. How reactions begin: nucleophiles and electrophiles

11.6.1. Electron-rich and electron-poor sites

When a new covalent bond forms in a reaction, both of its electrons often come from one of the two partners. The partner that supplies the electron pair is the **nucleophile** ("nucleus-loving"), and the partner that accepts it is the **electrophile** ("electron-loving") (IUPAC, n.d.-g).

- **Nucleophiles** are electron-rich. They carry a lone pair or a negative charge: the oxygen of water, of a hydroxy group or of a hydroxide ion; the nitrogen of an amine or of ammonia; the carbon of a cyanide ion.

- **Electrophiles** are electron-poor. They contain an atom with a partial or full positive charge: the carbon atom of a carbonyl group, the phosphorus atom of a phosphate group, or a proton.

How readily a nucleophile reacts is called its **nucleophilicity**, and it is measured by comparing rate constants (Chapter 10) for reactions with a common partner (IUPAC, n.d.-l). A negatively charged nucleophile, such as hydroxide, is generally more reactive than the corresponding neutral one, such as water.

11.6.2. Curved arrows

Chemists describe the movement of electrons in a reaction with **curved arrows** drawn on structures. Each full arrow represents the movement of one pair of electrons. It starts at the electrons that move, a lone pair or a bond, and ends where they go, at an atom (forming a new bond or lone pair) or between two atoms (forming a new bond). The step-by-step description of a reaction drawn in this way is called its **mechanism**.

Figure 11.4 shows the first step of the addition of a nucleophile to a carbonyl group. A lone pair of the nucleophile forms a new bond to the carbonyl carbon (first arrow). At the same time, one of the two bonds between carbon and oxygen breaks, and its electron pair moves onto the oxygen atom (second arrow), which becomes negatively charged. The carbon atom now carries four single bonds and has changed from a flat to a tetrahedral arrangement. In a final step, the negatively charged oxygen picks up a proton, usually from the solvent, to become a hydroxy group. The addition of water to formaldehyde, the addition of HCN to an aldehyde (Chapter 7) and the ring closure of a sugar (Section 11.3) all follow this pattern.

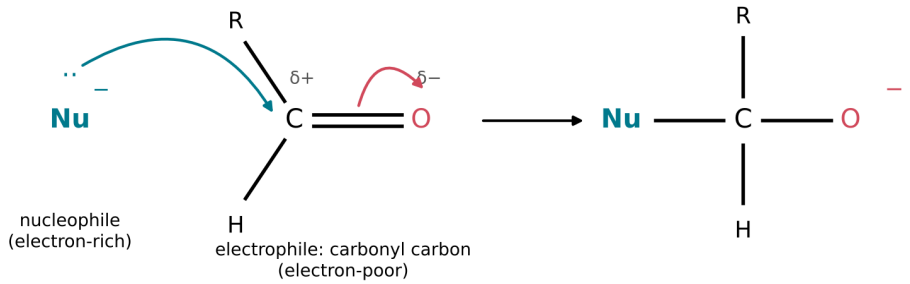


Figure 11.4.: Curved arrows for the addition of a nucleophile (Nu^-) to the carbonyl group of an aldehyde. The first arrow shows a lone pair of the nucleophile forming a bond to the electron-poor carbonyl carbon; the second shows one of the C=O bonds breaking, its electrons moving onto oxygen. A proton later adds to the negatively charged oxygen.

11.6.3. Substitution and leaving groups

In a carboxylic ester or a phosphate ester, the electrophilic atom, carbon or phosphorus, is bonded to an oxygen that can depart with its bonding electrons. When a nucleophile attacks such an atom, the attack can be followed by the departure of this group, called the **leaving group**. The overall result is that one group has replaced another, a **substitution**. Hydrolysis of an ester is a substitution in which water, or hydroxide, is the nucleophile and the alcohol part of the ester is the leaving group; the formation of an ester is the same process in reverse. The same pattern, with phosphorus as the electrophile, governs the joining of nucleotides into RNA and the cutting of RNA by water. How good a leaving group is, and how electrophilic the attacked atom is, largely determine how fast such reactions go, and therefore which bonds survive in water and which are easily formed.

Key points

- Organic molecules consist of a carbon skeleton carrying functional groups. Each functional group reacts in much the same way in whatever molecule it occurs.
- The main groups of the molecules of life are alcohols (–OH), ethers, aldehydes and ketones (C=O), carboxylic acids (–COOH), esters, amines ($–NH_2$), amides, phosphate esters and nitriles.
- At pH 7, carboxy groups are mostly negatively charged and amino groups mostly positively charged; a phosphate diester carries a negative charge.
- In a sugar, a hydroxy group can add to the carbonyl group of the same molecule to close a ring. In RNA, ribose is a five-membered ring (furanose) with the base in the β position.
- Isomers have the same formula but different structures. Constitutional isomers differ in the order of bonding; stereoisomers differ in the arrangement in space.
- A carbon atom with four different groups is a stereocentre. Non-superimposable mirror-image molecules are enantiomers. Sugars of RNA are D; amino acids of proteins are L. A molecule with n stereocentres has up to 2^n stereoisomers.
- Reactions often begin when an electron-rich nucleophile donates a pair of electrons to an electron-poor electrophile, such as a carbonyl carbon or a phosphorus atom. Curved arrows show the movement of electron pairs.

Self-test

1. Name the functional groups in the following condensed formulas: (a) CH_3CH_2CHO; (b) CH_3COCH_3; (c) CH_3CONH_2; (d) CH_3COOCH_3; (e) H_2NCH_2COOH.
2. Is the central carbon atom of propan-2-ol, $CH_3CH(OH)CH_3$, a stereocentre? Explain.

3. Glucose in its open-chain form, an aldohexose, has four stereocentres. What is the maximum number of stereoisomers? How many of them are enantiomer pairs?
4. In the reaction of hydroxide ion with the carbonyl group of formaldehyde, which species is the nucleophile and which atom is the electrophile? Describe the two curved arrows of the first step.
5. A sample of glyceraldehyde made in the laboratory from formaldehyde does not rotate polarised light. Explain why.

12. The Building Blocks of Life in Miniature

The question of this chapter

The previous chapters introduced atoms and bonds, drawings, water, equilibrium, energy, rates and functional groups. This last chapter puts these tools to work on the molecules at the centre of research into the origins of life. It asks: *how are RNA, DNA and proteins built from small parts, and what does their chemistry demand?* The answers are short descriptions of the nucleobases, the sugar, the phosphate group and the amino acids, of the bonds that join them, and of the base pairs that allow one chain of RNA or DNA to specify another. At the end, the chapter gathers the chemical questions that any account of the origin of these molecules must answer.

12.1. Polymers and monomers

The large molecules of life are **polymers**: long chains built from many smaller units, called **monomers**, joined by covalent bonds. Each monomer, once it is part of the chain, is called a **residue**. Three families of biological polymers are central:

- **Nucleic acids**, RNA and DNA, are chains of **nucleotides**.
- **Proteins** are chains of **amino acids**.
- **Polysaccharides**, such as starch and cellulose, are chains of sugars.

In all three, each new link is formed by a condensation that releases a molecule of water, and broken by hydrolysis (Chapter 7). In water, hydrolysis is generally favoured at equilibrium (Chapters 8 and 9), so in living cells the links are made with the help of activated monomers and catalysts. A short chain of a few units is called an **oligomer**: an oligonucleotide or, for amino acids, a peptide.

The order of the different kinds of monomer along a chain is its **sequence**. A polymer made from a few kinds of monomer can be arranged in an enormous number of sequences: with four kinds of nucleotide, a chain of length n can have 4^n different sequences, already more than 10^{60} for n = 100 (Chapter 1). The sequence is what makes a nucleic acid a carrier of information, and what gives each protein its particular shape and function.

12.2. Nucleobases

The **nucleobases** (or simply bases) are flat, aromatic rings (Chapter 4) of carbon and nitrogen atoms, carrying amino groups and carbonyl groups (Chapter 11). Five are common in nucleic acids (Figure 12.1):

- The **purines**, **adenine** (A) and **guanine** (G), have two fused rings, one of six and one of five atoms.
- The **pyrimidines**, **cytosine** (C), **uracil** (U) and **thymine** (T), have a single six-membered ring.

RNA contains A, G, C and U; DNA contains A, G, C and T. Thymine differs from uracil only by a methyl group ($-CH_3$) on carbon 5 of the ring.

The atoms of each ring are numbered by a fixed convention. Purines are attached to the sugar through their nitrogen atom N9, and pyrimidines through N1. The other N–H groups, the ring nitrogen atoms with lone pairs, and the C=O groups on the edges of the bases are the donors and acceptors of the hydrogen bonds (Chapter 5) that make base pairing possible (Section 12.5).

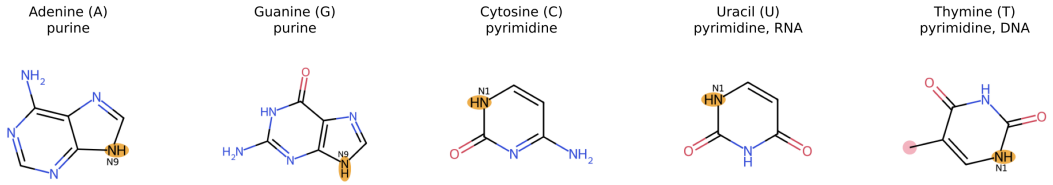


Figure 12.1.: The five common nucleobases. Adenine and guanine are purines; cytosine, uracil and thymine are pyrimidines. The nitrogen atom that bonds to the sugar in a nucleotide (N9 in purines, N1 in pyrimidines) is highlighted. Thymine, found in DNA in place of uracil, carries an extra methyl group (pink).

12.3. Nucleosides and nucleotides

12.3.1. Base plus sugar

A **nucleoside** consists of a nucleobase bonded to a sugar. In RNA the sugar is D-ribose in its five-membered ring form (Chapter 11). The bond joins carbon 1′ of the sugar to N9 of a purine or N1 of a pyrimidine, and it is formed, at least formally, by a condensation between the base and the hydroxy group on carbon 1′ of the sugar. As noted in Chapter 4, the carbon atoms of the sugar are numbered with primes (1′ to 5′) to distinguish them from the atoms of the base. In all the nucleosides of RNA, the base is attached

in the β arrangement. The four nucleosides of RNA are adenosine, guanosine, cytidine and uridine.

12.3.2. Adding phosphate

A **nucleotide** is a nucleoside carrying one or more phosphate groups, usually on the hydroxy group of carbon 5′, attached as a phosphate ester (Chapter 11). Adenosine with one phosphate is **adenosine monophosphate** (AMP, Figure 12.2); with a chain of two or three phosphates linked by P–O–P bonds it is adenosine diphosphate (ADP) or adenosine triphosphate (ATP), the activated molecule of Chapter 9. Every nucleotide therefore has three parts: a nucleobase, a sugar and a phosphate group, joined by two bonds that are both formed by condensation.

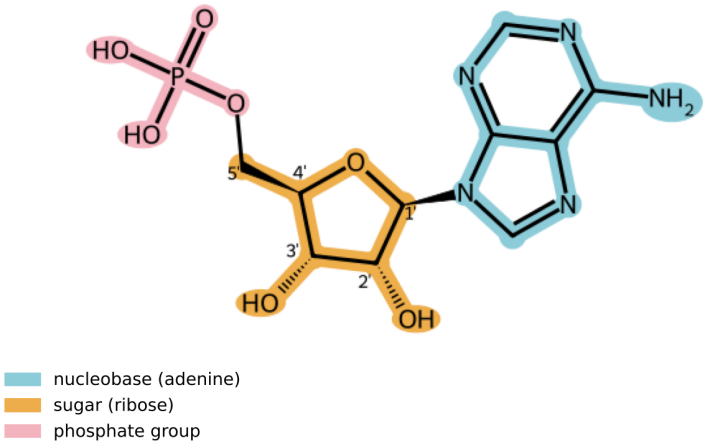


Figure 12.2.: Adenosine monophosphate (AMP), a nucleotide, with its three parts highlighted: the nucleobase adenine (blue), the sugar ribose (orange), with its carbon atoms numbered 1′ to 5′, and a phosphate group (pink) on the 5′ oxygen. The base is bonded to carbon 1′ through N9. In water at pH 7, the phosphate group carries negative charge (Chapter 8); it is drawn here in its uncharged form.

12.4. Chains of nucleotides

12.4.1. The phosphodiester backbone

In RNA, nucleotides are joined by **phosphodiester linkages**: a single phosphate group forms an ester with the 3′ hydroxy group of one nucleotide and another with the 5′ hydroxy group of the next, giving the sequence of bonds 3′-O–P–O-5′ (Figure 12.3). Each new link is a condensation between the phosphate of one unit and a hydroxy group of the other. Repeated, these links form a **backbone** of alternating sugars and phosphates, from which the bases project to the side.

Each phosphodiester linkage keeps one acidic O–H group, which at neutral pH has lost its proton (Chapter 8). The backbone therefore carries one negative charge per nucleotide. This charge keeps RNA dissolved in water, and it repels negatively charged nucleophiles such as the hydroxide ion, which helps to make the phosphodiester linkage resistant to hydrolysis (Westheimer 1987).

12.4.2. Direction

A chain of nucleotides has two different ends. At one end, the 5′ position of the first sugar carries a free phosphate or hydroxy group (the **5′ end**); at the other, the 3′ position of the last sugar carries a free hydroxy group (the **3′ end**). By convention, sequences are written from the 5′ end to the 3′ end. The sequence 5′-ACG-3′ and the sequence 5′-GCA-3′ are therefore different molecules, even though they contain the same nucleotides.

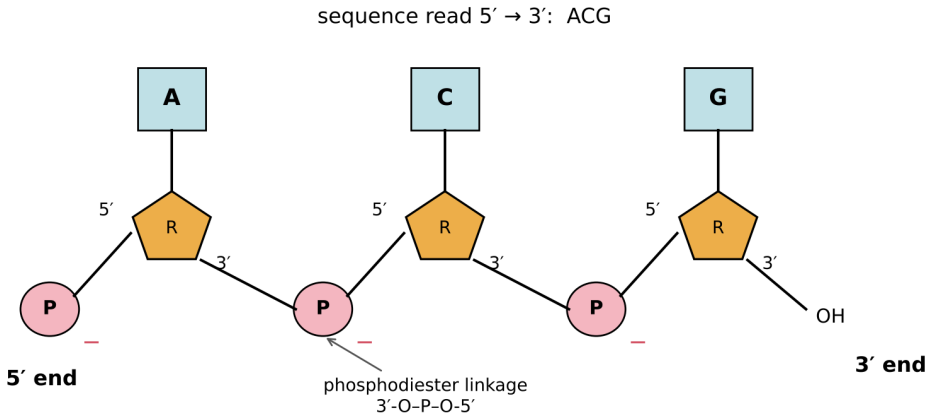


Figure 12.3.: Schematic of a short RNA chain with the sequence 5′-ACG-3′. Each ribose (R, orange pentagon) carries a base (blue square) on carbon 1′. Phosphate groups (P, pink), each carrying a negative charge, link the 3′ carbon of one sugar to the 5′ carbon of the next. The chain has a 5′ end and a 3′ end.

12.5. Base pairing

12.5.1. Complementary pairs

The bases on two chains of nucleotides can face each other and form hydrogen bonds between their edges. In 1953, in their model of the structure of DNA, James Watson and Francis Crick proposed that adenine pairs with thymine and guanine with cytosine (Watson and Crick 1953). In RNA, uracil takes the place of thymine. These **Watson–Crick base pairs** are:

- **A–U** (or A–T in DNA), held by two hydrogen bonds;
- **G–C**, held by three hydrogen bonds.

In each pair, a purine faces a pyrimidine, so that both kinds of pair have nearly the same width, and a double helix built from any sequence of them has a regular shape. Figure 12.4 shows which atoms form the hydrogen bonds.

12.5.2. Complementary and antiparallel strands

Two sequences that can form Watson–Crick pairs along their whole length are called **complementary**. Because of the geometry of the backbone, two paired strands run in opposite directions, one from 5′ to 3′ and the other from 3′ to 5′; they are **antiparallel**. The complement of 5′-GACUUCG-3′, read in its own 5′ to 3′ direction, is therefore 5′-CGAAGUC-3′.

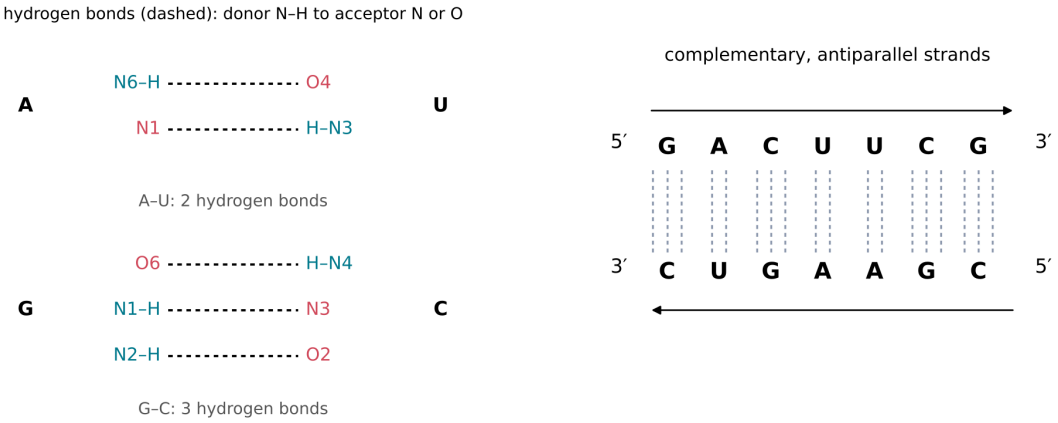


Figure 12.4.: Watson–Crick base pairs. Left: the hydrogen bonds (dashed) between adenine and uracil (two) and between guanine and cytosine (three), with the donor N–H groups and the acceptor N and O atoms labelled by their ring positions. Right: two complementary strands; every A faces a U and every G faces a C, and the two strands run in opposite directions.

Complementarity is the chemical basis of heredity. If the sequence of one strand is known, the sequence of its partner is fixed. A strand can therefore serve as a **template**: nucleotides line up along it according to the pairing rules and, once joined together, form a complementary copy. Copying the copy regenerates the original sequence. In cells, this process is carried out by enzymes; whether, and how fast, it can happen without enzymes is one of the central questions of research into the origin of life.

The same pairing also occurs within a single strand. A chain of RNA folds back on itself wherever two of its segments are complementary, forming short double-stranded regions separated by loops. In this way a single strand of RNA can adopt a specific three-dimensional shape, determined by its sequence.

12.6. RNA and DNA

RNA and DNA are built on the same plan but differ in three respects:

Table 12.1.: The chemical differences between RNA and DNA.

	RNA	DNA
Sugar	ribose, with an OH group on carbon 2′	2′-deoxyribose, with H in place of the 2′ OH
Bases	A, G, C, U	A, G, C, T

	RNA	DNA
Usual form in cells	mostly single strands, often folded	double helix of two complementary strands

The 2′ hydroxy group has important chemical consequences. It lies next to the phosphodiester linkage, and under basic conditions its oxygen can act as a nucleophile (Chapter 11) and attack the neighbouring phosphorus atom, cutting the chain (Li and Breaker 1999). DNA, which lacks this group, is much more resistant to cleavage, which suits its role as a long-term store of genetic information. The 2′ hydroxy group also gives RNA a greater variety of shapes and chemical abilities. Some RNA molecules act as catalysts, the ribozymes of Chapter 10 (Kruger et al. 1982; Guerrier-Takada et al. 1983). Because RNA can both store information in its sequence and catalyse reactions, many researchers consider that an "RNA world", in which RNA performed both roles, preceded the present system of DNA and proteins (Gilbert 1986).

12.7. Amino acids and proteins

12.7.1. Amino acids

An **amino acid**, in the sense used in biology, is a molecule in which an amino group and a carboxy group are attached to the same carbon atom, called the α **carbon**, which also carries a hydrogen atom and a variable group called the **side chain**. Twenty kinds of amino acid are commonly found in proteins. They differ only in their side chains, which range from a single hydrogen atom in glycine and a methyl group in alanine to larger groups that are polar, charged or hydrophobic (Chapter 5). As described in Chapter 11, at pH 7 the amino group of a free amino acid is protonated and the carboxy group deprotonated. Except for glycine, whose α carbon carries two hydrogen atoms, the α carbon is a stereocentre, and the amino acids of proteins are the L enantiomers (Chapter 11).

12.7.2. Peptide bonds and proteins

Two amino acids join by a condensation between the carboxy group of one and the amino group of the other, releasing water and forming an amide (Chapter 11). In this context the amide bond is called a **peptide bond** (Figure 12.5). A chain of amino acids joined in this way is a peptide; a long chain folded into a specific shape is a **protein**. There is no sharp boundary between the two, and chains of more than a few dozen residues are usually called proteins. Like nucleic acids, peptide chains have a direction, from the end with a free amino group (the N-terminus) to the end with a free carboxy group (the C-terminus).

The function of a protein depends on its shape, and its shape on its sequence of side chains, which pack together, with hydrophobic side chains generally buried inside and polar ones on the surface. Enzymes, the catalysts of cells (Chapter 10), are proteins whose shapes bring reacting molecules together and stabilise their transition states.

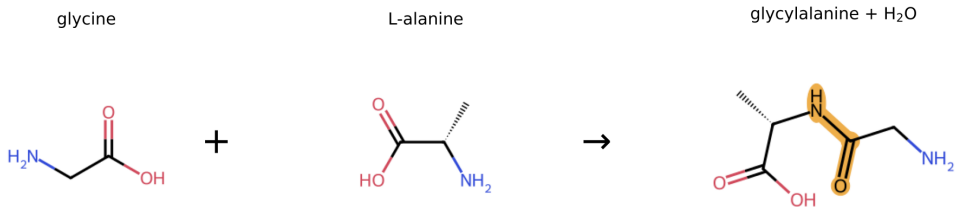


Figure 12.5.: Formation of a peptide bond. The carboxy group of glycine and the amino group of L-alanine condense, releasing water, to give the dipeptide glycylalanine. The new amide (peptide) bond is highlighted.

12.8. The chemistry of the origin of life, in outline

The descriptions in this chapter allow the chemical problems of the origin of life to be stated precisely. Each draws on earlier chapters.

- **Making the building blocks.** Nucleobases, ribose and amino acids must form from simple molecules such as hydrogen cyanide, formaldehyde, ammonia and water, through additions and other reactions (Chapters 7 and 11), with suitable sources of energy (Chapter 9).
- **Selectivity.** Ribose is one of many sugars and one of eight stereoisomeric aldopentoses; only D-ribose, in the β-furanose form, is found in RNA (Chapter 11). Reactions that make building blocks usually make many other products too, in amounts set by competing rates (Chapter 10).
- **Stability.** Many building blocks decompose in water: ribose, for example, has a half-life of hours at 100 °C and of the order of a year near room temperature at neutral pH (Chapter 10). They must be used or protected before they are lost.
- **Joining in water.** Linking nucleotides into chains, and amino acids into peptides, are condensations, opposed by the large excess of water and by dilution (Chapters 8 and 9). They require removal of water, activated monomers or another supply of Gibbs energy.
- **Copying.** For sequences to be inherited, chains must serve as templates for complementary copies (Section 12.5), fast and accurately enough to outrun hydrolysis and copying errors.

These are the questions that research into the origins of life seeks to answer, with experiments in the laboratory and with models of the early Earth. Each of them is a question of chemistry, and each can be read with the tools introduced in this book.

Key points

- Nucleic acids, proteins and polysaccharides are polymers formed by condensation of monomers and broken by hydrolysis. The order of monomers is the sequence.
- The nucleobases are the purines adenine and guanine and the pyrimidines cytosine, uracil (RNA) and thymine (DNA).
- A nucleoside is a base bonded through N9 (purines) or N1 (pyrimidines) to carbon 1′ of a sugar; a nucleotide is a nucleoside with a phosphate group, usually on carbon 5′.
- In RNA, phosphodiester linkages join the 3′ position of one nucleotide to the 5′ position of the next. The backbone carries one negative charge per nucleotide. Sequences are written from 5′ to 3′.
- Watson–Crick pairs are A–U (A–T in DNA), with two hydrogen bonds, and G–C, with three. Complementary strands are antiparallel; one strand can act as a template for the other.
- RNA has ribose and uracil; DNA has 2′-deoxyribose and thymine. The 2′ hydroxy group makes RNA more easily cleaved and chemically more versatile.
- Amino acids carry an amino group, a carboxy group and a side chain on one carbon; those of proteins are L. Peptide bonds are amides formed by condensation.
- The chemistry of the origin of life involves making the building blocks, selecting the right ones, keeping them intact, joining them in water and copying their sequences.

Self-test

1. Name the three parts of a nucleotide and the two bonds that join them. Which of these bonds is formed by a condensation?
2. Write the sequence of the strand complementary to 5′-AUGGCA-3′, in the 5′ to 3′ direction. How many hydrogen bonds hold the resulting double helix together (counting only Watson–Crick pairs)?
3. How many different RNA sequences of length 10 are possible? Express the answer as a power of 4 and as a power of 10.
4. Why is DNA chemically more resistant to cleavage of its backbone than RNA?
5. A tripeptide is made from glycine, alanine and glycine. How many water molecules are released when it forms, and how many peptide bonds does it contain?

References

Allred, A. L. 1961. "Electronegativity Values from Thermochemical Data." *Journal of Inorganic and Nuclear Chemistry* 17 (3–4): 215–21. https://doi.org/10.1016/0022-1902(61)80142-5.

Arunan, E., G. R. Desiraju, R. A. Klein, et al. 2011. "Definition of the Hydrogen Bond (IUPAC Recommendations 2011)." *Pure and Applied Chemistry* 83 (8): 1637–41. https://doi.org/10.1351/PAC-REC-10-01-02.

Bureau International des Poids et Mesures. 2019. *The International System of Units (SI)*. SI Brochure, 9th edition (2019), updated 2025; https://www.bipm.org/en/measurement-units/si-defining-constants.

Bureau International des Poids et Mesures. n.d. *History of the SI: The Mole*. Https://www.bipm.org/en/history-si/mole.

Chase, Jr., M. W. 1998. "NIST-JANAF Thermochemical Tables, Fourth Edition." *Journal of Physical and Chemical Reference Data, Monograph 9*.

Chechev, V. P., and M.-M. Bé. 2012. *Carbon-14: Table of Radionuclide Data*. Laboratoire National Henri Becquerel (LNHB/CEA), http://www.lnhb.fr/nuclides/C-14_tables.pdf.

CIAAW. n.d. *Abridged Standard Atomic Weights 2024*. Commission on Isotopic Abundances and Atomic Weights. https://www.ciaaw.org/abridged-atomic-weights.htm.

Cox, J. D., D. D. Wagman, and V. A. Medvedev. 1984. *CODATA Key Values for Thermodynamics*. Hemisphere Publishing Corp.

Drew, Kenneth N., Jaroslav Zajicek, Gail Bondo, Bidisha Bose, and Anthony S. Serianni. 1998. "13C-Labeled Aldopentoses: Detection and Quantitation of Cyclic and Acyclic Forms by Heteronuclear 1D and 2D NMR Spectroscopy." *Carbohydrate Research* 307: 199–209. https://doi.org/10.1016/S0008-6215(98)00040-4.

Encyclopaedia Britannica. n.d. *Atom*. https://www.britannica.com/science/atom.

Feistel, R., and W. Wagner. 2006. "A New Equation of State for H_2O Ice Ih." *Journal*

of Physical and Chemical Reference Data 35: 1021–47.

Gilbert, W. 1986. "Origin of Life: The RNA World." *Nature* 319 (6055): 618. https://doi.org/10.1038/319618a0.

Goldberg, R. N., N. Kishore, and R. M. Lennen. 2002. "Thermodynamic Quantities for the Ionization Reactions of Buffers." *Journal of Physical and Chemical Reference Data* 31: 231–370.

Grossman, N., E. Z. Ron, and C. L. Woldringh. 1982. *Cell Length of Escherichia Coli b, 1.94 to 2.72 Micrometres*. J. Bacteriol. 152, 35–41 (1982); compiled in BioNumbers, BNID 100001, https://bionumbers.hms.harvard.edu/bionumber.aspx?id=100001.

Guerrier-Takada, C., K. Gardiner, T. Marsh, N. Pace, and S. Altman. 1983. "The RNA Moiety of Ribonuclease P Is the Catalytic Subunit of the Enzyme." *Cell* 35 (3 Pt 2): 849–57. https://doi.org/10.1016/0092-8674(83)90117-4.

Heldal, M., S. Norland, and O. Tumyr. 1985. "X-Ray Microanalytic Method for Measurement of Dry Matter and Elemental Content of Individual Bacteria." *Applied and Environmental Microbiology* 50 (5): 1251–57.

International Association for the Properties of Water and Steam. 2011. *Revised Supplementary Release on Properties of Liquid Water at 0.1 MPa, IAPWS SR6-08(2011)*. https://iapws.org/technical-guidance/release/LiquidWater.

International Association for the Properties of Water and Steam. 2024. *Revised Release on the Ionization Constant of H_2O, IAPWS R11-24*. https://iapws.org/technical-guidance/release/Ionization.

International Union of Pure and Applied Chemistry. n.d.-a. *Haworth Representation*. Compendium of Chemical Terminology (the Gold Book), https://goldbook.iupac.org/terms/view/H02749.

International Union of Pure and Applied Chemistry. n.d.-b. *Periodic Table of Elements*. https://iupac.org/what-we-do/periodic-table-of-elements/.

IUPAC. n.d.-a. *Alcohols. In: Glossary of Class Names of Organic Compounds and Reactivity Intermediates Based on Structure (IUPAC Recommendations 1995), Pure Appl. Chem. 67, 1307 (1995), p. 1312*. IUPAC Compendium of Chemical Terminology (Gold Book). https://goldbook.iupac.org/terms/view/A00204.

IUPAC. n.d.-b. *Aldehydes. In: Glossary of Class Names of Organic Compounds and Reactivity Intermediates Based on Structure (IUPAC Recommendations 1995), Pure Appl. Chem. 67, 1307 (1995), p. 1312*. IUPAC Compendium of Chemical Terminol-

ogy (Gold Book). https://goldbook.iupac.org/terms/view/A00208.

IUPAC. n.d.-c. *Amides*. IUPAC Compendium of Chemical Terminology (Gold Book). https://goldbook.iupac.org/terms/view/A00266.

IUPAC. n.d.-d. *Arrhenius Equation. In: Compendium of Chemical Terminology (the Gold Book); Sources Include a Glossary of Terms Used in Chemical Kinetics, Including Reaction Dynamics (IUPAC Recommendations 1996), Pure Appl. Chem. 68, 149 (1996)*. https://goldbook.iupac.org/terms/view/A00446.

IUPAC. n.d.-e. *Catalyst. In: Compendium of Chemical Terminology (the Gold Book); Source: A Glossary of Terms Used in Chemical Kinetics, Including Reaction Dynamics (IUPAC Recommendations 1996), Pure Appl. Chem. 68, 149 (1996)*. https://goldbook.iupac.org/terms/view/C00876.

IUPAC. n.d.-f. *Chirality. In: Basic Terminology of Stereochemistry (IUPAC Recommendations 1996), Pure Appl. Chem. 68, 2193 (1996), p. 2203*. IUPAC Compendium of Chemical Terminology (Gold Book). https://goldbook.iupac.org/terms/view/C01058.

IUPAC. n.d.-g. *Electrophile, Electrophilic. In: Glossary of Terms Used in Physical Organic Chemistry (IUPAC Recommendations 1994), Pure Appl. Chem. 66, 1077 (1994), p. 1111*. IUPAC Compendium of Chemical Terminology (Gold Book). https://goldbook.iupac.org/terms/view/E02020.

IUPAC. n.d.-h. *Energy of Activation. In: Compendium of Chemical Terminology (the Gold Book); Source: Green Book, 2nd Ed., p. 55*. https://goldbook.iupac.org/terms/view/A00102.

IUPAC. n.d.-i. *Esters*. IUPAC Compendium of Chemical Terminology (Gold Book). https://goldbook.iupac.org/terms/view/E02219.

IUPAC. n.d.-j. *Functional Group*. IUPAC Compendium of Chemical Terminology (Gold Book). https://goldbook.iupac.org/terms/view/F02555.

IUPAC. n.d.-k. *Isomer. In: Glossary of Terms Used in Physical Organic Chemistry (IUPAC Recommendations 1994), Pure Appl. Chem. 66, 1077 (1994), p. 1129*. IUPAC Compendium of Chemical Terminology (Gold Book). https://goldbook.iupac.org/terms/view/I03289.

IUPAC. n.d.-l. *Nucleophile, Nucleophilicity. In: Glossary of Terms Used in Physical Organic Chemistry (IUPAC Recommendations 1994), Pure Appl. Chem. 66, 1077 (1994), p. 1146*. IUPAC Compendium of Chemical Terminology (Gold Book). https://goldbook.iupac.org/terms/view/N04251.

IUPAC. n.d.-m. *Rate of Reaction. In: Compendium of Chemical Terminology (the Gold Book)*. https://goldbook.iupac.org/terms/view/R05156.

IUPAC Commission on Isotopic Abundances and Atomic Weights. n.d.-a. *Atomic Weight of Carbon*. https://www.ciaaw.org/carbon.htm.

IUPAC Commission on Isotopic Abundances and Atomic Weights. n.d.-b. *Atomic Weight of Hydrogen*. https://www.ciaaw.org/hydrogen.htm.

Kekulé, A. 1865. "Sur La Constitution Des Substances Aromatiques." *Bulletin de La Société Chimique de Paris* 3: 98–110.

Kelley, D. S., J. A. Karson, G. L. Früh-Green, et al. 2005. "A Serpentinite-Hosted Ecosystem: The Lost City Hydrothermal Field." *Science* 307 (5714): 1428–34. https://doi.org/10.1126/science.1102556.

Kramida, A., Yu. Ralchenko, J. Reader, and NIST ASD Team. 2024. *NIST Atomic Spectra Database (Ver. 5.12): Ionization Energies*. National Institute of Standards; Technology, Gaithersburg, MD. https://doi.org/10.18434/T4W30F.

Kruger, K., P. J. Grabowski, A. J. Zaug, J. Sands, D. E. Gottschling, and T. R. Cech. 1982. "Self-Splicing RNA: Autoexcision and Autocyclization of the Ribosomal RNA Intervening Sequence of Tetrahymena." *Cell* 31 (1): 147–57. https://doi.org/10.1016/0092-8674(82)90414-7.

Larralde, Rosa, Michael P. Robertson, and Stanley L. Miller. 1995. "Rates of Decomposition of Ribose and Other Sugars: Implications for Chemical Evolution." *Proceedings of the National Academy of Sciences of the USA* 92 (18): 8158–60. https://doi.org/10.1073/pnas.92.18.8158.

Li, Y., and R. R. Breaker. 1999. "Kinetics of RNA Degradation by Specific Base Catalysis of Transesterification Involving the 2′-Hydroxyl Group." *Journal of the American Chemical Society* 121: 5364–72. https://doi.org/10.1021/ja990592p.

Millero, F. J., R. Feistel, D. G. Wright, and T. J. McDougall. 2008. "The Composition of Standard Seawater and the Definition of the Reference-Composition Salinity Scale." *Deep-Sea Research I* 55: 50–72. https://doi.org/10.1016/j.dsr.2007.10.001.

NASA Space Science Data Coordinated Archive. 2024. *Earth Fact Sheet*. Https://nssdc.gsfc.nasa.gov/planetary/factsheet/earthfact.html.

National Institute of Standards and Technology. 2022. *NIST Computational Chemistry Comparison and Benchmark Database (CCCBDB), Release 22, Experimental Geometry Data*. https://cccbdb.nist.gov/.

National Institute of Standards and Technology. n.d.-a. *Fundamental Physical Constants: 2022 CODATA Adjustment*. https://physics.nist.gov/cuu/Constants/Table/allascii.txt.

National Institute of Standards and Technology. n.d.-b. *NIST Chemistry WebBook: Phase Change Data*. https://webbook.nist.gov/chemistry/.

Neidhardt, F. C., J. L. Ingraham, and M. Schaechter. n.d. *Physiology of the Bacterial Cell: A Molecular Approach*. P. 3; values as compiled in BioNumbers, BNID 100650-100653. https://bionumbers.hms.harvard.edu/bionumber.aspx?id=100653.

Pauling, L. 1932. "The Nature of the Chemical Bond. IV. The Energy of Single Bonds and the Relative Electronegativity of Atoms." *Journal of the American Chemical Society* 54 (9): 3570–82. https://doi.org/10.1021/ja01348a011.

Radzicka, Anna, and Richard Wolfenden. 1995. "A Proficient Enzyme." *Science* 267: 90–93. https://doi.org/10.1126/science.7809611.

Thompson, Ambler, and Barry N. Taylor. 2008. *Guide for the Use of the International System of Units (SI)*. NIST Special Publication 811. National Institute of Standards; Technology. https://www.nist.gov/pml/special-publication-811.

Wagner, W., and A. Pruß. 2002. "The IAPWS Formulation 1995 for the Thermodynamic Properties of Ordinary Water Substance for General and Scientific Use." *Journal of Physical and Chemical Reference Data* 31: 387–535.

Watson, J. D., and F. H. C. Crick. 1953. "Molecular Structure of Nucleic Acids: A Structure for Deoxyribose Nucleic Acid." *Nature* 171 (4356): 737–38. https://doi.org/10.1038/171737a0.

Westheimer, F. H. 1987. "Why Nature Chose Phosphates." *Science* 235 (4793): 1173–78. https://doi.org/10.1126/science.2434996.

Wikipedia contributors. n.d.-a. *Conservation of Mass*. Wikipedia, https://en.wikipedia.org/wiki/Conservation_of_mass.

Wikipedia contributors. n.d.-b. *Phosphoric Acid*. Wikipedia, https://en.wikipedia.org/wiki/Phosphoric_acid.

Wöhler, F. 1828. "Ueber künstliche Bildung Des Harnstoffs." *Annalen Der Physik Und Chemie* 88 (2): 253–56.

A. Glossary

This glossary lists the technical terms defined in the book, in alphabetical order. Each entry gives the term, its definition, and the chapter in which it is first defined.

Table A.1.: Glossary of terms.

Term	Definition	Chapter
Acid	A substance that donates a proton (H^+) to another substance.	7
Acid dissociation constant (K_a)	The equilibrium constant for the reaction of a weak acid with water, $K_a = [H_3O^+][A^-]/[HA]$; usually given as $pK_a = -\log_{10} K_a$.	8
Acidic	Describing a solution with a pH below that of neutral water.	8
Activated	Of a molecule: carrying a built-in source of Gibbs energy that can drive a coupled reaction.	9
Activation energy	E_a; the energy that reactants must acquire to reach the transition state.	10
Activity	The effective concentration of a dissolved species, corrected for interactions between particles, expressed relative to a standard concentration of 1 mol L^{-1}.	8
Addition reaction	A reaction in which two molecules combine to form a single product without the loss of any atoms.	7
Alcohol	A compound with a hydroxy group, –OH, on a carbon atom that forms only single bonds.	11
Aldehyde	A compound in which a carbonyl carbon is bonded to at least one hydrogen atom, R–CHO.	11
Aldose	A sugar whose carbonyl group is an aldehyde.	11
Alkane	A hydrocarbon with only single bonds between carbon atoms.	11
Alkene	A hydrocarbon with at least one carbon-carbon double bond.	11
Alkyne	A hydrocarbon with at least one carbon-carbon triple bond.	11

Term	Definition	Chapter
Amide	A compound derived from an acid by replacing an –OH with a nitrogen group; in a carboxylic amide, C=O is bonded to N.	11
Amine	A compound with a nitrogen atom bonded to one or more carbon atoms and carrying a lone pair; a base.	11
Amino acid	In biology, a molecule with an amino group, a carboxy group and a side chain on the same carbon atom.	12
Amount concentration	The amount of solute divided by the volume of the solution, usually in mol L^{-1} (M); often simply called the concentration.	6
Amount of substance (n)	The quantity that counts the number of specified entities in a sample, measured in moles.	6
Amphiphilic	Having a hydrophilic part and a hydrophobic part, like soap; such molecules form layers and membranes in water.	5
Anion	A negatively charged ion, formed when an atom or group of atoms gains electrons.	2
Anomers	The two forms, α and β, that differ in the arrangement at the carbon atom created when a sugar closes its ring.	11
Antiparallel	Of two paired strands: running in opposite 5′-to-3′ directions.	12
Aqueous solution	A solution in which water is the solvent.	5
Aromatic	Describing a flat ring, such as benzene, in which the electrons of the double bonds are spread evenly around the ring, giving unusual stability.	4
Arrhenius equation	$k = A\, e^{-E_a/RT}$; the dependence of a rate constant on temperature.	10
Arrhenius plot	A graph of ln k against 1/T, a straight line of slope $-E_a/R$.	10
Atom	The smallest unit of a chemical element, consisting of a nucleus surrounded by electrons.	2
Atomic number (Z)	The number of protons in the nucleus of an atom; it defines the element.	2
Atomic weight	The average mass of the atoms of an element in a natural sample, relative to one-twelfth of the mass of a carbon-12 atom; also called relative atomic mass. It has no unit.	2
ATP	Adenosine triphosphate, the main activated molecule used by cells to drive unfavourable reactions.	9
Autocatalysis	Catalysis of a reaction by one of its own products.	10

Term	Definition	Chapter
Avogadro constant (N_A)	The number of entities per mole, exactly 6.022 140 76 $\times 10^{23}$ mol^{-1}.	6
Backbone	The repeating chain of sugars and phosphates (in nucleic acids) or of amide-linked units (in proteins).	12
Balanced equation	A chemical equation with the same number of atoms of each element, and the same total charge, on both sides.	7
Ball-and-stick model	A model of a molecule in which atoms are spheres and bonds are rods, showing bond angles and three-dimensional arrangement.	4
Bar	A unit of pressure equal to 10^5 Pa, close to atmospheric pressure at sea level.	6
Base	A substance that accepts a proton (H^+) from another substance.	7
Base (of a power)	The number that is multiplied by itself in a power; in 10^3 the base is 10.	1
Basic	Describing a solution with a pH above that of neutral water; also called alkaline.	8
Benzene	The aromatic compound C_6H_6, a flat ring of six carbon atoms, each bonded to one hydrogen atom.	4
Boltzmann constant	$k_B = 1.380\,649 \times 10^{-23}$ J K^{-1} (exact); relates thermal energy per particle to temperature.	9
Boltzmann factor	$e^{-E/RT}$; approximately the fraction of molecules with an energy of at least E.	10
Bond enthalpy	The energy needed to break a particular bond in the gas phase, given per mole of bonds.	3
Bond length	The distance between the nuclei of two bonded atoms.	3
Bond order	The number of electron pairs shared between two bonded atoms: 1 for a single, 2 for a double, 3 for a triple bond.	3
Buffer	A solution containing a weak acid and its conjugate base, which resists changes in pH.	8
Calorie	An older unit of energy; the thermochemical calorie is exactly 4.184 J.	9
Carbonyl group	A carbon atom double-bonded to an oxygen atom, C=O.	11
Carboxy group	The group –COOH, a carbon atom double-bonded to one oxygen atom and single-bonded to an –OH group.	4
Carboxylic acid	A compound containing the carboxy group, –COOH.	11
Catalyst	A substance that increases the rate of a reaction without changing its standard Gibbs energy change.	10

Term	Definition	Chapter
Cation	A positively charged ion, formed when an atom or group of atoms loses electrons.	2
Chemical equation	A representation of a reaction, with the formulas of the reactants on the left, an arrow, and the formulas of the products on the right.	7
Chemical equilibrium	The state of a reversible reaction in which the forward and reverse rates are equal, so that the amounts of reactants and products no longer change.	8
Chemical formula	A combination of element symbols and subscript numbers giving the number of atoms of each element in a substance, such as H_2O.	2
Chemical reaction	A process in which substances are converted into different substances by the breaking and forming of chemical bonds.	7
Chiral	Not superimposable on its mirror image.	11
Coefficient	A number placed in front of a formula in a chemical equation, giving the number of molecules or moles that take part.	7
Complementary	Of two sequences: able to form Watson–Crick pairs along their whole length.	12
Compound	A pure substance made of two or more elements combined in fixed proportions.	2
Condensation	A reaction in which two molecules join to form a larger molecule, releasing a small molecule, usually water.	7
Condensed formula	A formula that lists atoms in the order in which they are connected, grouping each carbon with its hydrogen atoms, such as CH_3CH_2OH.	4
Conjugate base	The species that remains when an acid has lost a proton, such as acetate from acetic acid.	8
Conservation of mass	The law that the total mass of the products of a chemical reaction equals the total mass of the reactants.	7
Constitutional isomers	Isomers whose atoms are bonded in a different order.	11
Coupling	Driving an unfavourable reaction by linking it, through a shared intermediate, to a favourable one.	9
Covalent bond	A chemical bond formed by a pair of electrons shared between two atoms.	3
Curved arrow	A symbol showing the movement of an electron pair in a reaction mechanism.	11

Term	Definition	Chapter
D and L	Labels for the two enantiomeric series of sugars and amino acids, based on comparison with glyceraldehyde.	11
Dalton (Da)	The unit of atomic and molecular mass, one-twelfth of the mass of a carbon-12 atom, $1.660\ 539\ 068\ 92 \times 10^{-27}$ kg; also called the unified atomic mass unit (u).	2
Dashed bond	In a structural drawing, a bond that points behind the page, away from the viewer.	4
Deoxyribose	The sugar of DNA, ribose lacking the hydroxy group on carbon 2′.	12
Derived unit	A unit formed by multiplying or dividing base units, such as the joule, 1 kg m^2 s^{-2}.	1
Dipole	A separation of positive and negative charge; a molecule with a dipole has a partially positive and a partially negative end.	5
Dipole moment	A measure of the size of a dipole, usually given in debye (D); water has 1.857 D.	5
Dipole-dipole interaction	The attraction between the positive end of one polar molecule and the negative end of another.	5
Dispersion force	A weak attraction between all molecules, caused by temporary fluctuations in their electron clouds; also called the London force.	5
Double bond	A covalent bond in which two pairs of electrons are shared.	3
Dynamic equilibrium	An equilibrium in which forward and reverse processes continue at equal rates, so that no net change occurs.	8
e	The base of natural exponentials and logarithms, 2.718 28…; e^x is also written exp(x).	1
Electron	A particle with charge −e and mass 9.1094×10^{-31} kg, found outside the nucleus of an atom.	2
Electron configuration	A list of the number of electrons in each subshell of an atom, such as $1s^2\ 2s^2\ 2p^2$ for carbon.	3
Electronegativity	The tendency of an atom in a bond to attract the shared electrons towards itself; on the Pauling scale, H 2.20, C 2.55, N 3.04, O 3.44.	5
Electronvolt (eV)	The energy gained by an elementary charge moving through a potential difference of one volt; exactly $1.602\ 176\ 634 \times 10^{-19}$ J.	3
Electrophile	An electron-poor species that forms a bond by accepting both bonding electrons from its partner.	11

Term	Definition	Chapter
Element	A pure substance whose atoms all have the same atomic number; it cannot be broken down by chemical means.	2
Elementary charge (e)	The charge of a proton, exactly $1.602\,176\,634 \times 10^{-19}$ C; the electron has charge $-$e.	2
Enantiomers	Two molecules that are non-superimposable mirror images of each other.	11
5′ end and 3′ end	The two ends of a nucleic acid chain; sequences are written from 5′ to 3′.	12
Endergonic	Having $\Delta G > 0$; unfavourable.	9
Endothermic	Absorbing heat from the surroundings ($\Delta H > 0$).	9
Energy	The capacity to do work or to supply heat; SI unit the joule.	9
Energy level	One of the fixed energies that an electron in an atom can have.	3
Enthalpy	H; its change, ΔH, equals the heat exchanged by a process at constant pressure.	9
Entropy	S; a measure of the number of microscopic arrangements consistent with the observed state of a system.	9
Enzyme	A biological catalyst, usually a protein.	10
Equilibrium constant (K)	For $aA + bB \rightleftharpoons cC + dD$, the ratio $[C]^c[D]^d/([A]^a[B]^b)$ at equilibrium, with concentrations relative to 1 mol L^{-1}; fixed at a given temperature.	8
Ester	A compound formed from an acid and an alcohol with the loss of water.	11
Ether	A compound with an oxygen atom bonded to two carbon atoms, C–O–C.	11
Exergonic	Having $\Delta G < 0$; favourable.	9
Exothermic	Releasing heat to the surroundings ($\Delta H < 0$).	9
Exponent	The number that states how many times the base appears as a factor in a power; in 10^3 the exponent is 3. A negative exponent denotes division.	1
Exponential change	Change by a constant factor in each equal interval of time, so that the elapsed time appears in an exponent.	1
First law of thermodynamics	Energy is conserved: it can be transferred and converted but not created or destroyed.	9
First-order reaction	A reaction whose rate is proportional to the concentration of one reactant.	10

Term	Definition	Chapter
Fischer projection	A drawing convention for chains of stereocentres: vertical bonds point away from the viewer, horizontal bonds towards.	11
Formal charge	A charge assigned to an individual atom in a structure, such as O^- with one bond or N^+ with four.	4
Functional group	A group of atoms that gives a molecule characteristic properties and reactions, largely independent of the rest of the molecule.	11
Furanose	The five-membered ring form of a sugar (four carbons and one oxygen).	11
Gas	A state of matter in which particles are far apart and move freely, so that the substance fills its container.	2
Gibbs energy	$G = H - TS$; at constant temperature and pressure a change is spontaneous only if $\Delta G < 0$.	9
Ground state	The lowest-energy, most stable state of an atom or molecule.	3
Group (periodic table)	A column of the periodic table; elements in a group have the same number of valence electrons and similar chemistry.	2
Half-life ($t_{1/2}$)	The time in which a quantity that decays exponentially falls to half its value; $t_{1/2} = \ln 2 \,/\, k$.	1
Haworth projection	A drawing of a sugar ring seen obliquely, with the near edge drawn thick and groups placed above or below the plane of the ring.	4
Heat	Energy transferred from one body to another because of a difference in temperature.	9
Henderson-Hasselbalch equation	The relation $pH = pK_a + \log_{10}([A^-]/[HA])$ between pH and the ratio of the two forms of a weak acid.	8
Hess's law	The enthalpy change of a reaction is the same whatever the path, and can be calculated from enthalpies of formation.	9
Heterocycle	A ring containing at least one atom other than carbon, such as oxygen or nitrogen.	4
Homochirality	The use of only one enantiomer of each kind of building block, as in living organisms.	11
Hybrid orbital	A combination of s and p orbitals pointing in the direction of a bond; sp^3, sp^2 and sp hybrids correspond to tetrahedral, planar and linear geometry.	3
Hydration shell	The layer of oriented water molecules surrounding a dissolved ion or polar molecule.	5
Hydrocarbon	A compound containing only carbon and hydrogen.	11

Term	Definition	Chapter
Hydrogen bond	An attraction X–H $\cdot\cdot\cdot$ Y between a hydrogen atom bonded to an electronegative atom X (usually O or N) and an atom Y carrying a lone pair (usually O or N).	5
Hydrogen-bond acceptor	The atom with a lone pair, usually O or N, that receives a hydrogen bond.	5
Hydrogen-bond donor	The group X–H, usually O–H or N–H, that provides the hydrogen atom of a hydrogen bond.	5
Hydrolysis	A reaction in which water breaks a bond, its H going to one fragment and its OH to the other; the reverse of condensation.	7
Hydronium ion	The ion H_3O^+, a water molecule carrying an extra proton.	7
Hydrophilic	Interacting favourably with water; describing polar and charged groups.	5
Hydrophobic	Not interacting favourably with water; describing non-polar groups.	5
Hydrophobic effect	The tendency of non-polar molecules to cluster together in water, reducing their contact with it.	5
Hydroxide ion	The ion OH^-.	7
Hydroxy group	The group –OH bonded to a carbon atom.	4
Ideal gas law	The relation $pV = nRT$ between the pressure, volume, amount and absolute temperature of a gas.	6
Independent events	Events such that the outcome of one does not affect the probability of the other; the probability that all occur is the product of their probabilities.	1
Intermolecular force	An attraction between molecules, much weaker than a covalent bond.	5
International System of Units (SI)	The system of units used in science, built on seven base units including the metre, kilogram, second, kelvin and mole.	1
Ion	An atom or group of atoms that carries a net electric charge because it has gained or lost electrons.	2
Ion product of water (K_w)	The equilibrium constant for the self-ionisation of water, 1.0×10^{-14} at 25 °C ($pK_w = 13.99$).	8
Ionic bond	The electrical attraction between oppositely charged ions.	3
Ionic compound	A compound made of positive and negative ions held together by their electrical attraction, forming a crystal rather than separate molecules.	2
Ionisation energy (first)	The minimum energy needed to remove one electron from a neutral atom in its ground state.	3

Term	Definition	Chapter
Isomers	Compounds that have the same molecular formula but different structures.	4
Isotope	One of two or more forms of an element whose atoms have the same atomic number but different mass numbers.	2
Joule	The SI unit of energy, 1 kg m^2 s^{-2}.	9
Kelvin (K)	The SI unit of absolute temperature; T/K = t/°C + 273.15.	1
Ketone	A compound in which a carbonyl carbon is bonded to two carbon atoms.	11
Ketose	A sugar whose carbonyl group is a ketone.	11
Kinetic control	A situation in which the products are determined by the relative rates of competing reactions.	10
Le Chatelier's principle	The rule that a system at equilibrium, when disturbed, shifts in the direction that partly counteracts the disturbance.	8
Leaving group	The group that departs, with its bonding electrons, when a nucleophile attacks an atom.	11
Lewis structure	A drawing of a molecule in which shared electron pairs are shown as lines and lone pairs as pairs of dots.	3
Limiting reactant	The reactant that is used up first in a reaction and so determines the amount of product formed.	7
Liquid	A state of matter in which particles are close together but can move past one another, so that the substance takes the shape of its container but keeps its volume.	2
Logarithm ($\log_{10}$)	The power to which 10 must be raised to give a number: $\log_{10}\ 1000 = 3$.	1
Logarithmic axis	A graph axis on which equal distances represent equal factors, so that 1, 10, 100 and 1000 are equally spaced.	1
Logarithmic scale	A scale that reports a quantity by its logarithm, so that each step of 1 is a factor of ten; examples are pH and log epsilon.	1
Lone pair	A pair of valence electrons that belongs to one atom and is not shared in a bond.	3
Mass number (A)	The total number of protons and neutrons in a nucleus.	2
Mechanism	The step-by-step description of how a reaction occurs.	11
Metal	An element that conducts electricity and heat, has a lustre, and in compounds tends to lose electrons and form cations.	2

Term	Definition	Chapter
Methyl group	The group CH_3–.	4
Mixture	A sample of matter containing two or more pure substances in proportions that can vary.	2
Molar (M)	The unit mol L^{-1} of amount concentration.	6
Molar gas constant (R)	The constant in the ideal gas law, 8.314 J mol^{-1} K^{-1}; the product of the Avogadro and Boltzmann constants.	6
Molar mass (M)	The mass of one mole of a substance, usually in g mol^{-1}; numerically equal to the relative molecular mass.	6
Mole (mol)	The SI unit of amount of substance; one mole contains exactly $6.022\ 140\ 76 \times 10^{23}$ specified entities.	6
Molecular formula	A formula that gives only the number of atoms of each element in a molecule, such as C_2H_6O.	4
Molecule	A group of two or more atoms held together by chemical bonds; the smallest unit of a molecular substance.	2
Monomer	A small molecule that can be joined with others of its kind to form a polymer.	12
Natural logarithm (ln)	The logarithm to base e; $\ln x = 2.303 \log_{10} x$.	1
Neutral	Describing a solution in which the concentrations of H_3O^+ and OH^- are equal; pH 7.00 at 25 °C.	8
Neutron	An uncharged particle in the nucleus, with a mass of 1.6749×10^{-27} kg.	2
Nitrile	A compound containing a carbon atom triple-bonded to nitrogen, –C≡N.	11
Noble gas	An element of group 18 (helium, neon, argon and heavier), with a filled outer shell; noble gases hardly react.	2
Non-metal	An element that does not have metallic properties and in compounds tends to gain or share electrons; examples are H, C, N, O, P, S and Cl.	2
Nucleobase	A flat, nitrogen-containing ring compound (A, G, C, U or T) that is part of a nucleotide.	12
Nucleon	A proton or a neutron.	2
Nucleophile	An electron-rich species that forms a bond by donating an electron pair.	11
Nucleoside	A nucleobase bonded to carbon 1′ of a sugar.	12
Nucleotide	A nucleoside carrying one or more phosphate groups, usually on carbon 5′.	12

Term	Definition	Chapter
Nucleus	The small, dense, positively charged centre of an atom, made of protons and neutrons.	2
Octet rule	The tendency of atoms to form bonds until they are surrounded by eight valence electrons, the arrangement of a noble gas.	3
Oligomer	A short chain of a few monomer units.	12
Orbital	The region around a nucleus in which an electron of a given energy is likely to be found.	3
Order of magnitude	The power of ten nearest to a quantity; quantities that differ by a factor of about ten differ by one order of magnitude.	1
Order of reaction	The sum of the powers of the concentrations in the rate law.	10
Organic compound	A compound built on a framework of carbon atoms, usually bonded to hydrogen.	3
Oxidation	The loss of electrons, or an increase in oxidation number.	7
Oxidation number	The charge an atom would have if all its bonds were fully ionic, with shared electrons assigned to the more electronegative atom; also called oxidation state.	7
Oxidising agent	A substance that gains electrons and so oxidises another substance.	7
Partial pressure	The pressure that one gas in a mixture would exert if it alone occupied the volume; the partial pressures add up to the total pressure.	6
Pauli exclusion principle	The rule that no two electrons in an atom can be in the same state, so that an orbital holds at most two electrons, of opposite spin.	3
Peptide	A chain of amino acids joined by peptide bonds.	12
Peptide bond	The amide bond joining two amino acids in a peptide or protein.	12
Period (periodic table)	A row of the periodic table.	2
Periodic table	An arrangement of the elements in order of atomic number, in rows and columns such that elements with similar chemistry fall in the same column.	2
pH	Minus the logarithm to base 10 of the activity of hydronium ions; each unit is a factor of ten.	8
Phosphate	The polyatomic anion PO_4^{3-}, a phosphorus atom bonded to four oxygen atoms; also the corresponding group in compounds such as RNA.	2

Term	Definition	Chapter
Phosphate ester	A compound in which phosphoric acid is joined to an alcohol (monoester) or to two alcohols (diester).	11
Phosphodiester linkage	A phosphate group joined as an ester to the 3′ oxygen of one nucleotide and the 5′ oxygen of the next.	12
Polar bond	A covalent bond in which the electrons are shared unequally, giving partial charges $\delta+$ and $\delta-$ at its ends.	5
Polar molecule	A molecule whose bond dipoles do not cancel, so that it has a net dipole moment.	5
Polyatomic ion	An ion made of several atoms bonded together, such as OH^-, NH_4^+ or PO_4^{3-}.	2
Polymer	A long chain molecule built from many smaller units (monomers) joined by covalent bonds.	12
Power	A number multiplied by itself a given number of times, written with a base and an exponent, such as $2^5 = 32$.	1
Pre-exponential factor	A in the Arrhenius equation; reflects the frequency of suitably oriented collisions.	10
Prefix (SI)	A symbol placed before a unit to multiply it by a power of ten, such as k (10^3) or n (10^{-9}).	1
Pressure	The force per unit area exerted by a gas or liquid, measured in pascals (Pa).	6
Probability	A number between 0 and 1 expressing how likely an event is; 0 means never and 1 always.	1
Product	A substance formed in a chemical reaction.	7
Protein	A long chain of amino acids folded into a specific shape.	12
Proton	A particle in the nucleus with charge +e and a mass of 1.6726×10^{-27} kg. In chemistry, "proton" also means a hydrogen ion, H^+, which is a bare proton.	2
Pure substance	A sample of matter with a fixed composition and fixed properties.	2
Purine	A nucleobase with two fused rings: adenine or guanine.	12
Pyrimidine	A nucleobase with one six-membered ring: cytosine, uracil or thymine.	12
Quantity (physical)	A measurable property, expressed as a number multiplied by a unit.	1
Racemic mixture	A mixture of equal amounts of two enantiomers.	11
Radioactive decay	The spontaneous transformation of an unstable nucleus, which occurs with a fixed probability per unit time.	1

Term	Definition	Chapter
Rate constant	k; the proportionality factor in a rate law, independent of concentration but dependent on temperature.	10
Rate constant (k)	The constant of proportionality between a rate and the amounts on which it depends; for exponential decay, the fraction lost per unit time, with units of inverse time.	1
Rate law	An experimentally found equation relating the rate of a reaction to concentrations.	10
Rate of reaction	The change in concentration of a reactant or product per unit time, divided by its coefficient.	10
Reactant	A substance present at the start of a chemical reaction and consumed by it.	7
Reaction quotient	Q: the ratio of product to reactant concentrations written like K but with actual, not equilibrium, values.	9
Redox reaction	A reaction in which electrons are transferred, so that one substance is oxidised and another reduced.	7
Reducing agent	A substance that loses electrons and so reduces another substance.	7
Reduction	The gain of electrons, or a decrease in oxidation number.	7
Relative molecular mass	The sum of the atomic weights of the atoms in a molecule; numerically equal to the average molecular mass in daltons.	2
Relative permittivity	The factor by which a medium reduces the electrical force between charges compared with a vacuum; 78.4 for water at 25 °C.	5
Residue	A monomer unit as it is present within a polymer chain.	12
Resonance	The description of a molecule by two or more Lewis structures that differ only in the placement of electrons; the real molecule is a blend of them.	4
Reversible reaction	A reaction that proceeds in both directions, so that reactants and products coexist; written with $\rightleftharpoons$.	7
Ribose	The five-carbon sugar $C_5H_{10}O_5$ found in RNA; in RNA it forms a five-membered ring of four carbon atoms and one oxygen atom.	4
Ribozyme	An RNA molecule that acts as a catalyst.	10
Ring	A closed chain of atoms, drawn in skeletal formulas as a polygon.	4

Term	Definition	Chapter
RNA world	The hypothesis that RNA, acting both as genetic material and as catalyst, preceded DNA and proteins.	12
Saturated	Of an organic compound: having only single bonds between carbon atoms.	11
Scientific notation	A way of writing a number as a value between 1 and 10 multiplied by a power of ten, such as 5.29×10^{-11}.	1
Second law of thermodynamics	In any spontaneous change, the total entropy of a system and its surroundings increases.	9
Second-order reaction	A reaction whose rate is proportional to the product of two concentrations, or to the square of one.	10
Sequence	The order of the different kinds of monomer along a polymer chain.	12
Sequence space	The set of all possible sequences of a given length; for RNA of length n it contains 4^n sequences.	1
Shell	A set of orbitals with the same principal number n; electrons in higher shells are farther from the nucleus on average.	3
Side chain	The variable group that distinguishes one amino acid from another.	12
Significant figures	The digits of a reported value that carry information about its precision.	1
Skeletal formula	A drawing in which lines represent bonds, every corner and line end without a symbol is a carbon atom, and hydrogen atoms on carbon are omitted.	4
Skeleton	The framework of carbon atoms of an organic molecule: chains, branches and rings.	11
Slope	The change in the vertical quantity of a graph divided by the corresponding change in the horizontal quantity.	1
Solid	A state of matter in which particles are held in fixed positions and the substance keeps its own shape.	2
Solubility	The maximum amount of a solute that dissolves in a given amount of solvent at a given temperature.	5
Solute	A substance dissolved in a solvent.	5
Solution	A mixture in which one substance is spread evenly through another at the level of molecules or ions.	5
Solvent	The substance, usually a liquid, in which a solute is dissolved.	5
Space-filling model	A model of a molecule in which each atom is a sphere of roughly its real size, showing the overall shape and surface.	4

Term	Definition	Chapter
Spin	An intrinsic property of electrons that can take one of two values, “up” or “down”.	3
Spontaneous	Able to occur without being driven from outside; says nothing about speed.	9
Standard enthalpy of formation	$\Delta_f H^\circ$: the enthalpy change when one mole of a compound forms from its elements in their standard states.	9
Standard Gibbs energy change	ΔG°: the Gibbs energy change with all substances in their standard states; $\Delta G^\circ = -RT \ln K$.	9
Standard state	Reference conditions for tabulated values: 1 bar for gases, 1 mol L^{-1} for solutes, the pure substance for liquids and solids.	9
State function	A quantity whose value depends only on the present state of a system, not on how it was reached.	9
State of matter	One of the forms in which a substance can exist, chiefly solid, liquid and gas.	2
State symbol	A label after a formula giving its physical state: (s) solid, (l) liquid, (g) gas, (aq) dissolved in water.	7
Steady state	A state in which a substance’s concentration is constant because its formation and destruction balance.	10
Stereocentre	An atom, usually carbon bonded to four different groups, at which exchanging two groups gives a stereoisomer.	11
Stereoisomers	Isomers with the same order of bonding but different arrangements in space.	11
Stoichiometry	The calculation of the amounts of reactants and products from a balanced chemical equation.	7
Strong acid	An acid that donates its proton to water essentially completely.	8
Strong nuclear force	The force that holds protons and neutrons together in a nucleus; much stronger than the electrical repulsion between protons, but acting only over distances about the size of a nucleus.	2
Structural formula	A drawing that shows every atom of a molecule and every bond between them.	4
Subshell	A subdivision of a shell, labelled s, p, d or f, containing 1, 3, 5 or 7 orbitals.	3
Substitution	A reaction in which one group in a molecule is replaced by another.	11
Template	A strand whose sequence directs the assembly of a complementary copy.	12

Term	Definition	Chapter
Tetrahedron	A solid with four triangular faces; four electron pairs around an atom point towards its corners, 109.47° apart.	3
Thermodynamic control	A situation in which the products are determined by their relative stabilities at equilibrium.	10
Transition state	The arrangement of atoms of highest energy along the path from reactants to products.	10
Triple bond	A covalent bond in which three pairs of electrons are shared.	3
Unit	A reference amount of a quantity, such as the metre or the second, by which values of that quantity are expressed.	1
Unsaturated	Of an organic compound: having one or more carbon-carbon double or triple bonds.	11
Valence	The number of covalent bonds an atom usually forms: 1 for H, 2 for O, 3 for N, 4 for C.	3
Valence electrons	The electrons in the outermost occupied shell of an atom, which take part in bonding.	3
Van 't Hoff equation	The relation describing how K changes with temperature; heating lowers K of an exothermic reaction.	9
Van der Waals forces	The weak intermolecular forces between molecules, including dipole-dipole and dispersion forces.	5
VSEPR model	Valence shell electron pair repulsion: the principle that electron pairs around a central atom arrange themselves as far apart as possible, which predicts molecular shapes.	3
Watson–Crick base pair	The pairs A–U (A–T in DNA), with two hydrogen bonds, and G–C, with three.	12
Weak acid	An acid that donates its proton only partly, establishing an equilibrium with its conjugate base.	8
Wedge bond	In a structural drawing, a bond that points out of the page towards the viewer.	4
Ångström (Å)	A unit of length equal to 10^{-10} m, convenient for bond lengths.	3

B. Units and Constants

This appendix collects the units, physical constants and conversion factors used in the book, and lists characteristic magnitudes that appear in the chapters.

B.1. SI units and prefixes

The book uses the International System of Units (SI) (Bureau International des Poids et Mesures 2019). Table B.1 lists the units that occur most often, and Table B.2 the prefixes used to form multiples and submultiples (Chapter 1).

Table B.1.: SI units used in this book.

Quantity	Unit	Symbol	Expressed in other SI units
Length	metre	m	base unit
Mass	kilogram	kg	base unit
Time	second	s	base unit
Temperature	kelvin	K	base unit
Amount of substance	mole	mol	base unit
Energy	joule	J	$\mathrm{kg\ m^2\ s^{-2}}$
Pressure	pascal	Pa	$\mathrm{N\ m^{-2}} = \mathrm{kg\ m^{-1}\ s^{-2}}$
Electric charge	coulomb	C	A s
Concentration (amount)	mole per litre	$\mathrm{mol\ L^{-1}}$ (also written M)	$10^3\ \mathrm{mol\ m^{-3}}$

Table B.2.: SI prefixes used in this book (Bureau International des Poids et Mesures 2019).

Prefix	Symbol	Factor	Prefix	Symbol	Factor
kilo	k	10^3	milli	m	10^{-3}
mega	M	10^6	micro	µ	10^{-6}
giga	G	10^9	nano	n	10^{-9}

Prefix	Symbol	Factor	Prefix	Symbol	Factor
tera	T	10^{12}	pico	p	10^{-12}
			femto	f	10^{-15}

Several units outside the SI are used because they are standard in chemistry:

- **Degree Celsius** (°C). A temperature t in °C corresponds to the absolute temperature T = t + 273.15 K. A temperature difference of 1 °C equals a difference of 1 K.
- **Litre** (L), equal to 1 $dm^3 = 10^{-3}$ m^3.
- **Bar**, equal to 10^5 Pa exactly; the **standard atmosphere** (atm) is 101 325 Pa exactly (National Institute of Standards and Technology, n.d.-a).
- **Ångström** (Å), equal to 10^{-10} m = 0.1 nm, used for bond lengths.
- **Dalton** (Da), also called the unified atomic mass unit (u): one twelfth of the mass of an atom of carbon-12 (Table B.3).
- **Calorie**. The thermochemical calorie is exactly 4.184 J (Thompson and Taylor 2008); the "Calorie" of food labels is a kilocalorie.
- **Debye** (D), the unit of dipole moment used in Chapter 5, equal to about 3.335 64 $\times$ 10^{-30} C m.
- **Electronvolt** (eV), the energy gained by an elementary charge across a potential difference of 1 volt, exactly 1.602 176 634 $\times$ 10^{-19} J (National Institute of Standards and Technology, n.d.-a).

B.2. Physical constants

Table B.3 lists the constants used in the book, from the 2022 CODATA adjustment (National Institute of Standards and Technology, n.d.-a). Values marked exact follow from the definition of the SI; for the molar gas constant, which is a product of two exact constants, the digits shown are truncated. The last two values are measured, and the digits in parentheses give the standard uncertainty in the last digits.

Table B.3.: Physical constants, CODATA 2022 (National Institute of Standards and Technology, n.d.-a).

Constant	Symbol	Value
Avogadro constant	N_A	6.022 140 76 $\times$ 10^{23} mol^{-1} (exact)
Boltzmann constant	k_B	1.380 649 $\times$ 10^{-23} J K^{-1} (exact)
Molar gas constant	$R = N_A k_B$	8.314 462 618 J mol^{-1} K^{-1} (exact)

Constant	Symbol	Value
Elementary charge	e	$1.602\ 176\ 634 \times 10^{-19}$ C (exact)
Atomic mass constant (1 Da)	m_u	$1.660\ 539\ 068\ 92(52) \times 10^{-27}$ kg
Electron mass	m_e	$9.109\ 383\ 7139(28) \times 10^{-31}$ kg

B.3. Useful conversions and reference values

Chemical energies are given per mole, and energies of single particles per particle. The two are related by the Avogadro constant, $E_{molar} = N_A \times E_{particle}$. From the values of Table B.3 follow the reference values of Table B.4.

Table B.4.: Conversions and reference values.

Quantity	Value
1 eV per particle	96.485 kJ mol^{-1}
1 kcal mol^{-1}	4.184 kJ mol^{-1}
RT at 25 °C (298.15 K)	2.479 kJ mol^{-1}
RT at 37 °C (310.15 K)	2.579 kJ mol^{-1}
RT ln 10 at 25 °C (change in $\Delta G°$ for a tenfold change in K)	5.708 kJ mol^{-1}
Volume of one mole of ideal gas at 25 °C and 1 bar	24.8 L
ln 2 (half-life = ln 2 / k)	0.693

B.4. Orders of magnitude in this book

Table B.5 collects characteristic quantities from the chapters.

Table B.5.: Characteristic magnitudes discussed in this book.

Quantity	Value	Chapter
Size		
Covalent bond lengths in Chapter 3	0.96 to 1.54 Å	3
Energy		

Quantity	Value	Chapter
Thermal energy per mole at 25 °C, RT	2.479 kJ mol^{-1}	9
Bond enthalpies in Chapter 3	416 to 945 kJ mol^{-1}	3
Formation of liquid water from H_2 and O_2, $\Delta G°$	−237.14 kJ mol^{-1}	9
Evaporation of water at 25 °C, $\Delta H°$	+44.0 kJ mol^{-1}	9
Concentration and number		
Water in liquid water at 25 °C	about 55 mol L^{-1}	6
H_3O^+ in pure water at 25 °C	about 10^{-7} mol L^{-1} (pH 7.00)	8
RNA sequences of length 100	$4^{100} \approx 1.6 \times 10^{60}$	12
Time		
Half-life of carbon-14	5700 years	1
Half-life of ribose at 100 °C, pH 7	73 min	10

C. Answers to the Self-Test Questions

Chapter 1

1. (a) 4.5×10^{-4} m; (b) 6.371×10^{6} m. $(2 \times 10^{-3}) \times (4 \times 10^{5}) = 8 \times 10^{2} = 800$. $(9 \times 10^{4}) \,/\, (3 \times 10^{-2}) = 3 \times 10^{6}$.
2. $37 + 273.15 = 310.15$ K. 2.5 nm $= 2.5 \times 10^{-9}$ m.
3. 32 days is 4 half-lives, so $(1/2)^4 = 1/16$ of the sample remains. $k = \ln 2 \,/\, t_{1/2} = 0.693 \,/\, 8$ days $= 0.0866$ day^{-1}.
4. $10^{-3} \,/\, 10^{-6} = 10^{3}$: solution A has a thousand times the hydrogen-ion concentration. pH of A = 3; pH of B = 6.
5. $0.999^{500} = 0.606$, so about 61 % of copies are error-free.

Chapter 2

1. Oxygen gas: element. Sea water: mixture. Carbon dioxide: compound. Gold: element. Air: mixture.
2. Oxygen-18: 8 protons, $18 - 8 = 10$ neutrons, 8 electrons. $^{16}O^{2-}$: 8 protons, 8 neutrons, $8 + 2 = 10$ electrons.
3. Both have 7 protons (and 7 electrons in the neutral atom), so both are nitrogen with the same chemistry. Nitrogen-14 has 7 neutrons and nitrogen-15 has 8, so they differ in mass.
4. $MgCl_2$ (one 2+ charge balanced by two 1− charges). Na_3PO_4 (three 1+ charges balance one 3− charge).
5. CH_4: $12.011 + 4 \times 1.0080 = 16.043$. NH_3: $14.007 + 3 \times 1.0080 = 17.031$.

Chapter 3

1. $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^4$; six valence electrons. Oxygen is in the same group.
2. Removing one electron leaves Na^+ with the arrangement of neon, a filled shell. Removing a second electron would break into this filled shell, which, like the shell of neon itself (first ionisation energy 21.56 eV), holds its electrons very tightly. The second electron therefore costs far more energy than the first (5.14 eV).

3. H–S–H, with two lone pairs on sulfur, which has six valence electrons like oxygen. Four electron pairs around sulfur give a bent molecule, like water.
4. Formaldehyde: carbon in the centre, single bonds to two hydrogen atoms, a double bond to oxygen, and two lone pairs on oxygen; C–O bond order 2. Hydrogen cyanide: H–C≡N with one lone pair on nitrogen; C–N bond order 3.
5. Carbon forms 4 bonds (to three H and to O), oxygen 2 (to C and to H), each hydrogen 1. The structure is H_3C–O–H: a carbon bonded to three hydrogen atoms and to an oxygen, which carries the fourth hydrogen and two lone pairs.

Chapter 4

1. $CH_3CH_2CH_3$. The skeletal formula is a single bent line with two segments (a "V"): each end is a CH_3 group and the corner is a CH_2 group.
2. The carbon at the line end (the CH_3 group) carries three hydrogen atoms; the carbon bonded to both oxygen atoms carries none, since it already has four bonds (one to carbon, two to the double-bonded oxygen, one to OH). Adding the hydrogen of the OH group gives $C_2H_4O_2$.
3. A chain of three carbon atoms with an OH group on the middle carbon: H_3C–CH(OH)–CH_3, with every hydrogen drawn. Its molecular formula is C_3H_8O, the same as propan-1-ol, but the OH group is on a different carbon, so the two compounds are isomers. (This compound is propan-2-ol.)
4. In cyclohexane, each carbon is a corner with two single bonds and carries two hydrogen atoms: C_6H_{12}. In benzene, each carbon has one double and one single bond in the ring and carries one hydrogen atom: C_6H_6.
5. Above the ring: the CH_2OH group on carbon 4 and the OH group on carbon 1. Below the ring: the OH groups on carbons 2 and 3. The remaining positions carry hydrogen atoms (below on carbons 1 and 4, above on carbons 2 and 3).

Chapter 5

1. Differences in electronegativity: C–H 0.35, N–H 0.84, C–O 0.89, O–H 1.24. From least to most polar: C–H < N–H < C–O < O–H. The $\delta-$ end is C in C–H, N in N–H, and O in C–O and O–H.
2. Carbon dioxide is linear, so its two C=O bond dipoles point in exactly opposite directions and cancel. Sulfur dioxide is bent, so its two S–O bond dipoles do not cancel, and the molecule is polar.
3. Each of the three N–H groups is a hydrogen-bond donor; the lone pair on nitrogen is an acceptor. Yes: an N–H group of one molecule can point at the lone pair of another, forming N–H $\cdot\cdot\cdot$ N.
4. Ethanol has an O–H group that can donate and accept hydrogen bonds, so it fits into the hydrogen-bond network of water. Hexane has only C–C and C–H bonds;

it is non-polar and cannot form hydrogen bonds, so it is excluded from water by the hydrophobic effect.
5. Ice would sink as it formed. Lakes would freeze from the bottom upwards, and in a cold climate they could freeze solid, instead of being protected by a floating layer of ice with liquid water beneath.

Chapter 6

1. M = 6 × 12.011 + 12 × 1.0080 + 6 × 15.999 = 180.16 g mol^{-1}. 90 g / 180.16 g mol^{-1} = 0.50 mol.
2. 1.0 g / 18.015 g mol^{-1} = 0.0555 mol; 0.0555 mol × 6.022 × 10^{23} mol^{-1} = 3.3 × 10^{22} molecules.
3. 5.85 g / 58.44 g mol^{-1} = 0.100 mol; 0.100 mol / 0.250 L = 0.400 mol L^{-1} (0.400 M).
4. The concentration rises fiftyfold: 2 μM × 50 = 100 μM (0.1 mM).
5. V = nRT/p = 2.0 × 8.314 × 298.15 / 10^5 m^3 = 0.0496 m^3 = 49.6 L.

Chapter 7

1. $C_2H_5OH + 3\ O_2 \rightarrow 2\ CO_2 + 3\ H_2O$. (Check: 2 C, 6 H and 7 O on each side.)
2. $2\ H_2 + O_2 \rightarrow 2\ H_2O$, so one mole of water forms per mole of H_2. 4.0 g of H_2 (molar mass 2.016 g mol^{-1}) is 1.98 mol, giving 1.98 mol × 18.015 g mol^{-1} = 35.7 g, about 36 g of water.
3. HCl donates a proton and is the acid; NH_3 accepts it and is the base.
4. CO: oxygen −2, so carbon +2. NH_3: hydrogen +1 each, so nitrogen −3. N_2: an element, so 0.
5. Water is released. Glycine is $C_2H_5NO_2$ and alanine $C_3H_7NO_2$; together $C_5H_{12}N_2O_4$, minus H_2O, gives $C_5H_{10}N_2O_3$.

Chapter 8

1. $K = [NH_3]^2 / ([N_2]\ [H_2]^3)$.
2. [AB]/[A] = K[B]. At [B] = 0.1 M: 100 × 0.1 = 10, so AB : A = 10 : 1 and 10/11 ≈ 91 % of A is joined. At [B] = 10 μM = 10^{-5} M: 100 × 10^{-5} = 10^{-3}, so only about 0.1 % of A is joined.
3. pH = $-\log_{10}(2.0 \times 10^{-5})$ = 4.70. It is acidic (below 7).
4. $\log_{10}([HCOO^-]/[HCOOH])$ = 5.75 − 3.75 = 2, so the ratio is 10^2 = 100.

5. In the buffer, the added H_3O^+ ions are taken up by HPO_4^{2-}, forming $H_2PO_4^-$; the ratio of the two forms changes only slightly, so the pH barely moves. Pure water has no such base to take up the protons, so all the added H_3O^+ remains free and the pH falls sharply.

Chapter 9

1. $\Delta H^\circ = \Delta_f H^\circ(H_2O, l) - \Delta_f H^\circ(H_2O, g) = -285.830 - (-241.826) = -44.004$ kJ mol^{-1}. Condensation releases heat: it is exothermic, the reverse of evaporation.
2. $\Delta G = 20 - 298.15 \times 0.100 = -9.8$ kJ mol^{-1}, so yes, it is spontaneous at 25 °C. $\Delta G = 0$ at $T = \Delta H/\Delta S = 20\,000/100 = 200$ K; the reaction is spontaneous at all temperatures above 200 K.
3. -11.4 kJ mol^{-1} is about two steps of 5.71 kJ mol^{-1} in the favourable direction, so $K \approx 10^2 = 100$.
4. $Q > K$, so $\Delta G = RT \ln(Q/K)$ is positive and the reaction proceeds backwards, converting products into reactants until Q falls to K.
5. Joining two molecules into one reduces the number of independent particles, so ΔS is negative; if the joining releases heat, ΔH is also negative. This is the third case of Table 9.1: $\Delta G = \Delta H - T\Delta S$, and the term $-T\Delta S$ is positive and grows with T. The joining is therefore most favourable at low temperature and becomes less favourable as the temperature rises; equivalently, K of an exothermic reaction decreases on heating.

Chapter 10

1. $t_{1/2} = 0.693/0.01$ s^{-1} $= 69.3$ s. After 5 minutes (300 s), the fraction remaining is $e^{-0.01\times300} = e^{-3} \approx 0.050$, or 5 %.
2. Doubling [A] and halving [B] leaves the product [A][B] unchanged, so the rate does not change. Reducing both to one tenth multiplies the rate by $(1/10) \times (1/10) = 1/100$.
3. A factor of about 3.7, almost twice the factor of 2 given by the rule of thumb. The rule holds only for activation energies near 50 kJ mol^{-1}.
4. By the same factor, 1000. A catalyst does not change ΔG°, so K is unchanged; equilibrium is reached sooner.
5. The rate constant of destruction is inversely proportional to the half-life, so it increases tenfold. The steady-state concentration, P/k, falls to one tenth of its former value.

Chapter 11

1. (a) Aldehyde (propanal). (b) Ketone (acetone). (c) Amide (acetamide). (d) Ester (methyl acetate). (e) Amine and carboxylic acid; this is glycine, the simplest amino acid.
2. No. The central carbon carries –OH, –H and two identical $–CH_3$ groups. Because two of its four groups are the same, its mirror image can be turned to coincide with the original.
3. $2^4 = 16$ stereoisomers, forming 8 pairs of enantiomers. One of them is D-glucose and its mirror image is L-glucose.
4. Hydroxide, OH^-, is the nucleophile and the carbonyl carbon is the electrophile. The first arrow starts at a lone pair on the oxygen of hydroxide and ends at the carbonyl carbon, forming a new C–O bond. The second arrow starts at the C=O double bond and ends at the carbonyl oxygen, which receives the electron pair and a negative charge.
5. Formaldehyde is achiral, and without any chiral influence the synthesis produces equal amounts of D- and L-glyceraldehyde, a racemic mixture. The two enantiomers rotate polarised light by equal amounts in opposite directions, so their effects cancel.

Chapter 12

1. A nucleobase, a sugar (ribose in RNA) and a phosphate group. The base is bonded to carbon 1′ of the sugar, and the phosphate is joined as an ester to the hydroxy group on carbon 5′. Both bonds are formed, formally, by condensation, each releasing a molecule of water.
2. The complement, written 5′ to 3′, is 5′-UGCCAU-3′. The duplex contains three A–U pairs (2 hydrogen bonds each) and three G–C pairs (3 each), so $3 \times 2 + 3 \times 3 = 15$ hydrogen bonds.
3. $4^{10} = 1\,048\,576$, about 10^6.
4. DNA lacks the hydroxy group on carbon 2′. In RNA, the oxygen of this group lies next to the phosphodiester linkage and can attack the phosphorus atom as a nucleophile, cutting the chain. Without it, the DNA backbone can be cut only by much slower reactions.
5. Two peptide bonds join three amino acids, and each condensation releases one water molecule, so two molecules of water are released.